Introductory Notes on Mathematics of Deep Learning

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April 1, 2026

Preface

These introductory lecture notes on the mathematics of deep learning were prepared for a senior undergraduate and introductory graduate-level course taught at the University of Waterloo. We assume that students and readers have no background beyond advanced calculus, linear algebra, and introductory probability or statistics. While familiarity with real analysis and measure theory may enhance understanding of certain parts, it is not strictly required. While the book aims to be self-contained whenever possible, with all supplementary background material included in the appendix, any unproved claims will be accompanied by a reference.

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Notation

$\mathbb{R}$	the set of real numbers
$\mathbb{R}_{\geq 0}$	the set of non-negative real numbers
$\mathbb{R}^d$	the d-dimensional Euclidean space ($\mathbb{R}^1 = \mathbb{R}$)
$\mathbb{Z}$	the set of all integer numbers
$\mathbb{Z}_{\geq 0}$	the set of all non-negative integer numbers
$\mathbb{N}$	the set of natural numbers $\{1, 2, \ldots\}$
$\lvert x \rvert$	the 2-norm (Euclidean norm) of a vector $x \in \mathbb{R}^d$, defined as $\lvert x \rvert = \sqrt{\sum_{i=1}^d x_i^2}$
$\lVert x \rVert_\infty$	the infinity norm (maximum norm) of a vector $x \in \mathbb{R}^d$, defined as $\lVert x \rVert_\infty = \max_{1 \leq i \leq d} \lvert x_i \rvert$
$x^\top$	the transpose of a vector $x \in \mathbb{R}^d$
$x \cdot y$	the inner product of two vectors $x, y \in \mathbb{R}^d$, i.e., $x \cdot y = x^\top y$
$\lVert A \rVert$	the matrix 2-norm (spectral norm) of a matrix $A \in \mathbb{R}^{m \times n}$, defined as $\lVert A \rVert = \sup_{\lvert x \rvert = 1} \lvert Ax \rvert$
$\mathscr{C}(\mathbb{R}^d)$	the set of all real-valued continuous functions defined on $\mathbb{R}^d$
$\mathscr{C}(\mathbb{R}^d, \mathbb{R}^k)$	the set of all continuous functions mapping from $\mathbb{R}^d$ to $\mathbb{R}^k$
$\mathscr{C}^k(\mathbb{R}^d, \mathbb{R}^k)$	the set of all functions mapping from $\mathbb{R}^d$ to $\mathbb{R}^k$ that have continuous partial derivatives up to order k
$\mathscr{C}^\infty(\mathbb{R}^d, \mathbb{R}^k)$	the set of all functions mapping from $\mathbb{R}^d$ to $\mathbb{R}^k$ that have continuous partial derivatives of all orders
$\mathscr{C}_0^\infty(\mathbb{R}^d, \mathbb{R}^k)$	the set of all functions mapping from $\mathbb{R}^d$ to $\mathbb{R}^k$ that have continuous partial derivatives of all orders and compact support (i.e., they are zero outside a compact subset of $\mathbb{R}^d$).
$\lvert A \rvert$	the cardinality of a finite set A

A^n	the set of all n-tuples $(a_1, \ldots, a_n)$ with $a_i \in A$ (also called the Cartesian nth power of the set A)
$A \setminus B$	the set difference between A and B, defined as $A \setminus B = \{x \in A : x \notin B\}$
A^n	the set of all n-tuples $(a_1, \ldots, a_n)$ with $a_i \in A$
$\mathbb{E}[\cdot]$	the expected value of a random variable
$f_t = o(g_t)$	the little-o notation, which indicates that $f_t/g_t \to 0$ as $t \to \infty$, where $\{f_t\}$ and $\{g_t\}$ are positive real-valued sequences indexed by t
$f_t = O(g_t)$	the big-O notation, which indicates that there exists some $C > 0$ such that $f_t \leq C g_t$ for all t sufficiently large
$f_t = \Theta(g_t)$	the big-Θ notation, which indicates that $f_t = O(g_t)$ and $g_t = O(f_t)$

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Chapter 1

Introduction

Neural networks have a long history, originating from early efforts to model human cognition and develop early neuron-based computational models inspired by the brain (McCulloch and Pitts, 1943; Minsky and Papert, 1969). Over the decades, their development has evolved significantly, but recent advances in artificial intelligence have once again propelled them into the spotlight (Goodfellow et al., 2016). Today, neural networks are becoming a ubiquitous computing framework for both scientific and engineering applications.

In this set of notes, we aim to provide a self-contained introduction to the mathematical aspects of learning with (deep) neural networks. While many important theoretical questions remain unanswered, and a complete mathematical understanding of deep learning is still elusive, the main criteria for selecting topics are their fundamental relevance to understanding (deep) learning and the feasibility of providing a self-contained treatment of definitive results. Additional topics and references will be provided.

1.1 Neural Networks

We will focus almost exclusively on fully connected feedforward neural networks.

Definition 1.1. *A fully connected feedforward neural network is a function* $h : \mathbb{R}^d \to \mathbb{R}^k$,

where n is the input dimension and m is the output dimension. The function is constructed as a composition of layers:

$$y = h(x) = h^{(L+1)} \circ h^{(L)} \circ \cdots \circ h^{(1)}(x), \tag{1.1}$$

where

$$h^{(i)}(s) = \sigma_i(W_i s + b_i), \quad i = 1, \ldots, L, \tag{1.2}$$

represents the hidden layers, each of which applies an affine transformation with weight matrix $W_i \in \mathbb{R}^{n_i \times n_{i-1}}$ and bias vector $b_i \in \mathbb{R}^{n_i}$, followed by a nonlinear activation $\sigma_i : \mathbb{R} \to \mathbb{R}$ applied elementwise. The output layer $h^{(L+1)}$ typically applies an affine transformation $W_{L+1}s + b_{L+1}$ without a nonlinear activation function, or a task-dependent activation function for specific tasks.

Figure 1.1 depicts a typical feedforward neural network with three hidden layers.

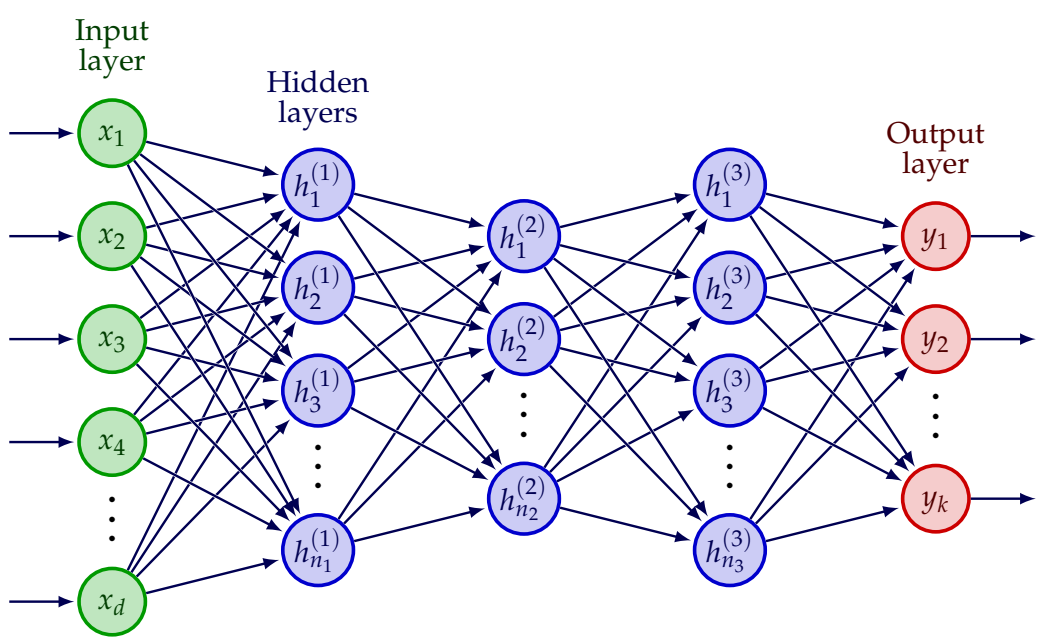

Figure 1.1: A feedforward neural network with three hidden layers.

Example 1.1. Typical activation functions include:

- **Sigmoid:** $\text{sigmoid}(x) = \frac{1}{1+e^{-x}}$, which maps inputs to the range $[0, 1]$.
- **Tanh:** $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$, which maps inputs to the range $[-1, 1]$.

- **ReLU (Rectified Linear Unit):** $\mathrm{ReLU}(x) = \max(0, x)$, commonly used due to its simplicity and effectiveness.
- **Leaky ReLU:** $\mathrm{LeakyReLU}(x) = \max(\alpha x, x)$ with a small slope $\alpha > 0$ for negative inputs.
- **Softmax:** $\mathrm{Softmax}(x)_i = \frac{e^{x_i}}{\sum_j e^{x_j}}$, used in the output layer for multi-class classification to normalize outputs into probabilities.

Some of these functions are illustrated in Figure 1.2.

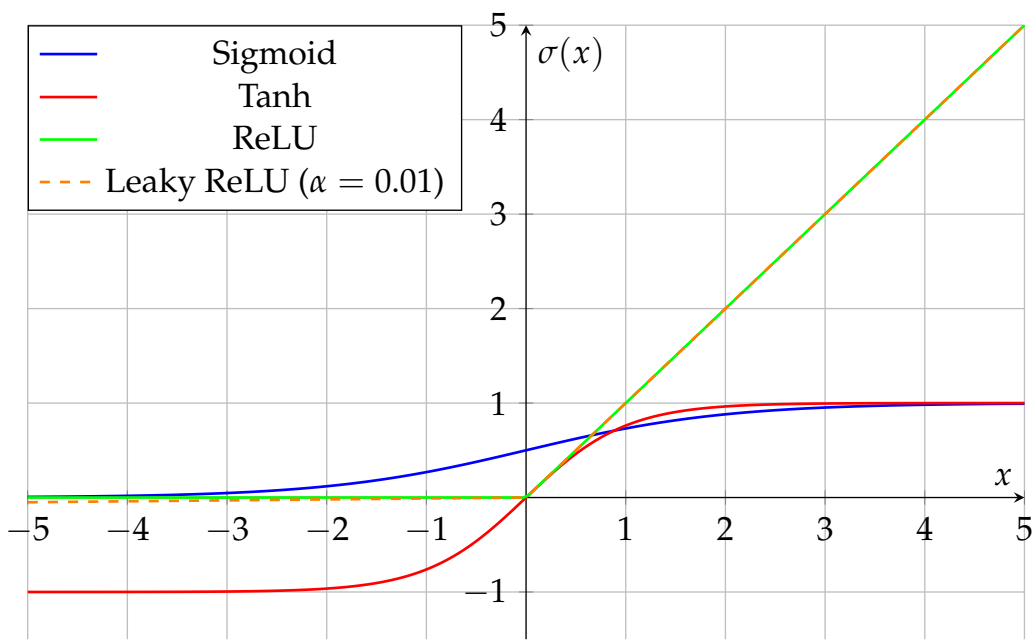

Figure 1.2: Typical activation functions.

Remark 1.1 (Deep versus shallow networks). The distinction between deep and shallow neural networks is based on the number of hidden layers. A network with a single hidden layer is typically referred to as shallow, while networks with two or more hidden layers are classified as deep. While the definition of "deep" can vary across different applications and contexts, we adopt this convention.

1.2 Neural Network Learning

Neural network learning has a long history. Classical works by Legendre (1805) and Gauss (1795) on linear least squares can be viewed as training a single-layer network with linear activation functions (i.e., linear regression), which is essentially a special case of neural network learning.

In general, neural network learning can be understood as the process of adjusting model parameters to capture patterns, structures, or distributions in the data, depending on the specific task and learning paradigm.

More specifically, learning involves updating the connection weights W and biases b within the network. After processing each data sample, the model computes the error between the predicted output and the expected result, often using a loss function. The gradients of the loss with respect to W and b are calculated using backpropagation, a method that efficiently propagates the error backward through the network layers by applying the chain rule for differentiation. These gradients guide the parameter updates, typically through a gradient-based optimization algorithm such as stochastic gradient descent (SGD). This process exemplifies supervised learning, where adjustments to model parameters are made iteratively to improve the network's performance on labeled data.

1.3 Measuring the Performance of Learning

We consider the following basic learning model.

Definition 1.2 (A Basic Learning Model)**.** *Consider the following basic ingredients:*

- X*: input set.*
- Y*: output set.*
- Z*: sample set (typically taken as* $Z = X \times Y$*).*
- $\mathscr{H}$*: a chosen class of functions from* X *to* Y*, called hypothesis class.*
- $L : \mathscr{H} \times Z \to \mathbb{R}_{\geq 0}$*: a loss function.*

A learning algorithm (learner) is a mapping

$$A : \bigcup_{m \in \mathbb{N}} Z^m \to \mathscr{H},$$

which uses training data $S = \{z_i\}_{i=1}^m \in Z^m$ *to select a model* $h = A(S) \in \mathscr{H}$.

Typically, the learning algorithm aims to minimize the **empirical risk** during training:

$$\hat{R}_S(h) = \frac{1}{m} \sum_{i=1}^{m} L(h, z_i).$$

One obvious algorithm for this is the **empirical risk minimization** (ERM) rule:

$$A_{\text{ERM}}(S) = \arg\min_{h \in \mathscr{H}} \hat{R}_S(h).$$

The performance of learning, however, should be measured on unseen data to evaluate the ability of the learned model to generalize beyond the training dataset. For this purpose, define the **expected risk**:

$$R(h) = \mathbb{E}_{z \sim \mathscr{D}}[L(h, z)],$$

where $\mathscr{D}$ is the underlying data distribution. The expected risk quantifies the average loss of the hypothesis h over the entire data distribution. The expected risk is also called the population risk or true error.

Let $\mathscr{M}(X, Y)$ be the set of all measurable[1] functions from X to Y. The best performance we can hope for a learning algorithm is to learn h^* such that

$$R(h^*) = R^* = \inf_{h \in \mathscr{M}(X,Y)} R(h),$$

which is known as the Bayes optimal function. This h^* minimizes the expected risk $R(h)$ over all measurable functions and represents the theoretically best function given complete knowledge of the data distribution.

To make training practical, we must limit the hypothesis class. One option is to use neural network functions of a specific architecture with learnable parameters.

[1] We will not be rigorous with the measurability of sets and functions in these notes. Roughly speaking, being measurable means that a function or set is compatible with the underlying structure of the probability space, allowing us to assign probabilities or compute integrals in a well-defined manner.

Given the hypothesis class $\mathcal{H}$, the best achievable approximation is:

$$h^*_{\mathcal{H}} = \arg\min_{h \in \mathcal{H}} R(h).$$

This leads to the gap:

$$\epsilon_{\text{approx}} = R(h^*_{\mathcal{H}}) - R(h^*) = R(h^*_{\mathcal{H}}) - R^*, \tag{1.3}$$

which defines an **approximation error**.

Without knowledge of the underlying data distribution, the learner faces the challenge of minimizing the expected risk using only the training data. This leads naturally to empirical risk minimization. However, even this empirical optimization problem is typically nonconvex and computationally intractable. Let $h_S = A(S) \in \mathcal{H}$ be the hypothesis returned by a learning algorithm and $\hat{h}^*_S \in \mathcal{H}$ be the hypothesis returned by the ERM rule. The gap

$$\epsilon_{\text{opt}} = \hat{R}_S(h_S) - \hat{R}_S(\hat{h}^*_S) \tag{1.4}$$

gives an **optimization error**.

The ultimate measure of the performance of the learned hypothesis is $R(h_S) - R^*$, the gap between the expected risk of the learned hypothesis and the theoretical best risk. To understand what contributes to this gap, we can write

$$\begin{aligned} R(h_S) - R^* &= (R(h_S) - \hat{R}_S(h_S)) + \underbrace{(\hat{R}_S(h_S) - \hat{R}_S(\hat{h}^*_S))}_{=\varepsilon_{\text{opt}}} \\ &\quad + \underbrace{(\hat{R}_S(\hat{h}^*_S) - \hat{R}_S(h^*_{\mathcal{H}}))}_{\leq 0} + (\hat{R}_S(h^*_{\mathcal{H}}) - R(h^*_{\mathcal{H}})) + \underbrace{(R(h^*_{\mathcal{H}}) - R^*)}_{=\varepsilon_{\text{approx}}} \\ &\leq \varepsilon_{\text{approx}} + \varepsilon_{\text{opt}} + (R(h_S) - \hat{R}(h_S)) + (\hat{R}_S(h^*_{\mathcal{H}}) - R(h^*_{\mathcal{H}})), \end{aligned} \tag{1.5}$$

where we used $\hat{R}_S(\hat{h}^*_S) - \hat{R}_S(h^*_{\mathcal{H}}) \leq 0$ because $\hat{h}^*_S$ is a minimizer of the empirical risk and the definitions of approximation and optimization errors. Now suppose that

we can uniformly bound

$$\epsilon_{\text{gen}} = \sup_{h \in \mathscr{H}} \left| R(h) - \hat{R}_S(h) \right|. \tag{1.6}$$

Then (1.5) reduces to

$$R(h_S) - R^* \leq \varepsilon_{\text{approx}} + \varepsilon_{\text{opt}} + 2\epsilon_{\text{gen}}. \tag{1.7}$$

The decomposition of the true error in (1.7) suggests that we can separately bound the approximation, optimization, and generalization errors to achieve good performance on the true expected risk. Some general remarks are in order:

- **Approximation Error (ϵ_{approx})**: This arises from the limitations of the hypothesis class $\mathscr{H}$. It represents the gap between the best achievable expected risk within the hypothesis class (i.e., $R(h^*_{\mathscr{H}})$) and the theoretical best risk (i.e., R^*). Reducing the approximation error typically involves selecting a more expressive model class or architecture.
- **Optimization Error (ϵ_{opt})**: This reflects the inability to perfectly minimize the empirical risk $\hat{R}_S$ during training. Contributing factors include suboptimal optimization algorithms, local minima, or poor initialization. Improving the optimization process or using better training techniques can reduce this error.
- **Generalization Error (ϵ_{gen})**: This error arises from the discrepancy between the empirical risk and the expected risk due to finite sampling. It quantifies how well performance on the training data reflects performance on unseen data drawn from the same distribution. Reducing the generalization error typically involves increasing the sample size, controlling the effective complexity of the learned hypothesis, or using learning algorithms that promote stability or regularization.

Remark 1.2 (Generalization error versus estimation error). What we refer to as generalization error here is sometimes called estimation error, while the term generalization error is often used to refer to the true error. When an ideal ERM rule is applied, $\epsilon_{\text{opt}} = 0$, and the true error simplifies to a decomposition of approximation error and estimation error (Shalev-Shwartz and Ben-David, 2014). Since the hypothesis class is typically chosen before training, the approximation error is essentially fixed, making it reasonable to interpret the estimation error as the generalization

error.

Remark 1.3 (Bias-complexity tradeoff). There is a tradeoff known as the bias-complexity tradeoff in classical statistical learning theory, which highlights the tension between underfitting and overfitting in model selection. Simpler models with lower complexity typically have higher bias but lower variance, while more complex models reduce bias at the cost of increased variance. Achieving good generalization often requires carefully balancing these factors.

1.4 How Do Deep Neural Networks Learn?

Despite the success of deep neural networks in a wide range of applications, understanding the underlying mechanisms of their learning process remains an active area of research. We formulate a few theoretical questions inspired by the discussions in the previous section:

- **Are deep neural networks more efficient function approximators?** Deep neural networks are hypothesized to be more efficient function approximators due to their ability to represent complex, high-dimensional functions with relatively fewer parameters compared to traditional methods such as polynomials or basis function representations. Determining the theoretical limits of this efficiency remains an open question.
- **Do deep neural networks generalize better, and how?** Despite being highly overparameterized, deep neural networks often generalize well to unseen data. This defies the classical wisdom of the bias-complexity tradeoff. Understanding how deep neural networks generalize remains one of the most important theoretical questions.
- **Does gradient-based optimization of deep neural networks contribute to their performance, and how?** Gradient-based optimization methods, particularly stochastic gradient descent (SGD), play a critical role in the practical success of deep learning. Does such optimization contribute to the observed generalization capabilities of deep neural networks? Does it implicitly bias models toward "simpler" solutions? These remain active areas of research.

In these introductory notes, we will not be able to explore all state-of-the-art theoretical results in the literature. However, by bearing these important theoretical

questions in mind, we aim to provide a foundational treatment of the theoretical principles that are crucial for understanding what definitive answers to these questions might look like.

Part I

Approximation

Chapter 2

Universal Approximation Theorem

We prove a universal approximation theorem for neural networks. The focus is on neural networks with continuous activation functions and approximation guarantees in the topology of uniform convergence on compact sets.

2.1 Density of One-Hidden-Layer Neural Networks

Let $\sigma : \mathbb{R} \to \mathbb{R}$ be an arbitrary function. Define the following set of functions from $\mathbb{R}^d$ to $\mathbb{R}$:

$$\mathcal{M}(\sigma) = \operatorname{span}\left\{x \mapsto \sigma(w \cdot x + b) : w \in \mathbb{R}^d,\ b \in \mathbb{R}\right\}, \tag{2.1}$$

i.e., $\mathcal{M}(\sigma)$ denotes the set of all finite linear combinations of functions of the form $x \mapsto \sigma(w \cdot x + b)$, with coefficients in $\mathbb{R}$.

Definition 2.1 (Density)**.** *A set of functions $\mathcal{M} \subseteq \mathcal{C}(\mathbb{R}^d)$ is said to be dense in $\mathcal{C}(\mathbb{R}^d)$ (with respect to the topology of uniform convergence on compact sets), denoted by $\overline{\mathcal{M}} = \mathcal{C}(\mathbb{R}^d)$ if, for every $f \in \mathcal{C}(\mathbb{R}^d)$, every $\varepsilon > 0$, and every compact set $K \subseteq \mathbb{R}^d$, there exists $g \in \mathcal{M}$ such that*

$$\sup_{x \in K} |f(x) - g(x)| < \varepsilon.$$

In the late 1980s, several influential papers considered the density of neural network approximations with general activation functions. A common requirement was

that the activation function σ is sigmoidal (Cybenko, 1989; Funahashi, 1989), i.e., $\sigma(t) \to 1$ as $t \to \infty$ and $\sigma(t) \to 0$ as $t \to -\infty$, or that it is bounded and monotone (Hornik et al., 1989). We introduce the following result, which assumes only that the activation function is non-polynomial, and show that this condition is both necessary and sufficient for a single-hidden-layer neural network to be dense in the topology of uniform convergence on compact sets.

Theorem 2.1 (Leshno et al. (1993))**.** *Suppose that $\sigma \in \mathscr{C}(\mathbb{R})$. Then $\overline{\mathscr{M}(\sigma)} = \mathscr{C}(\mathbb{R}^d)$ if and only if is σ is not a polynomial.*

Intuitively, the theorem above shows that we can approximate any function in $\mathscr{C}(\mathbb{R}^d)$ arbitrarily well on any compact set using a function in $\mathscr{M}(\sigma)$ of the form $\sum_{i=1}^{m} \beta_i \sigma(w_i x + b_i)$. Clearly, such a function corresponds to the output of a one-hidden-layer neural network with σ as the activation function.

Remark 2.1. It follows directly from Theorem 2.1 that a one-hidden-layer neural network can approximate vector-valued functions in $\mathscr{C}(\mathbb{R}^d, \mathbb{R}^k)$ arbitrarily well. Specifically, given $f \in \mathscr{C}(\mathbb{R}^d, \mathbb{R}^k)$, we can approximate each component f_i of f separately using a one-hidden-layer neural network and then combine all k such networks in parallel to approximate f. Therefore, we focus solely on approximation results for real-valued functions.

Definition 2.2. *A **multivariate polynomial** $p(x_1, x_2, \ldots, x_d)$ is a finite linear combination of monomials, given by:*

$$p(x_1, x_2, \ldots, x_d) = \sum_{\alpha} c_\alpha x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_d^{\alpha_d},$$

*where $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_d)$ is a multi-index, c_α are coefficients, and $x_i^{\alpha_i}$ represents powers of the variables. We say p is **homogeneous** of degree d if all its terms have the same total degree d, meaning:*

$$p(x_1, x_2, \ldots, x_d) = \sum_{|\alpha|=d} c_\alpha x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_d^{\alpha_d},$$

where $|\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_d$ is the total degree of each term.

We first prove the necessity part of Theorem 2.1.

Proof of Theorem 2.1 ("only if" part). Suppose that σ is a polynomial, then for every w and b, $\sigma(wx+b)$ is a multivariate polynomial of the same degree as σ. Thus $\mathscr{M}(\sigma)$ only contains polynomials of a bounded degree, which cannot be dense in $\mathscr{C}(\mathbb{R}^d)$ (see Exercise 2.1). □

To prove the sufficient part, we need a few preliminary results.

Definition 2.3 (Ridge function)**.** *A function from $\mathbb{R}^d$ to $\mathbb{R}$ is called a ridge function if it is of the form*

$$g(a_1x_1 + \cdots + a_dx_d) = g(a \cdot x), \tag{2.2}$$

where $g : \mathbb{R} \to \mathbb{R}$ and $a = (a_1, \ldots, a_d) \in \mathbb{R}^d \setminus \{0\}$.

Intuitively, ridge functions are multivariate functions that remain constant on parallel hyperplanes of the form $a \cdot x = c$, where $c \in \mathbb{R}$. These functions are relevant for one-hidden-layer neural networks because each unit $\sigma(w \cdot x + b)$ is a ridge function for any choice of σ, w, and b.

Let $A \subseteq \mathbb{R}^d$. Define

$$\mathscr{R}(A) = \operatorname{span}\{g(a \cdot x) : \ g \in \mathscr{C}(\mathbb{R}),\ a \in A\}. \tag{2.3}$$

Lemma 2.1 (Vostrecov and Kreines (1961))**.** *The set $\mathscr{R}(A)$ is dense in $\mathscr{C}(\mathbb{R}^d)$ if and only if there is no nontrivial homogeneous polynomial that vanishes on A.*

It is not difficult to see that if $A = S^{d-1}$, where S^{d-1} denotes the $(d-1)$-dimensional unit sphere in $\mathbb{R}^d$, defined as

$$S^{d-1} = \{x \in \mathbb{R}^d : |x| = \sqrt{\sum_{i=1}^{d} |x_i|^2} = 1\},$$

then no trivial homogeneous polynomial that vanishes on A (see Exercise 2.3). Lemma 2.1 implies that $\mathscr{R}(S^{d-1})$ is dense in $\mathscr{C}(\mathbb{R}^d)$. We shall not prove Lemma 2.1 now. For a proof, one can see Vostrecov and Kreines (1961) or Lin and Pinkus (1993). A more elementary proof can be constructed (see Exercise 3.5).

The following proposition allows us to focus on the single-variable case when

proving the sufficiency part of Theorem 2.1.

Proposition 2.1. *Let*

$$\mathcal{N}(\sigma) = span\left\{x \mapsto \sigma(\lambda x + b) : \lambda \in \mathbb{R},\, b \in \mathbb{R}\right\}. \tag{2.4}$$

If $\mathcal{N}(\sigma)$ is dense in $\mathcal{C}(\mathbb{R})$, then $\mathcal{M}(\sigma)$ defined in (2.1) is dense in $\mathcal{C}(\mathbb{R}^d)$.

Proof. Fix any $f \in \mathcal{C}(\mathbb{R}^d)$, any $\varepsilon > 0$, and any compact set $K \subseteq \mathbb{R}^d$. By Lemma 2.1, $\mathcal{R}(S^{d-1})$ is dense in $\mathcal{C}(\mathbb{R}^d)$. There exists $n > 0$, $g_i \in \mathcal{C}(\mathbb{R})$, and $u_i \in S^{d-1}$, $i = 1, \ldots, n$, such that

$$\left| f(x) - \sum_{i=1}^{n} g_i(u_i \cdot x) \right| \leq \frac{\varepsilon}{2}, \quad \forall x \in K.$$

Since S^{d-1} and K are both compact, there exists a finite interval $[\alpha, \beta]$ such that $u_i \cdot x \in [\alpha, \beta]$, for all $u_i \in S^{d-1}$ and $x \in K$. Since $\mathcal{N}(\sigma)$ is dense in $\mathcal{C}(\mathbb{R})$, for each $i = 1, \ldots, n$, there exist m_i, β_{ij}, w_{ij}, b_{ij}, $j = 1, \ldots, m_i$, such that

$$\left| g_i(t) - \sum_{j=1}^{m_i} \beta_{ij} \sigma(w_{ij} t + b_{ij}) \right| < \frac{\varepsilon}{2n}, \quad \forall t \in [\alpha, \beta].$$

It follows that

$$\left| f(x) - \sum_{i=1}^{n} \sum_{j=1}^{m_i} \beta_{ij} \sigma(w_{ij} u_i \cdot x + b_{ij}) \right| < \varepsilon, \quad \forall x \in K.$$

This shows $\overline{\mathcal{M}}(\sigma) = \mathcal{C}(\mathbb{R}^d)$. □

We need another technical lemma. Let $C^\infty(\mathbb{R})$ denote the set of smooth functions (i.e., functions with continuous derivatives of all orders) defined on $\mathbb{R}$.

Lemma 2.2. *If $\sigma \in \mathcal{C}^\infty(\mathbb{R})$ is not a polynomial on some interval I, then there exists some $b_0 \in I$ such that $\sigma^{(k)}(b_0) \neq 0$ for all $k = 0, 1, 2, \ldots$.*

For a proof of Lemma 2.2, see (Donoghue, 1969, p. 53).

We now prove the sufficiency part of Theorem 2.1.

Proof of Theorem 2.1 ("if" part). By Proposition 2.1, we only need to prove the case for $d = 1$.

Suppose that σ is not a polynomial. We first prove the case for $\sigma \in \mathscr{C}^\infty(\mathbb{R})$. By Lemma 2.2, there exists $b_0 \in \mathbb{R}$ such that $\sigma^{(k)}(b_0) \neq 0$ for all $k = 0, 1, 2, \ldots$. By the definition of $\mathscr{M}(\sigma)$, the finite difference form

$$\frac{\sigma((w+h)t + b_0) - \sigma(wt + b_0)}{h} \in \mathscr{M}(\sigma)$$

for all $h \neq 0$. Taking the limit as $h \to 0$, we have

$$t\sigma'(b_0) = \frac{d}{dw}\sigma(wt + b_0)\Big|_{w=0} \in \overline{\mathscr{M}(\sigma)}.$$

By the same argument,

$$t^k\sigma^{(k)}(b_0) = \frac{d^k}{dw^k}\sigma(wt + b_0)\Big|_{w=0} \in \overline{\mathscr{M}(\sigma)}.$$

Since $\sigma^{(k)}(b_0) \neq 0$ for all $k = 0, 1, 2, \ldots$, we have shown that $\overline{\mathscr{M}(\sigma)}$ contain monomials of all degrees and thus all polynomials. By the well-known Weierstrass approximation theorem (see Appendix A.1), we have $\overline{\mathscr{M}(\sigma)} = \mathscr{C}(\mathbb{R})$.

We now prove the general case for $\sigma \in \mathscr{C}(\mathbb{R})$. Define

$$\sigma_\phi(t) = \int_{-\infty}^{\infty} \sigma(t - y)\phi(y)dy,$$

where $\phi \in C_0^\infty(\mathbb{R})$, i.e., $\phi \in C^\infty(\mathbb{R})$ and ϕ has compact support (i.e., ϕ vanishes outside a compact set of $\mathbb{R}$). The integral is well-defined in the Riemann sense and can be approximated by a Riemann sum. Hence $\sigma_\phi \in \overline{\mathscr{M}(\sigma)}$. It also follows that $\overline{\mathscr{M}(\sigma_\phi)} \subseteq \overline{\mathscr{M}(\sigma)}$ because

$$\sigma_\phi(wt + b) = \int_{-\infty}^{\infty} \sigma(wt + b - y)\phi(y)dy.$$

If $\overline{\mathscr{M}(\sigma)} \neq \mathscr{C}(\mathbb{R})$, then $t^k \notin \overline{\mathscr{M}(\sigma)}$ for some k. It follows that $t^k \notin \overline{\mathscr{M}(\sigma_\phi)}$ for each $\phi \in C_0^\infty(\mathbb{R})$. It follows that $\sigma_\phi^{(k)}(b) = 0$ for every $b \in \mathbb{R}$ and every $\phi \in C_0^\infty(\mathbb{R})$ (otherwise, we can use the finite difference approximation above to show $t^k \in$

$\overline{\mathcal{M}(\sigma_\phi)}$). This implies that σ_ϕ is a polynomial of degree less than or equal to $k-1$ for each $\phi \in C_0^\infty(\mathbb{R})$. It follows from a well-known result (see Appendix A.2) in analysis that there exists a sequence $\{\phi_m\}_{m=1}^\infty$ in $C_0^\infty(\mathbb{R})$ such that $\sigma_{\phi_m} \to \sigma$ uniformly on any compact set (such ϕ_m's are known as mollifiers). Since each σ_{ϕ_m} is a polynomial of degree less than or equal to $k-1$, σ must also be a polynomial of degree less than or equal to $k-1$ (see Exercise 2.2). This is a contradiction. □

Remark 2.2 (Approximating univariate polynomials – a quantitative result). Let $\sigma \in \mathscr{C}^\infty(\mathbb{R})$ and define

$$\mathcal{M}_n(\sigma) = \left\{ \sum_{i=1}^n c_i \sigma(w_i x + b_i) : c_i, w_i, b_i \in \mathbb{R} \right\},$$

i.e., $\mathcal{M}_n(\sigma)$ consists of linear combinations of n terms of the form $\sigma(w_i x + b_i)$. In the proof of the sufficient part of Theorem 2.1 for $\sigma \in \mathscr{C}^\infty(\mathbb{R})$, we see that $\{1, t, \ldots, t^k\}$ are all included in $\overline{\mathcal{M}_{k+1}(\sigma)}$. This is because we need two terms to approximate a derivative, which means one additional term for each higher order derivative.

Remark 2.3. It follows immediately from Theorem 2.1 that deep neural works with any bounded number of hidden layers also possess universal approximation properties. We just need the additional layers to approximate the identity function.

Corollary 2.1. *Given any $L \in \mathbb{N}$, the set of neural networks with L hidden layers and activation function σ is dense in $\mathscr{C}(\mathbb{R}^d)$ if and only if σ is not a polynomial.*

We leave the proof as an exercise (Exercise 2.4).

2.2 Density of Width-Bounded Deep Neural Networks

In the previous section, we proved that a one-hidden-layer neural network can act as a universal approximator, meaning it can adjust its weights to approximate any continuous function arbitrarily well, uniformly on any given compact set. This is achieved by allowing the width to grow unboundedly. Additionally, we showed, as a corollary, that a neural network with bounded depth and unbounded width can achieve the same universal approximation guarantees.

In this section, we prove that a neural network with bounded width but unbounded

depth can also achieve universal approximation properties. The main result presented here slightly improves upon that of Kidger and Lyons (2020). The proof is new and simplifies the one provided in Kidger and Lyons (2020).

We first prove a lemma stating that a neural activation followed by an affine transformation can uniformly approximate the identity function arbitrarily well on compact sets. This result was originally proved in Kidger and Lyons (2020), but we relax the assumption that the activation function is continuously differentiable on an interval to differentiability at a single point.

Lemma 2.3. *Suppose that $\sigma : \mathbb{R} \to \mathbb{R}$ is differentiable at a point α and $\sigma'(\alpha) \neq 0$. Then a function of the form $t \mapsto w\sigma(\lambda t + b) + c$, where $w, \lambda, b, c \in \mathbb{R}$, can uniformly approximate the identity function arbitrarily well on any compact set $K \subseteq \mathbb{R}$.*

Proof. Fix any compact set $K \subseteq \mathbb{R}$ and any $\varepsilon > 0$. Let $C > 0$ be such that $|t| < C$ for all $t \in K$. By the definition of differentiability, there exists $\delta > 0$ such that

$$|\sigma(s + \alpha) - \sigma(\alpha) - s\sigma'(\alpha)| < \hat{\varepsilon}\,|s|$$

for all s such that $|s| < \delta$, where $\hat{\varepsilon} = \frac{\varepsilon|\sigma'(\alpha)|}{C} > 0$.

For $h > 0$, define

$$\sigma_h(t) = \frac{\sigma(ht + \alpha) - \sigma(\alpha)}{h\sigma'(\alpha)}.$$

By the choice of C, we have $|ht| < hC < \delta$ for all $h < \frac{\delta}{C}$ and all $t \in K$. It follows that

$$|\sigma_h(t) - t| = \frac{|\sigma(ht + \alpha) - \sigma(\alpha) - ht\sigma'(\alpha)|}{h\,|\sigma'(\alpha)|} < \frac{\hat{\varepsilon}\,|ht|}{h\,|\sigma'(\alpha)|} < \frac{C\hat{\varepsilon}}{|\sigma'(\alpha)|} = \varepsilon$$

for all $t \in K$ and all $h < \frac{\delta}{C}$. Note that $\sigma_h(t)$ is of the form $w\sigma(\lambda t + b) + c$. The proof is complete. □

Let $\mathcal{J}_{d,k,n}(\sigma)$ denote the set of functions representable by neural networks with d inputs, k outputs, and an unlimited number of hidden layers, each containing n neurons, of which at most one neuron uses σ as the activation function, while the remaining neurons use the identity function as the activation (i.e., no activation function is applied).

Let $\mathcal{N}_{d,k,n}(\sigma)$ denote the set of functions representable by neural networks with d inputs, k outputs, and an unlimited number of hidden layers, each containing n neurons, all of which use σ as the activation function.

The main result to be proved in this section is stated as follows.

Theorem 2.2. *Suppose that $\sigma \in \mathscr{C}(\mathbb{R})$ is differentiable at some point α with $\sigma'(\alpha) \neq 0$ and is not a linear function. Then $\overline{\mathcal{N}_{d,k,d+k+1}(\sigma)} = \mathscr{C}(\mathbb{R}^d, \mathbb{R}^k)$.*

We will prove Theorem 2.2 by first establishing the following result.

Proposition 2.2. *Suppose that $\sigma \in \mathscr{C}(\mathbb{R})$ is not a linear function. Then $\overline{\mathcal{I}_{d,k,d+k+1}(\sigma)} = \mathscr{C}(\mathbb{R}^d, \mathbb{R}^k)$.*

We now provide a proof of Theorem 2.2 based on Proposition 2.2.

Proof of Theorem 2.2. Given any $f \in \mathscr{C}(\mathbb{R}^d, \mathbb{R}^k)$, any compact set $K \subseteq \mathbb{R}^d$, and any $\varepsilon > 0$, let $g \in \mathcal{I}_{d,k,d+k+1}(\sigma)$ be such that $|f(x) - g(x)| < \varepsilon/2$ for all $x \in K$. The existence of such a g is guaranteed by Proposition 2.2.

Since σ is differentiable at some point α with $\sigma'(\alpha) \neq 0$, we can replace each occurrence of the identity activation function in g with σ as the activation function, followed by an affine transformation that can uniformly approximate the identity function arbitrarily well (Lemma 2.3). The affine transformation is absorbed into the affine transformations already present in g. Denote this new function by g_h, noting that $g_h \in \mathcal{N}_{d,k,d+k+1}(\sigma)$.

Since uniform continuity preserves uniform convergence, and the composition of uniformly convergent sequences of functions uniformly converges to the composition of their limits, g_h can uniformly approximate g arbitrarily well by replacing the identity activations with σ-activations. Specifically, we can ensure that $|g(x) - g_h(x)| < \varepsilon/2$ for all $x \in K$, with $g_h \in \mathcal{N}_{d,k,d+k+1}(\sigma)$.

It follows that:

$$|f(x) - g_h(x)| \leq |f(x) - g(x)| + |g(x) - g_h(x)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

for all $x \in K$, with $g_h \in \mathcal{N}_{d,k,d+k+1}(\sigma)$. This demonstrates that $\overline{\mathcal{N}_{d,k,d+k+1}(\sigma)} = \mathscr{C}(\mathbb{R}^d, \mathbb{R}^k)$. □

The rest of this section is devoted to proving Proposition 2.2. The following lemma is proved first.

Given $\sigma : \mathbb{R} \to \mathbb{R}$, define

$$\mathcal{M}^\circ(\sigma) = \operatorname{span}\left\{x \mapsto \sigma^{\circ k}(w \cdot x + b) : \, k \in \mathbb{N}, \, w \in \mathbb{R}^d, \, b \in \mathbb{R}\right\},$$

where $\sigma^{\circ k}$ indicates the k-th composition of the same function σ.

Lemma 2.4. *The set $\mathcal{F}_{d,d+1,d+2}(\sigma)$ contains all functions from $\mathbb{R}^d$ to $\mathbb{R}^{d+1}$ of the form*

$$x \mapsto (x, g(x)),$$

where $g \in \mathcal{M}^\circ(\sigma)$.

Proof. We prove this by induction. Let g be of the form

$$g(x) = \sum_{i=1}^{n} c_i \sigma^{\circ k_i}(w_i \cdot x + b_i), \quad k_i \in \mathbb{N}, \quad w_i \in \mathbb{R}^d, \quad b_i \in \mathbb{R}, \quad c_i \in \mathbb{R}.$$

We want to show that $x \mapsto (x, g(x))$ can be exactly computed by a function in $\mathcal{F}_{d,d+1,d+2}(\sigma)$. For $n = 1$, with one layer, we can compute $\sigma(w_1 \cdot x + b_1)$. For the remaining layers, let x be passed on exactly with d neurons with the identity activation function and apply σ to $\sigma^{\circ k}(w_1 \cdot x + b_1)$ for each $k < k_1$. In the final layer, we can retain x and compute $c_1 \sigma^{\circ k_1}(w_1 \cdot x + b_1)$.

Now suppose that $\mathcal{F}_{d,d+1,d+2}(\sigma)$ can exactly compute $x \mapsto (x, g_{n-1}(x))$, where $g_{n-1}(x) = \sum_{i=1}^{n-1} c_i \sigma^{\circ k_i}(w_i \cdot x + b_i))$. Given such a function in $\mathcal{F}_{d,d+1,d+2}(\sigma)$, note that the output layer has $d + 1$ neurons. We can continue the computation by modifying the output layer to a hidden layer of $d + 2$ neurons. We use the additional hidden neuron to save the value of $g_{n-1}(x)$. We then use the next layers to compute $c_n \sigma^{\circ k_n}(w_n \cdot x + b_n)$ as we did in the base case of the induction. The value of $g_{n-1}(x)$ is passed on identically with the $(d + 2)$-th neuron with the identity activation function. In the final output layer, we sum the value of g_{n-1} and $c_n \sigma^{\circ k_n}(w_n \cdot x + b_n)$ to obtain $\sum_{i=1}^{n} \sigma^{\circ k_i}(w_i \cdot x + b_i)$. □

The next lemma is a consequence of Lemma 2.4.

Lemma 2.5. *The set $\mathcal{I}_{d,k,d+k+1}(\sigma)$ contains all functions from $\mathbb{R}^d$ to $\mathbb{R}^k$ of the form*

$$x \mapsto (g_1(x), \dots, g_k(x)),$$

where $g_i \in \mathcal{M}^\circ(\sigma)$ for each $i = 1, 2, \dots, k$.

Proof. Following Lemma 2.4, we use a function in $\mathcal{I}_{d,d+1,d+2}(\sigma)$ to compute $(x, g_1(x))$ exactly. This computation is embedded in a function within $\mathcal{I}_{d,d+k,d+k+1}$, utilizing only $d+2$ hidden neurons, with the output $g_1(x)$ stored in the $(d+2)$-th neuron. Subsequently, we use the next layers to compute $(x, g_2(x))$ from x, while passing $g_1(x)$ forward via the identity activation function. We then store $g_2(x)$ in the $(d+3)$-th neuron and proceed to compute $g_3(x)$. This process is repeated iteratively until $g_i(x)$ has been computed for each $i = 1, 2, \dots, k$. Finally, we construct the output as $(g_1(x), \dots, g_k(x))$. □

Lemma 2.6. *If $\sigma \in \mathcal{C}(\mathbb{R})$ is not a linear function, then $\overline{\mathcal{M}^\circ(\sigma)} = \mathcal{C}(\mathbb{R}^d)$.*

Proof. We consider two cases: (1) σ is not a polynomial; (2) σ is a nonlinear polynomial.

(1) For this case, recall that

$$\mathcal{M}(\sigma) = \operatorname{span}\left\{x \mapsto \sigma(w \cdot x + b) : \ k \in \mathbb{N},\ w \in \mathbb{R}^d,\ b \in \mathbb{R}\right\}$$

and that $\overline{\mathcal{M}(\sigma)} = \mathcal{C}(\mathbb{R}^d)$ (Theorem 2.1). Since $\mathcal{M}(\sigma) \subseteq \mathcal{M}^\circ(\sigma)$, the conclusion follows.

(2) Suppose that σ is a polynomial of degree $m \geq 2$. By the same argument as in the proof of Proposition 2.1, we only need to show that

$$\mathcal{N}^\circ(\sigma) := \operatorname{span}\left\{t \mapsto \sigma^{\circ k}(\lambda t + b) : \ k \in \mathbb{N},\ \lambda \in \mathbb{R},\ b \in \mathbb{R}\right\}$$

is dense in $\mathcal{C}(\mathbb{R})$. For each $k \geq 1$, $\sigma^{\circ k}$ is a polynomial of degree m^k. To simplify notation, let $\rho = \sigma^{\circ k}$. We can pick b_0 such that $\rho^{(l)}(b_0) \neq 0$ for all $l = 1, 2, \dots, m^k$. Following the same argument as in the proof of Theorem 2.1 (the "if" part), we can show that $\left\{1, t, \dots, t^{m^k}\right\}$ is contained in $\overline{\mathcal{N}^\circ(\sigma)}$. Since this holds for all $k \geq 1$, we have that $\overline{\mathcal{N}^\circ(\sigma)}$ contains all polynomials and thus also $\mathcal{C}(\mathbb{R})$ by the Weierstrass approximation theorem (Appendix A.1). □

Remark 2.4. If $\sigma \in \mathscr{C}(\mathbb{R})$ is a nonlinear polynomial, we can directly show that $\mathcal{N}^\circ(\sigma)$ contains all polynomials (see Exercise 2.5).

Proof of Proposition 2.2. It follows directly from Lemmas 2.5 and 2.6. □

Remark 2.5. It is proved in Kidger and Lyons (2020) that either $\mathcal{N}_{d,k,d+k+1}(\sigma)$ or $\mathcal{N}_{d,k,d+k+2}(\sigma)$ is dense in $\mathscr{C}(\mathbb{R}^d, \mathbb{R}^k)$ for σ that is continuously differentiable on an interval and has at least one point where its derivative does not vanish. The proof by Kidger and Lyons (2020) for the case where σ is a nonlinear polynomial is quite convoluted. We considerably simplified the argument by proving Lemmas 2.4–2.6. Theorem 2.2 slightly strengthens the conclusion, showing that $\overline{\mathcal{N}_{d,k,d+k+1}(\sigma)} = \mathscr{C}(\mathbb{R}^d, \mathbb{R}^k)$ for all $\sigma \in \mathscr{C}(\mathbb{R})$ that is not a linear function and has a non-zero derivative at at least one point.

Remark 2.6. To establish the minimum width n required for $\mathcal{N}_{d,k,n}(\sigma)$ to have the universal approximation property remains an active area of research; see Lu et al. (2017); Hanin and Sellke (2017); Kidger and Lyons (2020); Park et al. (2021); Kim et al. (2024); Hwang (2023). For uniform approximation in $\mathscr{C}(\mathbb{R}^d, \mathbb{R}^k)$, the conclusion of Theorem 2.2 provides the current best-known upper bound on the required width, i.e., $d + k + 1$. By combining Lemmas 2.4 and 2.6 with the main result of Hwang (2023), we can also show that $\mathcal{N}_{d,k,\max\{2d+1,k\}+2}(\sigma)$ is dense in $\mathscr{C}(\mathbb{R}^d, \mathbb{R}^k)$ under the same assumptions as in Theorem 2.2. This bound is better than $d + k + 1$, provided $k > d + 2$.

2.3 Exercises

2.1. Show that the set of polynomials of a bounded degree cannot be dense in $\mathscr{C}(\mathbb{R}^d)$.

2.2. Let P_k denote the set of polynomials of degree at most k. Prove that $\overline{P_k} = P_k$.

2.3. (1) Show that, if a homogeneous polynomial $p(x_1, x_2, \ldots, x_d)$ of n variables vanishes on S^{d-1}, the $(d-1)$-dimensional unit sphere in $\mathbb{R}^d$, then p is identically zero.
(2) Show that the same conclusion holds of S^{d-1} is replaced by any open subset

of S^{d-1} (a set $U \subseteq S^{d-1}$ is open if there exists an open set $V \subseteq \mathbb{R}^d$ such that $U = V \cap S^{d-1}$).

2.4. Prove Corollary 2.1.

2.5. Suppose that $\sigma \in \mathscr{C}(\mathbb{R})$ is a nonlinear polynomial. Define

$$\mathcal{N}^\circ(\sigma) := \operatorname{span}\left\{t \mapsto \sigma^{\circ k}(\lambda t + b) : \ k \in \mathbb{N},\ \lambda \in \mathbb{R},\ b \in \mathbb{R}\right\},$$

where $\sigma^{\circ k}$ indicates the k-th composition of the same function σ. Show that $\mathcal{N}^\circ(\sigma)$ contains all univariate polynomials.

Chapter 3

Degree of Approximations

In this chapter, we introduce more quantitative results on approximating functions and their derivatives using neural networks. We first introduce some standard notation and define the Sobolev space in which we consider the approximation.

3.1 Sobolev Space

Let B_d denote the unit ball in $\mathbb{R}^d$, i.e.,

$$B_d = \{x \in \mathbb{R}^d : |x| = \sqrt{\sum_{i=1}^{d} x_i^2} \leq 1\}.$$

We consider approximation of functions defined on B_d.

Let $\mathscr{C}^m(B_d)$ denote the set of all functions f defined on B_d for which all partial derivatives of f up to order m are continuous. Specifically, $\mathscr{C}^m(B_d)$ is given by:

$$\mathscr{C}^m(B_d) = \{f : B_d \to \mathbb{R} \mid D^\alpha f \text{ exists and is continuous for all } \alpha \text{ with } |\alpha| \leq m\}.$$

Here,

$$D^\alpha f = \frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1} \cdots \partial x_d^{\alpha_d}}$$

represents the partial derivative of f corresponding to the multi-index $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_d)$, and $|\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_d$ denotes the total order of differentiation.

The Sobolev space $W^{m,p}(B_d)$ is the completion[1] of $\mathscr{C}^m(B_d)$ with respect to the norm

$$\|f\|_{W^{m,p}(B_d)} = \begin{cases} \left(\sum_{|\alpha|\le m} \|D^\alpha f\|^p_{L^p(B_d)}\right)^{1/p}, & \text{if } 1 \le p < \infty, \\ \max_{|\alpha|\le m} \operatorname{ess\,sup}_{x\in B_d} |D^\alpha f(x)|, & \text{if } p = \infty. \end{cases}$$

where the L^p-norm of the weak derivative $D^\alpha f$ is given by

$$\|D^\alpha f\|_{L^p(B_d)} = \begin{cases} \left(\int_{B_d} |D^\alpha f(x)|^p \, dx\right)^{1/p}, & \text{if } 1 \le p < \infty, \\ \operatorname{ess\,sup}_{x\in B_d} |D^\alpha f(x)|, & \text{if } p = \infty. \end{cases}$$

Here ess sup denotes the *essential supremum*, which intuitively means the smallest number M such that $|g(x)| \le M$ for all x except on a set of measure zero (i.e., negligible in size).

The unit ball with respect to the Sobolev norm in $W^{m,p}(B_d)$ is defined as

$$\mathscr{B}_d^{m,p} = \{f \in W^{m,p}(B_d) : \|f\|_{W^{m,p}(B_d)} \le 1\}.$$

We aim to approximate functions in $\mathscr{B}_d^{m,p}$ with neural networks and investigate the rates of approximation in L^p-norm. Since $\mathscr{C}(B_d)$ is dense in $L^p(B_d) := \{f : B_d \to \mathbb{R} \mid \|f\|_{L^p(B_d)} < \infty\}$, where $\|\cdot\|_{L^p(B_d)}$ is the L^p-norm defined above[2], we know from Chapter 2 that $\mathscr{M}(\sigma)$ is dense in $L^p(B_d)$. In this chapter, we focus on quantitative results concerning the rates of approximation, referred to as the degree of approximation.

[1] Intuitively, the completion of a space involves adding all "limit points" of sequences from the original space with respect to the given norm.

[2] Formally, $L^p(B_d)$ consists of Lebesgue integrable functions (identified up to equality almost everywhere).

3.2 Approximation by Polynomial and Ridge Functions

Define the set of functions that are a linear combination of n ridge functions

$$\mathcal{R}_n := \left\{ \sum_{i=1}^{n} g_i(a_i \cdot x) : \ a_i \in \mathbb{R}^d, \ g_i \in \mathcal{C}(\mathbb{R}) \right\} \tag{3.1}$$

Given an activation function $\sigma \in \mathcal{C}(\mathbb{R})$, define

$$\mathcal{M}_n(\sigma) := \left\{ \sum_{i=1}^{n} c_i \sigma(w_i \cdot x + b_i) : w_i \in \mathbb{R}^d, \ c_i, b_i \in \mathbb{R} \right\}. \tag{3.2}$$

The set $\mathcal{M}_n$ represents the functions that can be expressed by a one-hidden-layer neural network with at most n hidden units.

One reason to investigate approximation by ridge functions is that $\mathcal{M}_n \subseteq \mathcal{R}_n$. Consequently, any degree of approximation achievable by $\mathcal{M}_n$ is also achievable by $\mathcal{R}_n$. Conversely, any degree of approximation not achievable by $\mathcal{R}_n$ cannot be achieved by $\mathcal{M}_n$.

More precisely, for $1 \le q \le p \le \infty$ and any set of functions $\mathcal{M} \subseteq L^q(B_d)$, define the degree of approximation of functions in $\mathcal{B}_d^{m,p}$ by functions in $\mathcal{M}$ in L^q-norm as

$$E(\mathcal{B}_d^{m,p}, \mathcal{M}, L^q) := \sup_{f \in \mathcal{B}_d^{m,p}} \inf_{g \in \mathcal{M}} \|f - g\|_{L^q(B_d)}. \tag{3.3}$$

Since $\mathcal{M}_n \subseteq \mathcal{R}_n$, we have

$$E(\mathcal{B}_d^{m,p}, \mathcal{R}_n, L^q) \le E(\mathcal{B}_d^{m,p}, \mathcal{M}_n, L^q). \tag{3.4}$$

We first establish the following result on the degree of approximation to functions in $\mathcal{B}_d^{m,p}$ by ridge functions with respect to the L^p-norm.

Theorem 3.1. *For every $p \in [1, \infty]$, every $m \ge 1$, and every $d \ge 2$, we have*

$$E(\mathcal{B}_d^{m,p}, \mathcal{R}_n, L^p) \le C n^{-\frac{m}{d-1}}, \tag{3.5}$$

where the constant $C > 0$ is independent of n.

The following proof, taken from Pinkus (1999), is based on representing polynomials as linear combinations of ridge functions and then relying on classical results regarding the degree of approximation by polynomial functions.

We first prove the following results on representing polynomials as linear combinations of ridge functions. Let H_k denote the linear space of homogeneous polynomials of degree k and with d variables, i.e.,

$$H_k = \left\{ \sum_{|\alpha|=k} c_\alpha x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_d^{\alpha_d} : c_\alpha \in \mathbb{R} \right\},$$

where $|\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_d$. Then the space of polynomials of degree at most k is given by $P_k = \cup_{i=0}^k H_i$. Since there are $\binom{k+d-1}{d-1}$ distinct monomials of degree k (see Exercise 3.1), the dimension of H_k, $\dim H_k$, is given by

$$n = \dim H_k = \binom{k+d-1}{d-1} \asymp k^{d-1}, \tag{3.6}$$

where $\binom{k+d-1}{d-1} \asymp k^{d-1}$ means the quantities on both sides grow asymptotically at the same rate[3] as k^{d-1} (see Exercise 3.2).

Proposition 3.1. *Let P_k denote the set of multivariate polynomials with d variables and of degree at most k in d variables. Then there exist points $a_1, \ldots, a_n \in \mathbb{R}^d$ such that*

$$P_k = \left\{ \sum_{i=1}^n p_i(a_i \cdot x) : \ p_i \in \pi_k, \ i = 1, 2, \ldots, n \right\},$$

where n is defined in (3.6) and π_k represents the space of univariate polynomials of degree at most k. This implies $P_k \subseteq \mathcal{R}_n$.

Proof. The set of monomials $\{x^\alpha : |\alpha| = k\}$ forms a basis for H_k. Let α_1 and α_2 be

[3] Precisely, this means there exist positive constants C_1 and C_2, independent of k and d, such that $C_1 k^{d-1} \le \binom{k+d-1}{d-1} \le C_2 k^{d-1}$.

multi-indices for d variables such that $|\alpha_1| = |\alpha_2|$. Note that

$$D^{\alpha_1} x^{\alpha_2} = \alpha_1!\, \delta_{\alpha_1,\alpha_2}, \tag{3.7}$$

where $\delta_{\alpha_1,\alpha_2}$ is the Kronecker delta, defined as

$$\delta_{\alpha_1,\alpha_2} = \begin{cases} 1, & \text{if } \alpha_1 = \alpha_2, \\ 0, & \text{otherwise.} \end{cases}$$

This implies that $\{D^\alpha : |\alpha| = k\}$ forms a basis for H_k^* (see Exercise 3.3), where H_k^* represents the dual space of H_k consisting of all linear functionals on H_k, meaning all linear maps from H_k to $\mathbb{R}$. It follows that any linear functional on H_k can be written as a linear combination $\sum_{|\alpha|=k} c_\alpha D^\alpha$.

Let $H_k = \operatorname{span}\{q_1, \ldots, q_n\}$. Pick points $a_1, \ldots, a_n$ in $\mathbb{R}^d$ such that the rank of

$$V := \begin{bmatrix} q_1(a_1) & q_1(a_2) & \cdots & q_1(a_n) \\ q_2(a_1) & q_2(a_2) & \cdots & q_2(a_n) \\ \vdots & \vdots & \ddots & \vdots \\ q_n(a_1) & q_n(a_2) & \cdots & q_n(a_n) \end{bmatrix}$$

is n (see Exercise 3.4). For any non-trivial polynomial q in H_k, write $q = \sum_{i=1}^n c_i q_i$. The evaluation of q at the points $A = \{a_1, \ldots, a_n\}$ is given by $V^\top c$, where $c = (c_1, \ldots, c_n)^\top$. Since $\operatorname{rank}(V) = n$ and $c \neq 0$, $V^\top c \neq 0$, i.e., the evaluation of any non-trivial polynomial q in H_k at the points in A cannot be all zero.

Claim 1: $\left\{(a_i \cdot x)^k : i = 1, \ldots, n\right\}$ spans H_k. Suppose that this does not hold. Then there exists a non-trivial linear functional on H_k that annihilates every $(a_i \cdot x)^k$, $i = 1, \ldots, n$. This functional can be represented as $\sum_{|\alpha|=k} c_\alpha D^\alpha = q(D)$, where $q(x) = \sum_{|\alpha|=k} c_\alpha x^\alpha$ is a non-trivial polynomial in H_k. Then $q(D)$ annihilating every $(a_i \cdot x)^k$ means

$$0 = q(D)(a_i \cdot x)^k = k! q(a_i),$$

which implies the evaluation of q at all points in A is zero. This contradicts the fact that q is a non-zero polynomial in H_k.

Claim 2: $\{(a_i \cdot x)^s : i = 1, \ldots, n\}$ spans H_s for each $s = 0, \ldots, k$. Similar to the

argument above, there exists a non-trivial $q \in H_s$ such that $q(a_i) = 0$ for each $i = 1, \ldots, n$. For any non-trivial $p \in H_{k-s}$, the product $pq \in H_k$ is also non-trivial. However, the evaluation of pq on the set A is identically zero. This means pq is a zero polynomial. This is a contradiction.

By the two claims above, we have

$$\begin{aligned} P_k &= \operatorname{span}\{(a_i \cdot x)^s : i = 1, \ldots, n,\ s = 0, 1, \ldots, k\} \\ &= \left\{ \sum_{i=1}^{n} p_i(a_i \cdot x) : p_i \in \pi_k,\ i = 1, 2, \ldots, n \right\}, \end{aligned}$$

where π_k is the space of univariate polynomials of degree at most k. □

We now state a classical result on approximation with polynomial functions.

Lemma 3.1. *For every $p \in [1, \infty]$, every $m \geq 1$, and every $d \geq 1$, we have*

$$E(\mathscr{B}_d^{m,p}, P_k, L^p) := \sup_{f \in \mathscr{B}_d^{m,p}} \inf_{g \in P_k} \|f - g\|_{L^p(B_d)} \leq C k^{-m}, \tag{3.8}$$

where the constant $C > 0$ is independent k.

In fact, we can even choose P_k using a linear functional $L : W^{m,p}(B_d) \to P_k$ as $P_k = L(f)$ to achieve the approximation degree above. For a proof, see Mhaskar (1996).

We can now prove Theorem 3.1.

Proof of Theorem 3.1. By Proposition 3.1, $P_k \subseteq \mathscr{R}_n$. By Lemma 3.1, We have

$$E(\mathscr{B}_d^{m,p}, \mathscr{R}_n, L^p) \leq E(\mathscr{B}_d^{m,p}, P_k, L^p) \leq C_1 k^{-m} \leq C_2 n^{-\frac{m}{d-1}},$$

where we used $n \asymp k^{d-1}$ from (3.6). □

Remark 3.1 (Best approximation by ridge functions). The result of Theorem 3.1 essentially states that with linear combinations of ridge functions, we can approximate at least as well as polynomial functions. Unfortunately, the degree of approximation by ridge functions cannot be improved. As proved in Maiorov (1999); Gordon et al.

(2002); Maiorov (2010), for every $d \geq 2$, every $m \geq 1$, and all $1 \leq q \leq p \leq \infty$, we have

$$E(\mathscr{B}_d^{m,p}, \mathscr{R}_n, L^q) \asymp n^{-\frac{m}{d-1}}. \tag{3.9}$$

By (3.4), this implies

$$E(\mathscr{B}_d^{m,p}, \mathscr{M}_n(\sigma), L^q) \geq C n^{-\frac{m}{d-1}}$$

for any $\sigma \in \mathscr{C}(\mathbb{R})$.

3.3 Approximation by One-Hidden-Layer Neural Networks

The results in Section 3.2 show that ridge functions in $\mathscr{R}_n$ can achieve optimal approximation rates of $n^{-\frac{m}{d-1}}$. Given $\sigma \in \mathscr{C}(\mathbb{R})$, since one-hidden-layer neural network functions in $\mathscr{M}_n(\sigma)$ are a subset of $\mathscr{R}_n$, they cannot perform better than functions in $\mathscr{R}_n$.

A natural question is whether an arbitrary $\sigma \in \mathscr{C}(\mathbb{R})$ that is not a polynomial can achieve this optimal rate of approximation. Unfortunately, the answer is no, as shown by the following result from Maiorov and Meir (2000).

Theorem 3.2 (Maiorov and Meir (2000))**.** *Let σ be the standard sigmoid function $\sigma(t) = \frac{1}{1+e^{-t}}$. For any $1 \leq p \leq \infty$, we have*

$$E(\mathscr{B}_d^{m,p}, \mathscr{M}_n(\sigma), L^p) \geq C(n \log n)^{-\frac{m}{d}}.$$

The proof of this result is quite complicated, and we refer readers to (Maiorov and Meir, 2000, Section 5) for its proof. Note that when σ is the standard sigmoid function, $\tanh(t) = 2\sigma(2t) - 1$. Hence, $\mathscr{M}_n(\tanh) \subseteq \mathscr{M}_{n+1}(\sigma)$. The lower bound in Theorem 3.2 holds for the tanh activation function as well.

The following result by Maiorov and Pinkus (1999) shows that there exists a $\sigma \in \mathscr{C}^\infty(\mathbb{R})$ that is strictly increasing and *sigmoidal*, i.e., $\lim_{t\to\infty} \sigma(t) = 1$ and $\lim_{t\to-\infty} \sigma(t) = 0$, such that $\mathscr{M}_n(\sigma)$ achieves the optimal degree of approximation $n^{-\frac{m}{d-1}}$.

Theorem 3.3 (Maiorov and Pinkus (1999))**.** *There exists a $\sigma \in \mathscr{C}^\infty(\mathbb{R})$ that is strictly*

increasing and sigmoidal such that

$$E(\mathcal{B}_d^{m,p}, \mathcal{M}_n(\sigma), L^p) \leq Cn^{-\frac{m}{d-1}}$$

holds for every $p \in [1, \infty]$, every $m \geq 1$, every $d \geq 2$, and every $n \geq 1$, where C is a constant independent of n.

The proof of Theorem 3.3 relies on the following result, which demonstrates that any linear combination of n ridge functions, i.e., any function in $\mathcal{R}_n$, can be approximated arbitrarily well by a function of the form $\sum_{i=1}^{n+d+1} c_i \sigma(w_i \cdot x + b_i)$, i.e., a function in $\mathcal{M}_{n+d+1}(\sigma)$ with respect to the uniform norm (or L^∞-norm) on $\mathcal{C}(B_d)$. This result is formally stated below.

Proposition 3.2 (Maiorov and Pinkus (1999))**.** *There exists a $\sigma \in \mathcal{C}^\infty(\mathbb{R})$ that is strictly increasing and sigmoidal such that, for every $g \in \mathcal{R}_n$ and every $\varepsilon > 0$, there exists $h \in \mathcal{M}_{n+d+1}(\sigma)$ such that $\sup_{x \in B_d} |h(x) - g(x)| < \varepsilon$.*

Proof. The proof essentially uses the fact that $\mathcal{C}([-1, 1])$ is separable, i.e., there exists a countable dense subset of $\mathcal{C}([-1, 1])$ with respect to the uniform norm. Let $\{u_k\}_{k=1}^\infty$ be such a subset in $\mathcal{C}^\infty(\mathbb{R})$ (e.g., all polynomials with rational coefficients). Then $\overline{\{u_k\}_{k=1}^\infty} = \mathcal{C}([-1, 1])$.

Claim: We can construct a function σ strictly increasing and sigmoidal such that any $u_k \in \mathcal{C}([-1, 1])$ can be expressed as a linear combination of the form:

$$c_1^k \sigma(t - 3) + c_2^k \sigma(t + 1) + c_3^k \sigma(t + 4k + 1) = u_k(t), \tag{3.10}$$

for all $t \in [-1, 1]$ and all $k = 1, 2, \ldots$, where $\sigma(t - 3)$ and $\sigma(t + 1)$ are linear functions for $t \in [-1, 1]$ and linearly independent of each other.

We prove the claim as follows. Let β be any smooth, strictly increasing and sigmoidal function. For example, $\beta(t) = 1/(1 + e^{-t})$. We define σ in a piecewise fashion but ensure it is smooth and satisfies the desired properties.

- On intervals $[4k, 4k + 2]$, $k = 1, 2, \ldots$, set

$$\sigma(t + 4k + 1) = b_k + c_k t + d_k u_k(t), \quad t \in [-1, 1].$$

We can pick the coefficients b_k, c_k, and d_k such that $\sigma(4k) = \beta(4k)$ and $0 < \sigma'(s) \leq \beta'(s)$ for $s \in [4k, 4k+2]$. To see why this is always possible, note that

$$\sigma(4k) = b_k - c_k + d_k u_k(-1), \quad \sigma'(s) = c_k + d_k u_k'(s - 4k - 1), \quad s \in [4k, 4k+2].$$

Let $\delta_k = \min_{s\in[4k,4k+2]} \beta'(s) > 0$ and $\varepsilon_k = \sup_{t\in[-1,1]} |u_k'(t)|$. Pick $d_k > 0$ such that $d_k \varepsilon_k < \frac{\delta_k}{2}$ and $c_k = \frac{\delta_k}{2}$. Then it is easy to verify that σ has the desired properties.

- On $[-4,-2]$ and $[0,2]$, we construct σ to be linear functions that are linearly independent but still satisfy $\sigma(-4) = \beta(-4)$, $\sigma(-2) = \beta(-2)$, and $0 < \sigma'(s) \leq \beta'(s)$ on $[-4,-2]$ and $[0,2]$.
- The rest of σ on $(-\infty,-4)$, $(-2,0)$, and $(4k+2, 4k+4)$, $k = 1,2,\ldots$, can be constructed such that $\sigma \in \mathscr{C}^\infty(\mathbb{R})$, it is strictly increasing, $\lim_{t\to\infty} \sigma(t) = 1$ and $\lim_{t\to-\infty} \sigma(t) = 0$.

By the construction above, we can choose c_1^k, c_2^k, and c_3^k such that (3.10) holds. The claim is proved.

We can now use this claim to construct an arbitrarily close approximation of a function $g \in \mathscr{R}_n$ using a function $h \in \mathscr{M}_{n+d+1}(\sigma)$. Without loss of generality, write $g(x) = \sum_{i=1}^n g_i(a_i \cdot x)$, where $g_j \in \mathscr{C}(\mathbb{R})$ and $a_i \in S^{d-1}$. Since $a_i \cdot x \in [-1,1]$ for all $x \in B_d$, we only need to consider $g_i \in \mathscr{C}([-1,1])$.

Recall that $\overline{\{u_k\}_{k=1}^\infty} = \mathscr{C}([-1,1])$. Fix any $\varepsilon > 0$ and each $i = 1,2,\ldots,n$, let u_{k_i} be such that $\sup_{t\in[-1,1]} \left|g_i(t) - u_{k_i}(t)\right| < \varepsilon$. By (3.10), this gives

$$\left|g_i(t) - c_1^{k_i}\sigma(t-3) + c_2^{k_i}\sigma(t+1) + c_3^{k_i}\sigma(t+4k_i+1)\right| < \varepsilon \tag{3.11}$$

for all $t \in [-1,1]$ and $i = 1,2,\ldots,n$. It follows that

$$\left|g(x) - \sum_{i=1}^n \left[c_1^{k_i}\sigma(a_i \cdot x - 3) + c_2^{k_i}\sigma(a_i \cdot x + 1) + c_3^{k_i}\sigma(a_i \cdot x + 4k_i + 1)\right]\right| < \varepsilon$$

for all $x \in B_d$. Due to the linearity of $\sigma(t-3)$ and $\sigma(t+1)$ for $t \in [-1,1]$, the sum $\sum_{i=1}^n c_1^{k_i}\left[\sigma(a_i \cdot x - 3) + c_2^{k_i}\sigma(a_i \cdot x + 1)\right]$ reduces to a linear combinations of $1, x_1, \ldots, x_d$, which can be rewritten a sum of a total of $d+1$ similar terms (e.g., a sum of the form $\sum_{j=1}^{d+1} c_j \sigma(e_i \cdot x + 1)$, where $\{e_i\}_{i=1}^d$ is the standard basis in $\mathbb{R}^d$). This demonstrates that we can select $h \in \mathscr{M}_{n+d+1}(\sigma)$ to approximate g arbitrarily well.

□

Theorem 3.3 is an immediate consequence of Proposition 3.2 and Theorem 3.1.

Proof of Theorem 3.3. Fix any $f \in \mathscr{B}_d^{m,p}$ and any $\varepsilon > 0$. By Theorem 3.1, there exists $g \in \mathscr{R}_n$ such that

$$\|f - g\|_{L^p(B_d)} \leq Cn^{-\frac{m}{d-1}} + \varepsilon.$$

By Proposition 3.2, there exists $h \in \mathscr{M}_{n+d+1}(\sigma)$ such that

$$\|h - g\|_{L^\infty(B_d)} \leq \varepsilon.$$

Let $\mu(B_d)$ denotes the Lebesgue measure (volume) of B_d. We have

$$\|h - g\|_{L^p(B_d)} \leq \mu(B_d)^{\frac{1}{p}} \|h - g\|_{L^\infty(B_d)} \leq \mu(B_d)^{\frac{1}{p}} \varepsilon.$$

It follows that

$$\|f - h\|_{L^p(B_d)} \leq Cn^{-\frac{m}{d-1}} + (1 + \mu(B_d)^{\frac{1}{p}})\varepsilon$$

Since $n^{-\frac{m}{d-1}} \asymp (n+d+1)^{-\frac{m}{d-1}}$ asymptotically in n and $\varepsilon > 0$ is arbitrary, the conclusion follows. □

One caveat of this result is that the activation function σ constructed in this manner can be quite pathological and not practical for use in real-world applications. This naturally leads to the question: for any given $\sigma \in \mathscr{C}(\mathbb{R})$ that is not a polynomial, what degree of approximation can be achieved?

Theorem 3.4. *Assume that $\sigma \in \mathscr{C}^\infty(\mathbb{R})$ is not a polynomial. For every $p \in [1, \infty]$, we have*

$$E(\mathscr{B}_d^{m,p}, \mathscr{M}_n(\sigma), L^p) \leq Cn^{-\frac{m}{d}}.$$

Proof. By the argument in the proof universal approximation theorem (Theorem 2.1) (see Remark 2.2), the space of univariate polynomials of degree at most k, denoted by π_k, is contained in $\overline{\mathscr{M}_{k+1}(\sigma)}$ for $d = 1$.

By Proposition 3.1,

$$P_k = \left\{ \sum_{i=1}^{r} p_i(a_i \cdot x) : \, p_i \in \pi_k, \, i = 1, 2, \ldots, r \right\},$$

where $r \asymp k^{d-1}$. It follows that $P_k \in \overline{\mathcal{M}_{r(k+1)}(\sigma)}$. Let $n = r(k+1)$. Then $n \asymp k^d$. By Lemma 3.1, we have

$$E(\mathcal{B}_d^{m,p}, \mathcal{M}_n(\sigma), L^p) = E(\mathcal{B}_d^{m,p}, \overline{\mathcal{M}_{r(k+1)}(\sigma)}, L^p) \leq E(\mathcal{B}_d^{m,p}, P_k, L^p) \leq C_1 k^{-m} \leq C n^{-\frac{m}{d}},$$

where C_1 is independent of k and C is independent of n. □

3.4 Lower Bounds

In previous sections, we have shown that the degree of approximation by ridge functions is $E \asymp n^{-\frac{m}{d-1}}$. We also see that one-hidden-layer neural networks with a potentially pathological activation function σ can match this asymptotic rate. With any given smooth activation function, we have shown that the upper bound $E \leq C n^{-\frac{m}{d}}$ can be achieved. It is natural to ask what lower bounds on the degree of approximation can be obtained.

We introduce a now classical result by DeVore et al. (1989), which states that if we restrict the methods of approximation to those that depend continuously on the function being approximated, then the lower bound on the degree of approximation is $E \geq C n^{-\frac{m}{d}}$. This result implies that the bounds achieved by one-hidden-layer neural networks with a given smooth activation function cannot be improved.

The result proved in DeVore et al. (1989) in fact applies to any nonlinear approximation, including approximations by shallow and deep neural networks. Let $(X, \|\cdot\|)$ be a Banach space[4]. Define

$$\theta_n : X \to \mathbb{R}^n, \quad M_n : \mathbb{R}^n \to X,$$

where θ_n selects n parameters and M_n selects a n-parameter approximation. Both

[4] A Banach space is a complete normed vector space. Intuitively, this is a space of functions (or vectors) where distances and limits are well-defined, ensuring convergence of sequences under the given norm $\|\cdot\|$.

θ_n and M_n are assumed to be continuous. Let K be a compact subset of X. For each integer $n \geq 1$, the *continuous nonlinear manifold n-width* of K is

$$\delta_n(K; X) = \inf_{\theta_n, M_n} E(K, \theta_n, M_n, X),$$

where

$$E(K, \theta_n, M_n, X) = \sup_{f \in K} \|f - M_n(\theta_n(f))\|.$$

Restricting to the Sobolev space $W^{m,p}(B_d)$ and functions in $\mathcal{B}_d^{m,p}$, the result of DeVore et al. (1989) gives the following theorem.

Theorem 3.5 (DeVore et al. (1989))**.** *For every $p \in [1, \infty]$, every $m \geq 1$, and every $d \geq 1$, we have*

$$\delta_n(\mathcal{B}_d^{m,p}; L^p(B_d)) \geq C n^{-\frac{m}{d}},$$

where C is a constant independent of n.

The following result, as a corollary of Theorem 3.5, establishes a lower bound on the degree of approximation for one-hidden-layer neural networks.

Corollary 3.1. *Let σ be any continuous activation function. If $M : L^p(B_d) \to \mathcal{M}_n(\sigma)$ denotes the output of a method of approximation, where the weights of $M(f)$ depend continuously on f, then*

$$\sup_{f \in \mathcal{B}_d^{m,p}} \|f - M(f)\|_{L^p(B_d)} \geq C n^{-\frac{m}{d}},$$

where C is a constant independent of n.

Proof. Recall that an element of $\mathcal{M}_n(\sigma)$ takes the form

$$g(x) = \sum_{i=1}^{n} c_i \sigma(w_i \cdot x + b_i),$$

where $c_i, b_i \in \mathbb{R}$ and $w_i \in \mathbb{R}^d$. Hence, there are a total of $n(d+2)$ parameters determining g. By Theorem 3.5,

$$\sup_{f \in \mathcal{B}_d^{m,p}} \|f - M(f)\|_{L^p(B_d)} \geq \delta_{n(d+2)}(\mathcal{B}_d^{m,p}; L^p(B_d)) \geq C(n(d+2))^{-\frac{m}{d}} \geq C_2 n^{-\frac{m}{d}},$$

where C_2 is independent of n. □

Let us draw a comparison between this lower bound and that provided by Theorem 3.2. In Theorem 3.2, no assumption is made on the method of approximation. One can potentially select weights in a non-continuous way, and the lower bound is given by $C(n \log n)^{-\frac{m}{d}}$. This is smaller than the bound in Corollary 3.1, obtained using continuous n-width, by only a log factor.

Another comparison can be made with the degree of approximation by ridge functions and one-hidden layer neural networks (with a potentially pathological activation function), where we established $E \asymp n^{-\frac{m}{d-1}}$. According to Theorem 3.5, this optimal rate of approximation cannot be achieved in a continuous way. Indeed, this reflects a general theme: the optimal choice of network weights to minimize errors is known to be discontinuous in general (Kainen et al., 1999).

3.5 Approximation by Two-Hidden-Layer Neural Networks

In this section, we show that a two-hidden-layer neural network can potentially overcome the lower bounds on the degree of approximation by ridge functions and one-hidden-layer neural networks.

We state the following version of a celebrated theorem due to Kolmogorov (also known as the Kolmogorov-Arnold representation theorem). It was originally developed by Kolmogorov (1957) when providing a negative answer to Hilbert's 13th problem. The version stated below, along with its proof, can be found in (Lorentz et al., 1996, Theorem 1.1, Chapter 17, p. 553).

Theorem 3.6 (Kolmogorov's Theorem)**.** *There exist d positive constants λ_i, $i = 1, \dots, d$, such that $\sum_{i=1}^{d} \lambda_i \leq 1$, and $2d + 1$ strictly increasing continuous functions ϕ_j, $j = 1, \dots, 2d+1$, which map $[0,1]$ to itself, such that every continuous function $f \in \mathscr{C}([0,1]^d)$ has the representation*

$$f(x) = f(x_1, \dots, x_d) = \sum_{j=1}^{2d+1} g\left(\sum_{i=1}^{d} \lambda_i \phi_j(x_i)\right), \tag{3.12}$$

where $g \in \mathscr{C}([0,1])$ depends on f.

We now present a theorem from Maiorov and Pinkus (1999) that demonstrates there are no theoretical lower bounds when using a two-hidden-layer neural network to approximate a continuous function on a compact set.

Theorem 3.7 (Maiorov and Pinkus (1999))**.** *There exists an activation function $\sigma \in \mathscr{C}^\infty(\mathbb{R})$ that is strictly increasing and sigmoidal and has the following property: for every $f \in \mathscr{C}([0,1]^d)$ and every $\varepsilon > 0$, there exist constants β_j, γ_j, c_{ij}, b_{ij}, and vectors $w_{ij} \in \mathbb{R}^d$ such that*

$$\left| f(x) - \sum_{j=1}^{4d+3} \beta_j \sigma \left(\sum_{i=1}^{2d+1} c_{ij} \sigma(w_{ij} \cdot x + b_{ij}) + \gamma_j \right) \right| < \varepsilon,$$

for all $x \in [0,1]^d$.

Proof. We use the same construction of σ as in the proof of Proposition 3.2. Similar to (3.11), the following approximation guarantee is established: for every $h \in \mathscr{C}([-1,1])$ and every $\eta > 0$, there exists constants a_1, a_2, a_3, and integer k such that

$$|h(t) - (a_1\sigma(t-3) + a_2\sigma(t+1) + a_3\sigma(t+k))| < \eta,$$

for all $t \in [-1,1]$. Furthermore, $\sigma(t-3)$ and $\sigma(t+1)$ are linear for $t \in [-1,1]$. We shall only use this approximation property on $[0,1]$.

Fix any $f \in \mathscr{C}([0,1]^d)$ and $\varepsilon > 0$. Let g be given by Kolmogrov's theorem (Theorem 3.6) such that (3.12) holds for f and g, which is recalled here for convenience

$$f(x) = f(x_1, \ldots, x_d) = \sum_{j=1}^{2d+1} g \left(\sum_{i=1}^{d} \lambda_i \phi_j(x_i) \right). \tag{3.13}$$

We first approximate g. Let constants a_1, a_2, a_3, and integer k be such that

$$|g(t) - (a_1\sigma(t-3) + a_2\sigma(t+1) + a_3\sigma(t+k))| < \frac{\varepsilon}{2(2d+1)},$$

for all $t \in [0,1]$. Substituting this into (3.13), we obtain

$$\left| f(x) - \sum_{j=1}^{2d+1} \left(a_1\sigma \left(\sum_{i=1}^{d} \lambda_i \phi_j(x_i) - 3 \right) + a_2\sigma \left(\sum_{i=1}^{d} \lambda_i \phi_j(x_i) + 1 \right) + a_3\sigma \left(\sum_{i=1}^{d} \lambda_i \phi_j(x_i) + k \right) \right) \right| < \frac{\varepsilon}{2}.$$

By linearity of $\sigma(t-3)$ and $\sigma(t+1)$, we can combine the linear terms in $\sum_{i=1}^{d} \lambda_i \phi_j(x_i)$, $j = 1, \ldots, 2d+1$, introduced by $\sigma(t-3)$ and $\sigma(t+1)$, and rewrite the sum

$$\sum_{j=1}^{2d+1} \left(a_1 \sigma \left(\sum_{i=1}^{d} \lambda_i \phi_j(x_i) - 3 \right) + a_2 \sigma \left(\sum_{i=1}^{d} \lambda_i \phi_j(x_i) + 1 \right) \right)$$

as a sum of the form

$$\sum_{j=1}^{2d+2} d_j \sigma \left(\sum_{i=1}^{d} \lambda_i \phi_j(x_i) + c_j \right),$$

where ϕ_{2d+2} is ϕ_j for some $j \in \{1, 2, \cdots, 2d+1\}$, c_j is either -3 or 1, and $d_j \in \mathbb{R}$. We conclude that there exist constants β_j and γ_j such that

$$\left| f(x) - \sum_{j=1}^{4d+3} \beta_j \sigma \left(\sum_{i=1}^{d} \lambda_i \phi_j(x_i) + \gamma_j \right) \right| < \frac{\varepsilon}{2}. \tag{3.14}$$

We now approximate each $\phi_j(x_i)$. Given any $\delta > 0$, there exists a_{1j}, a_{2j}, a_{3j}, and integer k_j such that

$$\left| \phi_j(t) - (a_{1j}\sigma(t-3) + a_{2j}\sigma(t+1) + a_{3j}\sigma(t+k_j)) \right| < \delta,$$

for all $t \in [0, 1]$. By uniform continuity of the left-hand side of the following expression with respect to each $\phi_j(x_i)$, if δ is sufficiently small, we have

$$\left| \sum_{j=1}^{4d+3} \beta_j \sigma \left(\sum_{i=1}^{d} \lambda_i \phi_j(x_i) + \gamma_j \right) - \sum_{j=1}^{4d+3} \beta_j \sigma \left(\sum_{i=1}^{d} \lambda_i (a_{1j}\sigma(x_i - 3) + a_{2j}\sigma(x_i + 1) + a_{3j}\sigma(x_i + k_j)) + \gamma_j \right) \right| < \frac{\varepsilon}{2}. \tag{3.15}$$

Similarly, by linearity of $\sigma(t-3)$ and $\sigma(t+1)$, we can combine the linear terms in x_i, $i = 1, \ldots, d$, and rewrite

$$\sum_{i=1}^{d} \lambda_i (a_{1j}\sigma(x_i - 3) + a_{2j}\sigma(x_i + 1))$$

as a sum of at most $d+1$ terms of the form

$$\sum_{i=1}^{d+1} d_i \sigma(x_i + c_i),$$

where $d_i \in \mathbb{R}$ and $c_i \in \{-3, 1\}$ can be chosen. It follows that there exist $c_{ij}, b_{ij} \in \mathbb{R}$ and $w_{ij} \in \mathbb{R}^d$ such that

$$\left| \sum_{j=1}^{4d+3} \beta_j \sigma \left(\sum_{i=1}^{d} \lambda_i \phi_j(x_i) + \gamma_j \right) - \sum_{j=1}^{4d+3} \beta_j \sigma \left(\sum_{i=1}^{2d+1} c_{ij} \sigma(w_{ij} \cdot x + b_{ij}) + \gamma_j \right) \right| < \frac{\varepsilon}{2}. \tag{3.16}$$

The conclusion follows by combining (3.16) and (3.14) and using the triangle inequality. □

In contrast to the lower bounds on the degree of approximation by one-hidden-layer neural networks in Theorem 3.2 (for a sigmoidal activation function), Theorem 3.7 demonstrates that, when using a two-hidden-layer neural network, there are no inherent theoretical lower bounds on the degree of approximation. However, in light of the continuous width lower bound in Theorem 3.5, the choice of weights cannot be continuous with respect to the function being approximated.

3.6 Exercises

3.1. Prove that the number of distinct monomials of degree k in d variables is $\binom{k+d-1}{d-1}$. Conclude that the dimension of the space H_k of homogeneous polynomials of degree k in d variables is $\dim H_k = \binom{k+d-1}{d-1}$.

3.2. Show that

$$\binom{k+d-1}{d-1} \asymp k^{d-1}.$$

3.3. Show that $\{D^\alpha : |\alpha| = k\}$ forms a basis for the dual space H_k^* in the proof of Proposition 3.1. Specifically, prove that these differential operators act as linear functionals on H_k and are linearly independent.

3.4. Prove that, if $\{q_1, \ldots, q_n\}$ is a basis for H_k, then one can choose $\{a_1, \ldots, a_n\} \subseteq$

$\mathbb{R}^d$ such that the matrix

$$V := \begin{bmatrix} q_1(a_1) & q_1(a_2) & \cdots & q_1(a_n) \\ q_2(a_1) & q_2(a_2) & \cdots & q_2(a_n) \\ \vdots & \vdots & \ddots & \vdots \\ q_n(a_1) & q_n(a_2) & \cdots & q_n(a_n) \end{bmatrix}$$

has rank n.

3.5. Using the argument in the proof of Proposition 3.1, show that the following holds. Let $A \subseteq \mathbb{R}^d$ and fix $k \in \mathbb{N}$. Assume there is no nontrivial homogeneous polynomial $q \in H_k$ such that $q(a) = 0$ for all $a \in A$. Prove that there exist points $a_1, \ldots, a_n \in A$ such that

$$P_k \;=\; \left\{ \sum_{i=1}^{n} p_i(a_i \cdot x) : \; p_i \in \pi_k, \; i = 1, \ldots, n \right\} \;\subseteq\; \mathcal{R}(A),$$

where H_k is the space of homogeneous polynomials of degree k, P_k is the space of polynomials of degree at most k, π_k denotes the univariate polynomials of degree at most k, and $\mathcal{R}(A)$ is the set of all finite linear combinations of ridge functions $x \mapsto g(a \cdot x)$ with $a \in A$. Prove Lemma 2.1 using this result.

Chapter 4

Approximation of Compositional Functions

In Chapter 3, we observed that neural network approximation, and nonlinear approximation in general, suffers from the curse of dimensionality if we restrict the neural network architecture (e.g., to one hidden layer), the method of approximation (e.g., continuous weight selection), or the specific choice of activation function. In such cases, it can be shown that there exists a lower bound on the degree of approximation that can be achieved. This lower bound implies that the size of the neural network would increase exponentially with respect to the dimension in order to achieve the same level of approximation.

In this chapter, we present a particular class of functions for which this curse of dimensionality can be mitigated.

4.1 Compositional Functions

We define the notion of compositional functions, which can be shown to have lower complexity than that implied by their input dimension.

Definition 4.1. A *directed graph* is a pair $G = (V, E)$, where:

- V is a finite set of *vertices* (or nodes), and

- $E \subseteq \{(u, v) \mid u, v \in V\}$ is a set of *edges*, i.e., ordered pairs of vertices.

A *path* in G is a sequence of vertices $v_1, v_2, \ldots, v_k$ such that $(v_i, v_{i+1}) \in E$ for all $i = 1, 2, \ldots, k-1$. A *cycle* in G is a path $v_1, v_2, \ldots, v_k$ with $k \geq 2$ such that $v_1 = v_k$ and the vertices $v_1, v_2, \ldots, v_{k-1}$ are distinct. A *directed acyclic graph (DAG)* is a directed graph with no cycles.

We can always sort the nodes of a DAG by their levels of dependence on other nodes. A node v depends on a node u if there is an edge from u to v.

Given a DAG $G = (V, E)$, an input node of G is a node without incoming edges. Similarly, an output node is a node without outgoing edges. We define the layer of a node in G as follows. The *predecessor* of a node $v \in V$ is defined as

$$\text{pred}(v) = \{u \mid (u, v) \in E\}.$$

Input nodes are defined to have layer 0. For all other nodes, we recursively define

$$\text{layer}(v) = \max_{u \in \text{pred}(v)} \text{layer}(u) + 1.$$

Clearly, the output nodes are at the highest layers.

For each node $v \in V$, we define the *in-degree* (or *incoming degree*) of v, denoted as $\deg_{\text{in}}(v)$, as the number of incoming edges to v. Formally,

$$\deg_{\text{in}}(v) = |\text{pred}(v)| = |\{u \mid (u, v) \in E\}|.$$

Clearly, $\deg_{\text{in}}(v) = 0$ for any input node v.

We are now ready to define compositional functions over a DAG.

Definition 4.2. Let $G = (V, E)$ be a DAG. Let the number of input nodes be d and the number of output nodes be k. A function $f : \mathbb{R}^d \to \mathbb{R}^k$ is said to be a *compositional function* on G if it can be computed as follows:

- the input nodes are assigned values x_1, x_2, ..., and x_d;
- each node v other than the input and output nodes compute

$$h_v(u_1, \ldots, u_{\deg_{\text{in}}(v)}),$$

where $h_v : \mathbb{R}^{\deg_{\text{in}}(v)} \to \mathbb{R}$ and $u_1, \ldots, u_{\deg_{\text{in}}(v)}$ are values computed by predecessors of v;

- an output node computes a linear combination of the values computed by its predecessors.

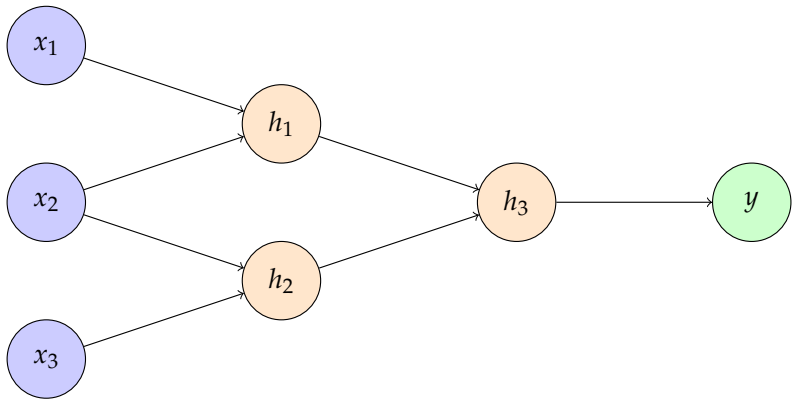

Figure 4.1: A compositional function defined on a directed acyclic graph. The graph can compute, for example, $f(x_1, x_2, x_3) = 4\cos(2\sin(x_1x_2) + x_2e^{-x_3}) + 1$.

We refer to the functions h_v as the constituent functions of the compositional function f.

Example 4.1. Figure 4.1 shows a DAG that can be used to define a compositional function such as $f(x_1, x_2, x_3) = 4\cos(2\sin(x_1x_2) + x_2e^{-x_3}) + 1$. For example, we can choose $h_3(s,t) = \cos(2s + t)$, $h_2(s,t) = se^{-t}$, and $h_1(s,t) = \sin(st)$. The choices of the functions h_1, h_2, h_3 to represent the same f over the DAG G is clearly not unique. For example, we can also choose $h_3(s,t) = \cos(2\sin(s) + t)$, $h_2(s,t) = se^{-t}$, and $h_1(s,t) = st$. Note that we allow the output node to apply another linear function $h(s) = 4s + 1$.

The definition of a function over a DAG clearly resembles that of a neural network. In fact, it includes neural network functions as special cases. A function defined over a DAG is allowed to have skip connections, as shown in Figure 4.2. We leave it as an exercise (Exercise 4.1) to define a compositional function that can use the DAG in Figure 4.2 as its computational graph.

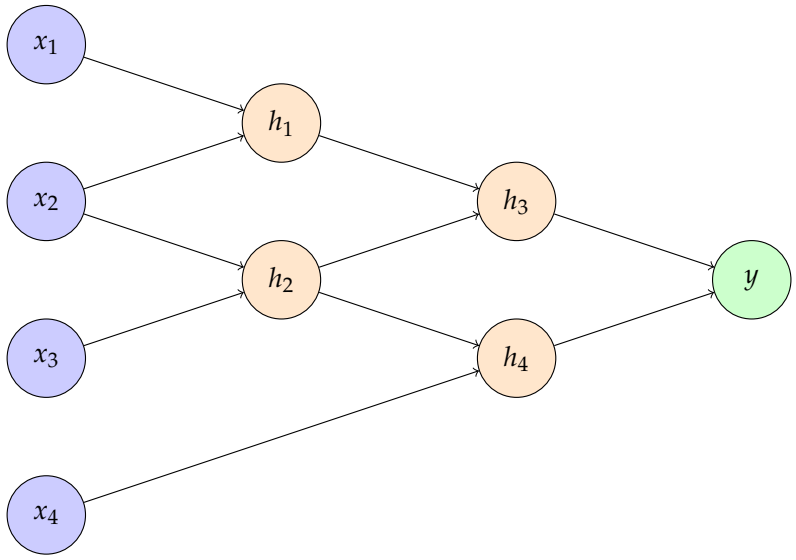

Figure 4.2: A directed acyclic graph with edges connecting non-adjacent layers.

4.2 Degree of Approximation for Compositional Functions

In Section 3.1, we defined the Sobolev space of functions that have m derivatives in L^p. Specifically, we defined the Sobolev space $W^{m,p}(B_d)$, where B_d is the domain of these functions, taken to be the unit Euclidean ball in $\mathbb{R}^d$.

The domain B_d is naturally not well-suited for decomposition, as its definition inherently couples the dimensions through the radial constraint. In this section, we consider the Sobolev space $W^{m,p}([-1,1]^d)$, where $[-1,1]^d$ is a hypercube. While hyperrectangular domains can also be considered, we restrict our focus to hypercubes for simplicity. We also focus on the case $p = \infty$ only, because we essentially require bounding error propagation from one layer to the next using the uniform/supremum norm of functions.

We consider the following assumption on a composition function f defined over a DAG $G = (V, E)$.

Assumption 4.1. *Let $G = (V, E)$ be a DAG and V_- denote the set of nodes excluding input and output nodes. Assume that $f : [-1,1]^d \to \mathbb{R}$ is a compositional function satisfying*

the following conditions:

(1) For each $v \in V_-$, it computes $h_v : [-1,1]^{d_v} \to \mathbb{R}$, where $d_v = \deg_{in}(v)$ and $h_v \in W^{m_v,\infty}([-1,1]^{d_v})$ and $\|h_v\|_{L^\infty([-1,1]^{d_v})} \leq 1$. We assume $d_v \geq 1$ and $m_v \geq 1$ for each $v \in V_-$.

(2) There exists a constant $\delta > 0$ such that each h_v is Lipschitz continuous with Lipschitz constant $L_v > 0$, i.e.,

$$|h_v(x) - h_v(y)| \leq L_v \|x - y\|_\infty, \quad \forall x, y \in [-1-\delta, 1+\delta]^{d_v},$$

where $\|\cdot\|_\infty$ is the infinity norm (or maximum norm) of a vector defined vector $x \in \mathbb{R}^d$, defined as $\|x\|_\infty = \max_{1\leq i\leq d} |x_i|$.

We provide some explanation of Assumption 4.1 as follows. Assumption 4.1(1) is similar to what we assumed for functions in

$$\mathcal{B}_d^{m,p} = \{f \in W^{m,p}(B_d) : \|f\|_{W^{m,p}(B_d)} \leq 1\},$$

in Section 3.1, but with the domain changed to the hypercube $[-1,1]^{d_v}$, where the dimension d_v is different for each constituent function h_v. Assumption 4.1(2) assumes Lipschitz continuity for each function h_v but on a slight larger domain than $[-1,1]^{d_v}$. This is used in the proof of the following theorem to ensure domains are compatible for composition when discussing Lipschitz continuity of the output of the compositional function f with respect to values computed by a constituent function h_v and its approximation.

Theorem 4.1. *Let $G = (V, E)$ be a DAG and $f : [-1,1]^d \to \mathbb{R}$ be a compositional function defined over G satisfying Assumption 4.1. Let V_- denote the set of nodes excluding input and output nodes. Let f_{NN} be a neural network function obtained by replacing each h_v, $v \in V_-$, with a one-hidden layer neural network with input dimension $\deg_{in}(v)$, output dimension 1, and width n. For each $v \in V_-$, there exists a constant C_v, independent of n, such that, for any $\varepsilon > 0$, there exists such a neural network f_{NN} such that*

$$|f(x) - f_{NN}(x)| \leq \sum_{v\in V_-} C_v(n^{-\frac{m_v}{d_v}} + \varepsilon) \tag{4.1}$$

for all $x \in [-1,1]^d$.

Proof. For each $\varepsilon > 0$ and each h_v, $v \in V_-$, by the degree of approximation[1] to functions in the unit ball of $W^{m_v,\infty}([-1,1]^{d_v})$ with respect to the infinity norm, i.e., functions satisfying $\|h_v\|_{L^\infty([-1,1]^{d_v})} \leq 1$, we can have a one-hidden layer neural network with input dimension $\deg_{in}(v)$, output dimension 1, and width n, denoted by $\hat{h}_v$, to achieve

$$\left|h_v(x) - \hat{h}_v(x)\right| \leq Cn^{-\frac{m_v}{d_v}} + \varepsilon, \quad \forall x \in [-1,1]^{d_v}, \tag{4.2}$$

where the constant C is independent of n and the function h_v. Let f_{NN} denote the neural network function obtained by replacing each h_v, $v \in V_-$, with $\hat{h}_v$. We assume that the right-hand side of the equation above is sufficiently small (by choosing n sufficiently large and ε sufficiently small) such that evaluation of $h_{v'}$ in f at any higher-layer node v' that depends on any lower-layer node h_v, after replacing all such node h_v with $\hat{h}_v$, still lie in the domain $[-1-\delta, 1+\delta]^{d_{v'}}$.

We show by induction that the error estimate (4.1) holds. We first show the base case. Write $V_- = \{v_1, \ldots, v_K\}$, $K > 0$ is the number of nodes in V_-. We also assume that the nodes are ranked by their layers, from lowest to highest. We first replace h_{v_1} with $\hat{h}_{v_1}$ described above and denote the resulting function by f^1_{NN}. We have

$$\left|f(x) - f^1_{\text{NN}}(x)\right| \leq \hat{L}_v(Cn^{-\frac{m_v}{d_v}} + \varepsilon), \quad \forall x \in [-1,1]^d, \tag{4.3}$$

where $\hat{L}_v > 0$ is a Lipschitz constant of f viewed as a function of h_v (see exercises 4.2 and 4.3).

Let $f^k_{\text{NN}}(x)$ be obtained by replacing each h_{v_i} with $\hat{h}_{v_i}$ for $i = 1, \ldots, k$, where $k < K$ and we have

$$\left|f(x) - f^k_{\text{NN}}(x)\right| \leq \sum_{i=1}^{k} \hat{L}_{v_i}(Cn^{-\frac{m_{v_i}}{d_{v_i}}} + \varepsilon), \quad \forall x \in [-1,1]^d. \tag{4.4}$$

We now construct $f^{k+1}_{\text{NN}}(x)$ from $f^k_{\text{NN}}(x)$ by further replacing $h_{v_{k+1}}$ with $\hat{h}_{v_{k+1}}$. Let

[1] While we proved the results only for functions defined on a unit Euclidean ball and the domain here is a hypercube, one can move between these domains by using restriction and extension operators that preserve equivalent Sobolev norms. This is a common technique in analysis of functions in Sobolev spaces.

$\hat{L}_{v_{k+1}} > 0$ be a Lipschitz constant of f viewed as a function of $h_{v_{k+1}}$, we have

$$\left|f_{\mathrm{NN}}^{k}(x) - f_{\mathrm{NN}}^{k+1}(x)\right| \leq \hat{L}_{v_{k+1}}(Cn^{-\frac{m_{v_{k+1}}}{d_{v_{k+1}}}} + \varepsilon), \quad \forall x \in [-1,1]^d. \tag{4.5}$$

Combining (4.5) and (4.4) using the triangle inequality, we obtain

$$\left|f(x) - f_{\mathrm{NN}}^{k+1}(x)\right| \leq \sum_{i=1}^{k+1} \hat{L}_{v_i}(Cn^{-\frac{m_{v_i}}{d_{v_i}}} + \varepsilon). \tag{4.6}$$

Note that $f_{\mathrm{NN}}^{K}(x) = f_{\mathrm{NN}}(x)$. By induction, the conclusion of the theorem follows by taking $C_v = \max(\hat{L}_v C, \hat{L}_v)$. □

Following Theorem 4.1, we can derive the following degree of approximation for compositional functions.

Corollary 4.1. *Let $\mathscr{H}_r^{N,L,M}$ denote the set of compositional functions satisfying Assumption 4.1 with at most N nodes in its computational graph (excluding input and output nodes), Lipschitz constants of each constitute functions bounded by L, the norm of weight matrix the output layer bounded by M, and $\frac{m_v}{d_v} \geq r$ for all $v \in V_-$. Let $\hat{\mathscr{H}}_n$ denote the set of neural network functions that are obtained by replacing each constitute function h_v with a one-hidden layer neural network with input dimension $\deg_{in}(v)$, output dimension 1, and width n. We have*

$$E(\mathscr{H}_r^{N,L,M}, \hat{\mathscr{H}}_n, L^\infty([-1,1]^d)) \leq Cn^{-r},$$

where $C > 0$ is independent of n but may depend on N, L, and M. Note that a function in $\hat{\mathscr{H}}_n$ has at most $\mathcal{O}(Nn)$ parameters.

Proof. The class of functions $\mathscr{H}_r^{N,L,M}$ limits the size of the DAG on which the compositional functions are defined, the Lipschitz constant of each constituent function, and the maximum value of the output of such functions. Under these assumptions, we can obtain an upper bound for the constant C_v in (4.1), independent of the individual function and the size of the individual neural network n. We can then define $C = \max_{v \in V_-} C_v N$. It follows that, for each $f \in \mathscr{H}_r^{N,L,M}$ and every $\varepsilon > 0$, there exists f_{NN} as derived in Theorem 4.1 such that

$$|f(x) - f_{\mathrm{NN}}(x)| \leq \sum_{v \in V_-} C_v(n^{-\frac{m_v}{d_v}} + \varepsilon) \leq C(n^{-r} + \varepsilon), \tag{4.7}$$

where $C > 0$ is independent of n and $\varepsilon > 0$. The conclusion of the corollary follows. Since each one-hidden-layer neural network of width n, as described in Theorem 4.1, for approximating h_v has $O(n)$ parameters, the total number of parameters in f_{NN} is $O(Nn)$, as there are N such nodes in total. □

Remark 4.1. The number r defined in Corollary 4.1 can be viewed as the effective dimension of the function class $\mathscr{H}_r^{N,L,M}$. By imposing a compositional structure on the function f, we avoid dependence on its actual input dimension. The smoothness of the individual functions and their input dimensions contribute to r, which determines the size of the network required to achieve a given approximation accuracy. The corollary concludes that $\mathcal{O}(\varepsilon^r)$ parameters are sufficient to achieve an approximation accuracy of ε. Recall that for functions in $\mathscr{B}_d^{m,p}$, we need $(\varepsilon^{\frac{d}{m}})$ parameters, assuming that weight selection is continuous, in view of Theorem 3.5. What we have discussed in this section is that we can exploit specific structures and properties of function classes to reduce the approximation complexity.

4.3 Exercises

4.1. Construct a compositional function that can use the DAG in Figure 4.2 as its computational graph. State the functions h_1, h_2, h_3, and h_4 clearly.

4.2. Let f_i, $i = 1, \ldots, k$, be vector-valued multivariate functions whose domains and ranges are compatible for the following composition:

$$y = f(x) = f_1 \circ \ldots \circ f_k(x).$$

Assume that each f_i is Lipschitz continuous with constant $L_i > 0$ with respect to the infinity norm, i.e.,

$$\|f_i(x) - f_i(y)\|_\infty \leq L_i \|x - y\|_\infty, \quad \forall x, y \in [-1, 1]^{d_i}.$$

Show that f is Lipschitz continuous with Lipschitz constant $L_1 L_2 \cdots L_k$.

4.3. Show that an affine map $x \mapsto f(x) = Ax + b$, where $x \in \mathbb{R}^d$, $A \in \mathbb{R}^{d \times k}$, and $b \in \mathbb{R}^k$, is Lipschitz continuous on $\mathbb{R}^d$ (with respect to any vector norms in $\mathbb{R}^d$ and $\mathbb{R}^k$).

Chapter 5

Approximation by ReLU Networks

In the previous chapters, we discussed the degree of approximation for different function classes using neural networks with smooth activation functions. In this chapter, we provide a brief introduction to the degree of approximation by neural networks with ReLU activation functions.

5.1 ReLU Networks

A ReLU network is a neural network using the rectified linear unit (ReLU)

$$\mathrm{ReLU}(x) = \max(0, x) = \begin{cases} x, & x \geq 0, \\ 0, & x < 0. \end{cases} \tag{5.1}$$

We recall and define the following notation:

- $\mathcal{M}(\mathrm{ReLU})$: the set of one-hidden-layer neural network functions using ReLU as the activation function.
- $\mathcal{M}_n(\mathrm{ReLU})$: the set of one-hidden-layer neural network functions using ReLU as the activation function, with n neurons in the hidden layer.
- $\mathcal{M}_W^L(\mathrm{ReLU})$: the set of neural network functions using ReLU as the activation function, with L hidden layers and W neurons in each hidden layer.

If we want to emphasize the input and output dimensions, we can explicitly write $\mathcal{M}(\text{ReLU}; d, k)$, $\mathcal{M}_n(\text{ReLU}; d, k)$, and $\mathcal{M}_W^L(\text{ReLU}; d, k)$, where d is the input dimension and k is the output dimension. By Theorem 2.1, we know that one-hidden-layer ReLU networks $\mathcal{M}(\text{ReLU}; d, 1)$ are dense in $\mathcal{C}(\mathbb{R}^d)$. From Theorem 2.2, we also know that narrow deep ReLU networks $\cup_{L=1}^{\infty} \mathcal{M}_{d+2}^L(\text{ReLU}; d, 1)$ are dense in $\mathcal{C}(\mathbb{R}^d)$. In fact, the minimum width to achieve density can be reduced to $d + 1$ for ReLU activation in this case (Hanin and Sellke, 2017). See Exercise 5.3, which shows that for $d = 1$, ReLU networks of width 1 cannot achieve universality.

The lower bound for the degree of approximation with continuous weight selection (Theorem 3.5 and Corollary 3.1) also applies to ReLU networks. Let n be the number of parameters in the neural network and d the input dimension. The degree of approximation obtained for functions in the Sobolev space $W^{m,p}(B_d)$ with unit norm is $n^{-\frac{m}{d}}$. Setting $m = d = 1$ (i.e., univariate functions with bounded derivatives), the degree of approximation simplifies to $1/n$. See Exercise 5.1, which shows that one can easily achieve such a degree of approximation for Lipschitz continuous functions with a fixed Lipschitz constant. The next section presents an intriguing result showing that deep ReLU networks can improve the degree of approximation for this class of functions to $1/n^2$.

We use the following example (Telgarsky, 2015) (see also (Yarotsky, 2017)) to illustrate how deep ReLU networks can exponentially increase the number of breakpoints in a function approximation via composition. This demonstrates the potential expressivity advantage of depth in neural networks.

Example 5.1. The "tooth" function is defined as

$$h(x) = \max(0, 1 - |1 - 2x|), \quad x \in \mathbb{R}. \tag{5.2}$$

This is a piecewise linear hat function centred at $x = \frac{1}{2}$, attaining its maximum value 1 at $x = \frac{1}{2}$ and decaying linearly to zero at $x = 0$ and $x = 1$. Restricted to $[0, 1]$, it simplifies to

$$h(x) = 1 - |1 - 2x|,$$

which can be expressed using a ReLU network as

$$h(x) = 1 - \text{ReLU}(1 - 2x) - \text{ReLU}(2x - 1). \tag{5.3}$$

Thus a single hidden-layer ReLU network can represent this function exactly.

Now consider composing h with itself. Define

$$h^{\circ L}(x) = \underbrace{h \circ h \circ \cdots \circ h}_{L \text{ times}}(x).$$

Each composition doubles the number of linear pieces. Consequently the function $h^{\circ L}$ forms a sawtooth function with 2^{L-1} teeth, as illustrated in Figure 5.1. Note that a ReLU network of depth L and fixed width has CL parameters for some constant C.

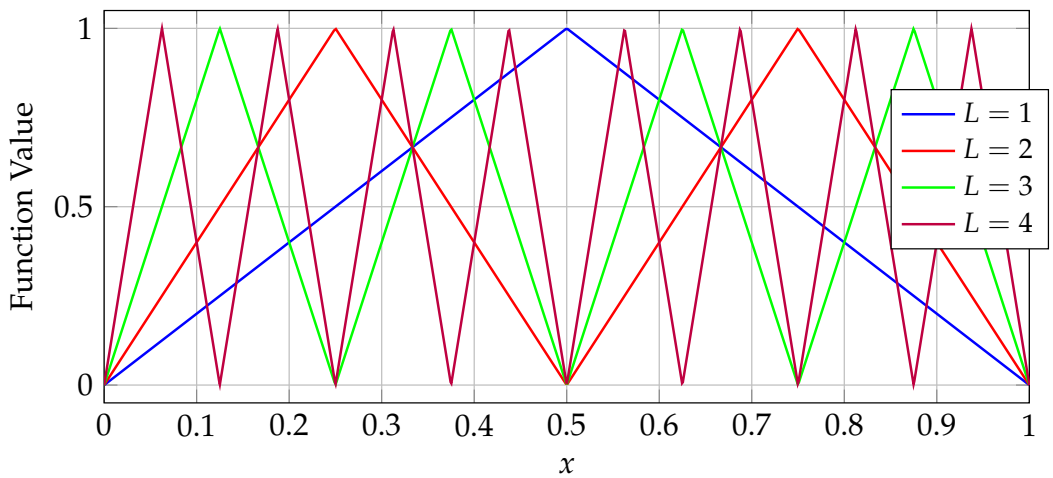

Figure 5.1: Sawtooth functions created by composing the hat function (Example 5.1).

One should note, however, that this example is far from demonstrating that a deep ReLU network with n parameters can, in general, interpolate an exponential number of data points in n. Nonetheless, using such sawtooth functions, Yarotsky (2017) was able to show that deep ReLU networks with unbounded depth can approximate the function $f(x) = x^2$ more efficiently than a depth-bounded ReLU network. Furthermore, we have the following lemma, which shows that with Cn parameters, where C is a constant, one can interpolate $n^2 + 1$ special types of data points known as bit data. The technique behind this lemma is known as bit extraction using ReLU networks (Yarotsky, 2018).

Lemma 5.1. *Let $N = n^2$ and $n \geq 4$ be an even integer. Let $t_i = i/N$, $i = 0, 1, \ldots, N$.*

Consider $N+1$ data points: y_i, $i = 0, 1, \ldots, N$, with the following properties

(1) $y_{jn=0}$, $j = 0, 1, \ldots, n$,
(2) $y_{i+1} = y_i + \varepsilon_i$, *with* $\varepsilon_i \in \{-1, 1\}$ *for* $i = 0, 1, \ldots, N-1$.

Then there exists a ReLU neural network $h \in \mathcal{M}_{11}^{15n+2}(ReLU)$ *such that*

$$h(t_i) = y_i, \quad i = 0, 1, \ldots, N,$$

and

$$|h(t) - h(t_i)| \leq 1, \quad t \in [t_i, t_{i+1}], \quad i = 0, 1, \ldots, N-1.$$

The result states that a deep ReLU network of width 11 and depth $15n + 2$ can interpolate n^2+1 bit data defined above. Note that such a network has Cn parameters for some constant C. The proof of this lemma is too complicated to be presented here and can be found in (DeVore et al., 2021, Theorem 3.13). The case for input dimension $d \geq 2$ can be found in Yarotsky (2018); Shen et al. (2019).

5.2 Degree of Approximation by Deep ReLU networks

In this section, we focus on degree of approximation for Lipschitz continuous functions defined on $[0, 1]$ with a Lipschitz constant bounded by 1. Extending to any bounded domain and Lipschitz constant is trivial. Extensions to multivariate functions with higher-order smoothness are beyond the scope of this discussion; see Yarotsky (2017); Lu et al. (2021). We shall also remark on the rate of approximation achieved by ReLU networks for smooth multivariate functions at the end of this chapter.

We state the main result of this section. The proof follows (DeVore et al., 2021, Theorem 8.11).

Theorem 5.1. *Let K denote the set of Lipschitz continuous functions on $[0, 1]$ with Lipschitz constant 1. Then*

$$E(K, \mathcal{M}_{11}^{16n+2}, L^\infty([0, 1])) \leq 6n^{-2}. \tag{5.4}$$

Proof. Define $N = n^2$ for an even integer $n \geq 4$. Set $t_i = i/N$, $i = 0, \ldots, N$ and $\xi_j = j/n$, $j = 0, \ldots, n$. Given a target function $f \in K$, we first construct h_0 that

interpolate f at the points ξ_j, $j = 0, \ldots, n$, using a continuous piecewise linear function. This function has $n-1$ breakpoints at ξ_j, $j = 1, \ldots, n-1$. We can implement this using a one-hidden-layer ReLU network of width n (Exercise 5.1). Similar to the proof of Lemma 2.4, one can also show that h_0 can be represented by a function in $\mathcal{M}_3^n(\text{ReLU})$ (Exercise 5.5).

Observe that h_0 has the following properties:

(1) $\sup_{t\in[0,1]} |f(t) - h_0(t)| \le \frac{1}{n}$,
(2) $|h_0(s) - h_0(t)| \le |s-t|$ for all $s, t \in [0,1]$, i.e., $h_0 \in K$.

Define $R = f - h_0$. Then $R(\xi_j) = 0$, $j = 0, \ldots, n$. We also have

$$|R(s) - R(t)| \le |f(s) - f(t)| + |h_0(s) - h_0(t)| \le 2\,|s-t|\,, \quad s, t \in [0,1],$$

i.e., R has Lipschitz constant 2 on $[0,1]$.

Claim: there exists a sequence $\varepsilon_i \in \{-1, 1\}$, $i = 0, \ldots, N-1$, such that the sequence y_i, $i = 0, \ldots, N$, defined as $y_0 = 0$ and

$$y_{i+1} = y_i + \varepsilon_i, \quad i = 0, \ldots, N-1$$

satisfies $y_{jn} = 0$, $j = 0, \ldots, n$, and $|R(t_i) - 2y_i/N| \le 2/N$, $i = 0, \ldots, N$.

Proof of the claim: We only need to prove the claim for $j = 0$ and $i = 0, \ldots, n-1$. The remaining case is just a repetition of the same argument. We prove by induction on i. For $i = 0$, clearly $y_0 = 0$ and $R(t_0) = 0$. Since R has Lipschitz constant 2, we have $|R(t_1)| \le 2/N$, we can pick $y_1 = \varepsilon_0 \in \{-1, 1\}$ (with the same sign of $R(t_1)$) such that $|R(t_1) - 2y_1/N| \le 2/N$ holds. Now assume that we have picked ε_i, $i = 0, \ldots, k-1$ such that $|R(t_i) - 2y_i/N| \le 2/N$ holds for $i = 1, \ldots, k$. Since $R(t_{k+1})$ changes from $R(t_k)$ by at most $2/N$, if we choose ε_k $\{-1, 1\}$ such that $2y_{k+1}/N = 2(y_k + \varepsilon_i)/N$ is closest to $R(t_{k+1})$, then we have $|R(t_{k+1}) - 2y_{k+1}/N| \le 2/N$ as well. We still need to show $y_n = 0$. Recall that $y_n = \sum_{i=0}^{n-1} \varepsilon_i$ is an even number because n is even. We also have $|R(t_n) - 2y_n/N| = |0 - 2y_n/N| = 2\,|y_n|\,/N \le 2/N$, which implies $|y_n| \le 1$. Hence we must have $y_n = 0$.

Let h_1 be a function given by Lemma 5.1, interpolating the data y_i at t_i, $i = 0, \ldots, N$, and satisfying $|h_1(t) - h_1(t_i)| \le 1$, for all $t \in [t_i, t_{i+1}]$, $i = 0, 1, \ldots, N-1$. Define $h =$

$h_0 + 2h_1/N$. Recall that $h_1 \in \mathscr{M}_{11}^{15n+2}(ReLU)$ and $h_0 \in \mathscr{M}_3^n(\text{ReLU})$. We can construct a deeper neural network in $\mathscr{M}_{11}^{16n+2}(ReLU)$ to perform the addition $h_0 + 2h_1/N$. The width is not increased because among the 11 channels there are already one for passing the input and one for collating the computed values.

We have

$$\begin{aligned}
|f(t) - h(t)| &= \left| f(t) - h_0(t) - \frac{2h_1(t)}{N} \right| \\
&= \left| R(t) - \frac{2h_1(t)}{N} \right| \\
&\le |R(t) - R(t_i)| + \left| R(t_i) - \frac{2h_1(t_i)}{N} \right| + \frac{2\,|h_1(t_i) - h_1(t)|}{N} \\
&\le \frac{2}{N} + \frac{2}{N} + \frac{2}{N} = \frac{6}{n^2}, \quad t \in [t_i, t_{i+1}], \quad i = 0, 1, \ldots, N-1,
\end{aligned}$$

where we used the triangle inequality for the first inequality and Lipschitz continuity of R, the claim above, and the property of h_1 for the second inequality. The proof is complete. □

We conclude this chapter with a brief discussion on degree of approximation by shallow and deep ReLU functions. A survey of results up to 2021 can be found in DeVore et al. (2021).

Remark 5.1 (Near-optimal degree of approximation for smooth functions). Lu et al. (2021) proved that deep ReLU networks of depth $\mathcal{O}(L \ln L)$ and width $\mathcal{O}(W \ln W)$ can achieve the approximation error

$$\mathcal{O}(W^{-\frac{2m}{d}} L^{-\frac{2m}{d}})$$

for functions in the unit ball of $\mathscr{C}^m([0,1]^d)$. They also showed that the upper bound is nearly optimal by establishing the following lower bound:

$$\Omega((W^2 L^2 \ln^3 W \ln^3 L)^{-\frac{m}{d}}),$$

modulo the ln factors.

Remark 5.2 (Lower bounds for shallow ReLU networks). Let K denote the set of

Lipschitz continuous functions on $[0,1]$ with Lipschitz constant 1. It is shown in Bach (2017) that the degree of approximation

$$E(K, \mathcal{M}_n(\text{ReLU}), L^\infty([0,1])) \leq Cn^{-\frac{1}{d}} \ln n.$$

A lower bound can also be proved (DeVore et al., 2021, p. 397)

$$E(K, \mathcal{M}_n(\text{ReLU}), L^\infty([0,1])) \geq C(n \ln n)^{-\frac{1}{d}}.$$

Again, the upper bound is optimal modulo the ln factor.

Remark 5.3 (Benefit of depth: ReLU networks and beyond). From the remarks above, it seems that deeper ReLU networks offer advantages over shallow ones. In particular, using an $\mathcal{O}(1)$-width ReLU network with $Cn \ln n$ layers results in $\mathcal{O}(n \ln n)$ parameters. With this setup, we achieve an approximation error of $\mathcal{O}((n \ln n)^{-\frac{2m}{d}})$. In contrast, using a shallow network of width n, with $\mathcal{O}(n)$ parameters, only achieves an approximation error of $\mathcal{O}(n^{-\frac{m}{d}})$.

It appears intriguing to know whether this benefit is specific to the use of ReLU activations, especially since the proof of Lemma 5.1 (see (DeVore et al., 2021, Theorem 3.13)) seems to crucially rely on the ReLU activation function. A recent result by Zhang et al. (2024) proves that a diverse class of activation functions, including most commonly used ones such as LeakyReLU, tanh, sigmoid, and softplus, allows a ReLU network of width W and depth L to be approximated to arbitrary precision on any bounded set by a network using any of the aforementioned activations, with width $3W$ and depth $2L$. A consequence of this result is that the upper bound in Remark 5.1 also holds for deep neural networks using a broad class of activation functions.

5.3 Exercises

5.1. Let Σ_n denote the set of continuous piecewise linear functions defined on $[0,1]$ with $n-1$ break points. Let $\mathcal{M}_n(\text{ReLU})$ denotes the set of functions represented by a one-hidden-layer neural network with ReLU activation and n neurons in

the hidden layer. Show that

$$\Sigma_n \subseteq \mathcal{M}_n(\text{ReLU}) \subseteq \Sigma_{n+1}.$$

5.2. Let K denote the set of Lipschitz continuous functions on $[0, 1]$ with Lipschitz constant 1. Show that the degree of approximation for K by one-hidden-layer ReLU networks of width n, i.e., $\mathcal{M}_n(\text{ReLU})$, in the infinity norm is upper bounded by C/n, where C is a constant independent of n.

5.3. Show that a neural network of width 1 and unbounded depth with the ReLU activation function cannot be dense in $\mathcal{C}(\mathbb{R})$.

5.4. Let $\sigma : \mathbb{R} \to \mathbb{R}$ be continuous and monotone. Show that any neural network on $\mathbb{R}$ of width 1 and arbitrary depth with activation σ represents a monotone function. Conclude that this class cannot be dense in $\mathcal{C}(\mathbb{R})$.

5.5. Consider a one-hidden-layer neural network with ReLU activation, width n, one input, and one output, of the form

$$x \mapsto \sum_{i=1}^{n} a_i \,\text{ReLU}(b_i x + c_i) + d,$$

where $a_i, b_i, c_i, d \in \mathbb{R}$, i.e., an element of $\mathcal{M}_n(\text{ReLU})$. Show that any such function can be represented by an element of $\mathcal{M}_3^n(\text{ReLU})$. In other words, prove that $\mathcal{M}_n(\text{ReLU}) \subseteq \mathcal{M}_3^n(\text{ReLU})$. (Hint: see the proof of Lemma 2.4.)

Part II

Optimization

Chapter 6

Unconstrained Smooth Optimization

Optimization is the backbone of neural network learning. At its core, deep learning is an optimization problem: finding the best set of weights that enable the network to generalize well beyond the training data. Without efficient optimization algorithms to train the millions—or even billions or trillions—of parameters in large models, the rapid advancements in modern AI would not have been possible.

In this chapter, we provide a basic mathematical introduction to the optimization of a smooth objective function.

6.1 Unconstrained Optimization

Most optimization problems in deep learning are unconstrained, which enables efficient gradient-based methods while allowing regularization and reparameterization to handle constraints implicitly.

Consider an objective function $f : \mathbb{R}^n \to \mathbb{R}$. The unconstrained minimization problem seeks to

$$\min_{x \in \mathbb{R}^n} f(x). \tag{6.1}$$

In the context of neural network training, f is understood as a loss function, and n represents the number of parameters to be trained.

Definition 6.1 (Global and local minima)**.** Consider a function $f : \mathbb{R}^n \to \mathbb{R}$.

- A point $x^* \in \mathbb{R}^n$ is called a *global minimum* of f if

$$f(x^*) \leq f(x), \quad \forall x \in \mathbb{R}^n.$$

- A point $x^* \in \mathbb{R}^n$ is called a *local minimum* of f if there exists $r > 0$ such that

$$f(x^*) \leq f(x), \quad \forall x \in B_r(x^*).$$

  where $B_r(x^*) = \{x \in \mathbb{R}^n \mid |x - x^*| < r\}$ denotes an open ball centered at x^* with radius r.
- A point $x^* \in \mathbb{R}^n$ is called a *strict local minimum* of f if there exists $r > 0$ such that

$$f(x^*) < f(x), \quad \forall x \in B_r(x^*) \setminus \{x^*\}.$$

To enable the efficient use of gradient-based methods and analyze their convergence, we require some basic assumptions about the regularity of the objective function. We consider different assumptions on the objective that will be used in this and subsequent chapters.

The following notation will be used. Given a function $f : \mathbb{R}^n \to \mathbb{R}$, its gradient is defined as

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \ldots, \frac{\partial f}{\partial x_n}\right)^\top.$$

Here, we follow the convention of interpreting the gradient as a column vector. The Hessian matrix of f is defined as

$$\nabla^2 f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}.$$

Definition 6.2 (L-smoothness)**.** A continuously differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ is said to be *L-smooth*, if its gradient is Lipschitz continuous with Lipschitz constant

L, i.e.,

$$|\nabla f(x) - \nabla f(y)| \leq L\,|x - y|\,,$$

for all $x, y \in \mathbb{R}^n$.

Definition 6.3 (Convexity)**.** *A differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ is said to be convex on $\mathbb{R}^n$, if it satisfies*

$$f(y) \geq f(x) + \nabla f(x) \cdot (y - x), \tag{6.2}$$

for all $x, y \in \mathbb{R}^n$.

Definition 6.4 (Strong convexity)**.** *A differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ is said to be μ-strongly convex on $\mathbb{R}^n$ for some $\mu \geq 0$, if it satisfies*

$$f(y) \geq f(x) + \nabla f(x) \cdot (y - x) + \frac{\mu}{2}\,|y - x|^2\,, \tag{6.3}$$

for all $x, y \in \mathbb{R}^n$. The number of μ is a strong convexity parameter for f.

Clearly, when $\mu = 0$, μ-strong convexity reduces to convexity.

We establish a few necessary and sufficient conditions for optimality.

Proposition 6.1 (First-order necessary condition for optimality)**.** *Let x^* be a local minimum of $f : \mathbb{R}^n \to \mathbb{R}$. Assume that f is differentiable at x^*. Then $\nabla f(x^*) = 0$.*

Proof. By the definition of differentiability, we have

$$\lim_{y \to x^*} \frac{|f(y) - f(x^*) - \nabla f(x^*) \cdot (y - x^*)|}{|y - x^*|} = 0.$$

Suppose that $\nabla f(x^*) \neq 0$. Let $r > 0$ be such that

$$\frac{|f(y) - f(x^*) - \nabla f(x^*) \cdot (y - x^*)|}{|y - x^*|} \leq \frac{|\nabla f(x^*)|}{2},$$

for all y such that $0 < |y - x^*| < r$. It follows that

$$f(y) - f(x^*) - \nabla f(x^*) \cdot (y - x^*) \leq \frac{1}{2}\,|\nabla f(x^*)|\,|y - x^*| \tag{6.4}$$

for all y such that $|y - x^*| < r$. Take $y = -s\nabla f(x^*) + x^*$, where $0 < s < \frac{r}{|\nabla f(x^*)|}$. Then

$|y - x^*| = s\,|\nabla f(x^*)| < r$. By (6.4), we have

$$f(y) - f(x^*) + s\,|\nabla f(x^*)|^2 \leq \frac{1}{2} s\,|\nabla f(x^*)|^2\,,$$

which implies

$$f(x^*) \geq f(y) + \frac{1}{2} s\,|\nabla f(x^*)|^2 > f(y),$$

where $y = -s\nabla f(x^*) + x^*$ and this holds for all $s \in (0, r/|\nabla f(x^*)|)$. This contradicts that x^* is a local minimum. Hence, we must have $\nabla f(x^*) = 0$. □

A point x^* satisfying $\nabla f(x^*) = 0$ is said to be a stationary point of f. The above first-order necessary condition for optimality says that a local minimum has to a stationary point of f, provided that f is differentiable at this point. A consequence of this result is that a local minimum for a convexity function has to be a global minimum.

Corollary 6.1. *Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is convex and x^* is a local minimum for f. Then x^* must be a global minimum.*

Proof. By Proposition 6.1, $\nabla f(x^*)$. By convexity, we have

$$f(y) \geq f(x^*) + \nabla f(x^*) \cdot (y - x) = f(x^*),$$

for all $y \in \mathbb{R}^n$, which shows that x^* is a global minimum. □

We recall the following definitions of positive definite and positive semi-definite matrices.

Definition 6.5. A symmetric matrix $A \in \mathbb{R}^{n\times n}$ is called *positive semi-definite*, denoted as $A \geq 0$, if, for all $x \in \mathbb{R}^n$,

$$x^\top A x \geq 0.$$

It is called *positive definite*, denoted as $A > 0$, if, for all nonzero $x \in \mathbb{R}^n$,

$$x^\top A x > 0.$$

Given symmetric matrices $A, B \in \mathbb{R}^{n\times n}$, we denote $A - B \geq 0$ by $A \geq B$ or equivalently

$B \leq A$. A similar meaning is assigned to $A > B$ and $A < B$.

From linear algebra, we know that a symmetric matrix A is positive definite (positive semi-definite) if and only if $\lambda_{\min}(A) > 0$ ($\lambda_{\min}(A) \geq 0$), where $\lambda_{\min}(A)$ denotes the smallest eigenvalue of A. This fact follows from the following lemma.

Lemma 6.1. *Let $A \in \mathbb{R}^{n\times n}$ be a symmetric matrix. Then we have*

$$\lambda_{\min}(A)\,|x|^2 \leq x^\top A x \leq \lambda_{\max}(A)\,|x|^2$$

for all $x \in \mathbb{R}^n$, where $\lambda_{\min}(A)$ and $\lambda_{\max}(A)$ denote the smallest and largest eigenvalues of A, respectively.

Proof. Since A is symmetric, there exists an orthogonal matrix S such that

$$S^T A S = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix},$$

where $\{\lambda_1, \cdots, \lambda_n\}$ are the real eigenvalues of A. Let $y = S^T x$ (or $x = Sy$). Then

$$x^T A x = y^T S^T A S y = \sum_{i=1}^{n} \lambda_i y_i^2 \leq \lambda_{\max}(A) y^T y = \lambda_{\max}(A)(S^T x)^T S^T x = \lambda_{\max}(A)\,|x|^2 .$$

Similarly, $x^T A x \geq \lambda_{\min}(A)\,|x|^2$. □

Proposition 6.2 (Second-order necessary condition for optimality)**.** *Let x^* be a local minimum of $f : \mathbb{R}^n \to \mathbb{R}$. Assume that f is twice differentiable at x^*. Then $\nabla^2 f(x^*) \geq 0$.*

Proof. By Taylor's theorem (see Corollary B.1 in Appendix B), we have

$$\lim_{y\to x^*} \frac{f(y) - f(x^*) - \nabla f(x^*)\cdot(y - x^*) - \frac{1}{2}(y - x^*)^\top \nabla^2 f(x^*)(y - x^*)}{|y - x^*|^2} = 0.$$

By the first-order optimality condition (Proposition 6.1), this reduces to

$$\lim_{y \to x^*} \frac{f(y) - f(x^*) - \frac{1}{2}(y - x^*)^\top \nabla^2 f(x^*)(y - x^*)}{|y - x^*|^2} = 0. \tag{6.5}$$

Let v be an eigenvalue vector of $\nabla^2 f(x^*)$ corresponding to $\lambda_{\min}(\nabla^2 f(x^*))$. Choose $y = sv + x^*$. Then $f(sv + x^*) \geq f(x^*)$ for all s sufficiently small because x^* is a local minimum of f. It follows that that

$$\begin{aligned} \lim_{s \to 0} \frac{f(sv + x^*) - f(x^*) - \frac{1}{2}s^2 v^\top \nabla^2 f(x^*) v}{s^2 |v|^2} &\geq \lim_{s \to 0} \frac{-\frac{1}{2}s^2 \lambda_{\min}(\nabla^2 f(x^*)) |v|^2}{s^2 |v|^2} \\ &= -\lambda_{\min}(\nabla^2 f(x^*)). \end{aligned} \tag{6.6}$$

By (6.5), we must have $-\lambda_{\min}(\nabla^2 f(x^*)) \leq 0$ or $\lambda_{\min}(\nabla^2 f(x^*)) \geq 0$. This implies $\nabla^2 f(x^*) \geq 0$. □

Proposition 6.3 (Second-order sufficient condition for optimality)**.** *Assume that f is twice differentiable at x^*, $\nabla f(x^*) = 0$, and $\nabla^2 f(x^*) > 0$. Then x^* be a strict local minimum of $f : \mathbb{R}^n \to \mathbb{R}$.*

Proof. By the assumption, we know $\lambda := \lambda_{\min}(\nabla^2 f(x^*)) > 0$. By Taylor's theorem,

$$\lim_{y \to x^*} \frac{f(y) - f(x^*) - \frac{1}{2}(y - x^*)^\top \nabla^2 f(x^*)(y - x^*)}{|y - x^*|^2} = 0.$$

Let $r > 0$ be sufficiently small such that

$$\frac{f(y) - f(x^*) - \frac{1}{2}(y - x^*)^\top \nabla^2 f(x^*)(y - x^*)}{|y - x^*|^2} \geq -\frac{1}{4}\lambda$$

holds for all y such that $0 < |y - x^*| < r$. It follows from Lemma 6.1 that

$$\begin{aligned} f(y) - f(x^*) &\geq \frac{1}{2}(y - x^*)^\top \nabla^2 f(x^*)(y - x^*) - \frac{1}{4}\lambda |y - x^*|^2 \\ &\geq \frac{1}{2}\lambda |y - x^*|^2 - \frac{1}{4}\lambda |y - x^*|^2 \\ &= \frac{1}{4}\lambda |y - x^*|^2, \end{aligned}$$

for all y such that $0 < |y - x^*| < r$, which clearly shows that x^* is a strict local minimum. □

Example 6.1. We use a few simple examples to illustrate the optimality conditions. Consider $f(x) = x^4$. It is easy to see that $x = 0$ is a strict local minimum of f. By the first-order necessary condition, we must have $f'(0) = 0$, which clearly holds. Furthermore, $f''(0) = 0$, which satisfies the second-order necessary condition but not the sufficient condition $f''(0) > 0$.

To show that neither the first-order nor the second-order necessary condition is sufficient, consider $f(x) = x^3$. We have $f'(0) = f''(0) = 0$, but $x = 0$ is neither a local minimum nor a local maximum of f.

Finally, consider $f(x) = e^x + x^4 - x$. Then $f'(0) = 0$ and $f''(0) = 1 > 0$. By the second-order sufficient condition for optimality, $x = 0$ is a strict local minimum for f.

6.2 Useful Inequalities

The convergence analysis of gradient descent methods crucially relies on inequalities that relate the values of the objective function, its gradient, and its inputs. We establish a few such inequalities in this section.

We first consider L-smooth functions.

Proposition 6.4 (Characterization of L-smooth functions)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be L-smooth. Then the following two inequalities hold:*

$$|f(y) - f(x) - \nabla f(x) \cdot (y - x)| \leq \frac{L}{2} |y - x|^2 \tag{6.7}$$

and

$$|(\nabla f(y) - \nabla f(x)) \cdot (y - x)| \leq L |y - x|^2 \tag{6.8}$$

for all $x, y \in \mathbb{R}^n$. Furthermore, if f is twice continuously differentiable on $\mathbb{R}^n$, then the following are equivalent:

(1) f is L-smooth.
(2) (6.7) holds for all $x, y \in \mathbb{R}^n$.

(3) (6.8) holds for all $x, y \in \mathbb{R}^n$.
(4) we have, for all $x \in \mathbb{R}^n$,

$$\left\|\nabla^2 f(x)\right\| \leq L. \tag{6.9}$$

Proof. We first consider the case where f is assumed to be continuous differentiable. Let $\phi(t) = f(x + t(y - x))$. Then ϕ is continuously differentiable and

$$\phi'(t) = \nabla f(x + t(y - x)) \cdot (y - x).$$

By the fundamental theorem of calculus,

$$f(y) - f(x) = \phi(1) - \phi(0) = \int_0^1 \phi'(t)dt = \int_0^1 \nabla f(x + t(y - x)) \cdot (y - x)dt.$$

It follows that

$$\begin{aligned}
|f(y) - f(x) - \nabla f(x) \cdot (y - x)| &= \left|\int_0^1 (\nabla f(x + t(y - x)) - \nabla f(x)) \cdot (y - x)dt\right| \\
&\leq \int_0^1 |\nabla f(x + t(y - x)) - \nabla f(x)|\, |y - x|\, dt \\
&\leq L\, |y - x|^2 \int_0^1 t dt = \frac{L}{2}\, |y - x|^2,
\end{aligned}$$

for all $x, y \in \mathbb{R}^n$. The proof of (6.8) is straightforward and left as an exercise (Exercise 6.5).

Now suppose that f is twice continuously differentiable on $\mathbb{R}^n$ and (6.9) holds for all $x \in \mathbb{R}^n$. By Taylor's theorem (Corollary B.2 in Appendix B), we have

$$\begin{aligned}
|\nabla f(y) - \nabla f(x)| &= \left|\int_0^1 \nabla^2 f(x + t(y - x))(y - x)dt\right| \\
&\leq \int_0^1 \left|\nabla^2 f(x + t(y - x))(y - x)\right| dt \\
&\leq \int_0^1 \left\|\nabla^2 f(x + t(y - x))\right\|\, |y - x|\, dt \\
&\leq \int_0^1 L\, |y - x|\, dt = L\, |y - x|.
\end{aligned}$$

Hence f is L-smooth and we have shown (4) $\Longrightarrow$ (1). We have also proved (1) $\Longrightarrow$ (2) in the first part of the proof. We can show (2) $\Longrightarrow$ (3) as follows. Write (6.7) as

$$-\frac{L}{2}|y-x|^2 \le f(y) - f(x) - \nabla f(x)\cdot(y-x) \le \frac{L}{2}|y-x|^2 .$$

Adding two copies of this inequality with x and y swapped gives (6.8).

We are left with the proof of (3) $\Longrightarrow$ (4). Suppose that $\left\|\nabla^2 f(x)\right\| > L$ for some $x \in \mathbb{R}^n$. Then at least one eigenvalue λ of $\nabla^2 f(x)$ satisfies $|\lambda| > L$. Let v be a corresponding eigenvector of λ and choose $y = sv + x$, where $s \in (0,1)$. By (6.7) and Taylor's theorem (Corollary B.2 in Appendix B), we have

$$\begin{aligned}
&|(\nabla f(y) - \nabla f(x))\cdot(y-x)| = \left|\int_0^1 sv^\top \nabla^2 f(x+tsv)svdt\right| \\
&= \left|\int_0^1 sv^\top \nabla^2 f(x)svdt + \int_0^1 sv^\top(\nabla^2 f(x+tsv) - \nabla^2 f(x))svdt\right| \\
&\ge \left|\int_0^1 sv^\top \nabla^2 f(x)svdt\right| - \left|\int_0^1 sv^\top(\nabla^2 f(x+tsv) - \nabla^2 f(x))svdt\right| \\
&\ge |\lambda|\,|sv|^2 - \int_0^1 \left\|\nabla^2 f(x+tsv) - \nabla^2 f(x))\right\| |sv|^2\, dt
\end{aligned}$$

By continuity of $\nabla^2 f$, for any $\varepsilon > 0$, we can choose $r > 0$ such that, for all $s \in (0,r)$, we have $\left\|\nabla^2 f(x+tsv) - \nabla^2 f(x)\right\| < \varepsilon$ for all $t \in [0,1]$. It follows from the inequality above that

$$|(\nabla f(y) - \nabla f(x))\cdot(y-x)| \ge |\lambda|\,|sv|^2 - \varepsilon\,|sv|^2 .$$

Since $\varepsilon > 0$ is arbitrary, we can choose it sufficiently small such that $|\lambda| - \varepsilon > L$. Recall that $sv = y - x$. Then we have

$$|(\nabla f(y) - \nabla f(x))\cdot(y-x)| \ge (|\lambda| - \varepsilon)\,|y-x|^2 > L\,|y-x|^2 ,$$

which contradicts (6.8). Hence, we must have $\left\|\nabla^2 f(x)\right\| \le L$ for all $x \in \mathbb{R}^n$. □

Example 6.2. Consider a quadratic function

$$f(x) = b + a^\top x + \frac{1}{2}x^\top Q x.$$

We have $\nabla f(x) = a + Qx$. Clearly, ∇f is Lipschitz continuous with Lipschitz constant $\|Q\|$. This can also be verified by computing $\nabla^2 f(x) = Q$.

The L-smoothness condition essentially limits how fast the function can change compared to quadratic functions. More precisely, if f is L-smooth, then the bound (6.7) indicates that the graph of f always lies between the graph of the following two quadratic functions:

$$y = f(x_0) + \nabla f(x_0) \cdot (x - x_0) \pm \frac{L}{2} |x - x_0|^2 ,$$

where $x_0 \in \mathbb{R}^n$ is fixed.

We now characterize smooth convex functions. To treat strongly convex and general convex functions in a unified manner, we recall that μ-strong convexity reduces to standard convexity when $\mu = 0$.

Proposition 6.5 (Characterization of convex functions)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable and $\mu \geq 0$. The following are equivalent:*

(1) f is μ-strongly convex on $\mathbb{R}^n$.
(2) we have

$$(\nabla f(y) - \nabla f(x)) \cdot (y - x) \geq \mu |y - x|^2 \tag{6.10}$$

for all $x, y \in \mathbb{R}^n$.

Furthermore, if f is twice continuously differentiable on $\mathbb{R}^n$, then the above is also equivalent to

$$\lambda_{\min}(\nabla^2 f(x)) \geq \mu, \tag{6.11}$$

for all $x \in \mathbb{R}^n$.

Proof. By the definition of strong convexity, we have

$$f(y) - f(x) - \nabla f(x) \cdot (y - x) \geq \frac{\mu}{2} |y - x|^2 .$$

Swapping x and y gives

$$f(x) - f(y) - \nabla f(y) \cdot (x - y) \geq \frac{\mu}{2} |y - x|^2 .$$

Adding the above two inequalities give (6.10).

Now suppose that (6.10) holds. Let $\phi(t) = f(x + t(y - x))$. Then ϕ is continuously differentiable and

$$\phi'(t) = \nabla f(x + t(y - x)) \cdot (y - x).$$

By the fundamental theorem of calculus,

$$f(y) - f(x) = \phi(1) - \phi(0) = \int_0^1 \phi'(t)dt = \int_0^1 \nabla f(x + t(y - x)) \cdot (y - x)dt.$$

It follows from (6.10) that

$$\begin{aligned} f(y) - f(x) - \nabla f(x) \cdot (y - x) &= \int_0^1 (\nabla f(x + t(y - x)) - \nabla f(x)) \cdot (y - x)dt \\ &\geq \int_0^1 \mu t \left|y - x\right|^2 dt = \frac{\mu}{2} \left|y - x\right|^2, \end{aligned}$$

for all $x, y \in \mathbb{R}^n$. In fact, this is very similar to the proof of Proposition 6.4.

The rest of the proof is also very similar to Proposition 6.4 and left as an exercise (Exercise 6.6). □

So far, we have characterized (strong) convexity and L-smoothness separately. Combining them naturally leads to the following result, which follows immediately from Propositions 6.4 and 6.5.

Proposition 6.6 (Characterization of L-smooth strongly convex functions)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable and $L \geq \mu \geq 0$. Then the following are equivalent:*

(1) f is L-smooth and μ-strongly convex.

(2) We have

$$\mu \left|y - x\right|^2 \leq (\nabla f(y) - \nabla f(x)) \cdot (y - x) \leq L \left|y - x\right|^2 \tag{6.12}$$

for all $x, y \in \mathbb{R}^n$.

(3) We have

$$\frac{\mu}{2} \left|y - x\right|^2 \leq f(y) - f(x) - \nabla f(x) \cdot (y - x) \leq \frac{L}{2} \left|y - x\right|^2 \tag{6.13}$$

for all $x, y \in \mathbb{R}^n$.

Furthermore, if f is twice continuously differentiable on $\mathbb{R}^n$, then the above is also equivalent to

$$\mu I_n \leq \nabla^2 f(x) \leq L I_n, \tag{6.14}$$

for all $x \in \mathbb{R}^n$. Here I_n is the n-dimensional identity matrix.

Proof. Left as an exercise (Exercise 6.9). □

From the characterization of strongly convex and L-smooth functions, we see that if a function $f(x)$ is simultaneously μ-strongly convex and L-smooth, then the graph of f lies between the graph of the following two quadratic functions:

$$y = f(x_0) + \nabla f(x_0) \cdot (x - x_0) + \frac{\mu}{2}\|x - x_0\|^2,$$

and

$$y = f(x_0) + \nabla f(x_0) \cdot (x - x_0) + \frac{L}{2}\|x - x_0\|^2,$$

where $x_0 \in \mathbb{R}^n$ is fixed.

We derive some additional properties for functions that are L-smooth and (strongly) convex.

Proposition 6.7 (Characterization of L-smooth convex functions)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable. Then the following are equivalent:*

(1) f is L-smooth and convex.

(2) We have

$$0 \leq f(y) - f(x) - \nabla f(x) \cdot (y - x) \leq \frac{L}{2}\,|y - x|^2 \tag{6.15}$$

for all $x, y \in \mathbb{R}^n$.

(3) We have

$$0 \leq (\nabla f(y) - \nabla f(x)) \cdot (y - x) \leq L\,|y - x|^2 \tag{6.16}$$

for all $x, y \in \mathbb{R}^n$.

(4) We have

$$f(y) - f(x) - \nabla f(x) \cdot (y - x) \geq \frac{1}{2L}\,|\nabla f(y) - \nabla f(x)|^2 \tag{6.17}$$

for all $x, y \in \mathbb{R}^n$.

(5) We have

$$(\nabla f(y) - \nabla f(x)) \cdot (y - x) \geq \frac{1}{L} |\nabla f(y) - \nabla f(x)|^2 \tag{6.18}$$

for all $x, y \in \mathbb{R}^n$.

Proof. Equivalence between (1), (2), and (3) is already established by Proposition 6.6 with $\mu = 0$.

We have (4) $\Longrightarrow$ (5) by adding two copies of the inequality (6.17) with x and y swapped.

Inequality (6.18) implies (6.10) with $\mu = 0$. It follows that f is convex. It also implies, by the Cauchy-Schwartz inequality, that

$$|\nabla f(y) - \nabla f(x)|^2 \leq L(\nabla f(y) - \nabla f(x)) \cdot (y - x) \leq L |\nabla f(y) - \nabla f(x)| |y - x| ,$$

which shows f is also L-smooth. Hence, we have (5) $\Longrightarrow$ (1).

We are left to prove (1) $\Longrightarrow$ (4). Fix any $x \in \mathbb{R}^n$. Consider the function $\phi(y) = f(y) - \nabla f(x) \cdot y$. Then ϕ is convex and L-smooth. Clearly, x is a stationary point for ϕ and hence a global minimum of ϕ. It follows that

$$\begin{aligned} \phi(x) &= \min_{z \in \mathbb{R}^n} \phi(z) \\ &\leq \min_{z \in \mathbb{R}^n} \left[\phi(y) + \nabla \phi(y) \cdot (z - y) + \frac{L}{2} |z - y|^2 \right] \\ &= \phi(y) - \frac{1}{2L} \left| \nabla \phi(y) \right|^2 , \end{aligned} \tag{6.19}$$

where the only inequality is based on (6.7) from Proposition 6.4 and the least equation is because the convex quadratic function in z is minimized at $z = y - \frac{1}{L} \nabla \phi(y)$. What we obtained in (6.19) is precisely (6.17). □

A similar technique to the proof of Proposition 6.7(4) can be used to prove the following additional properties for strongly convex functions.

Proposition 6.8 (Additional properties of strongly convex functions)**.** *Let* $f : \mathbb{R}^n \to \mathbb{R}$ *be continuously differentiable. Assume that* f *is* μ*-strongly convex with* $\mu > 0$*. Then the*

following inequalities hold for all $x, y \in \mathbb{R}^n$:

$$f(y) - f(x) - \nabla f(x) \cdot (y - x) \leq \frac{1}{2\mu} |\nabla f(x) - \nabla f(y)|^2 , \tag{6.20}$$

$$(\nabla f(y) - \nabla f(x)) \cdot (y - x) \leq \frac{1}{\mu} |\nabla f(x) - \nabla f(y)|^2 . \tag{6.21}$$

Proof. Inequality (6.21) follows from (6.20) by adding two copies of (6.20) with x and y swapped.

The proof of (6.20) is similar to that of (6.17) in Proposition 6.7. Fix any $x \in \mathbb{R}^n$. Consider the function $\phi(y) = f(y) - \nabla f(x) \cdot y$. Then ϕ is also μ-strongly convex. Clearly, x is a stationary point for ϕ and hence a global minimum of ϕ. It follows that

$$\begin{aligned} \phi(x) &= \min_{z \in \mathbb{R}^n} \phi(z) \\ &\geq \min_{z \in \mathbb{R}^n} \left[\phi(y) + \nabla \phi(y) \cdot (z - y) + \frac{\mu}{2} |z - y|^2 \right] \\ &= \phi(y) - \frac{1}{2\mu} \left| \nabla \phi(y) \right|^2 , \end{aligned} \tag{6.22}$$

where the only inequality is based on the definition of strong convexity and the least equation is because the convex quadratic function in z is minimized at $z = y - \frac{1}{\mu} \nabla \phi(y)$. What we obtained in (6.22) is precisely (6.20). □

Finally, combining μ-strong convexity and L-smoothness gives the following interesting bound, which sometimes can lead to improved bounds in convergence analysis of gradient descent methods.

Corollary 6.2. *Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable. Assume that f is L-smooth and μ-strongly convex. Then*

$$(\nabla f(y) - \nabla f(x)) \cdot (y - x) \geq \frac{\mu L}{\mu + L} |x - y|^2 + \frac{1}{\mu + L} |\nabla f(x) - \nabla f(y)|^2 , \tag{6.23}$$

for all $x, y \in \mathbb{R}^n$.

Proof. Clearly, we must have $L \geq \mu$ (see Proposition 6.6). If $\mu = L$, by Proposition

6.5 and that ∇f is L-Lipschitz, we have

$$(\nabla f(y) - \nabla f(x)) \cdot (y - x) \geq L\,|x - y|^2 \geq \frac{L}{2}\,|x - y|^2 + \frac{1}{2L}\,|\nabla f(x) - \nabla f(y)|^2\,,$$

which proves (6.23).

Assume $L > \mu$. Consider $\phi(x) = f(x) - \frac{\mu}{2}\,|x|^2$. Then $\nabla\phi(x) = \nabla f(x) - \mu x$. It follows that

$$0 \leq (\nabla\phi(x) - \nabla\phi(y)) \cdot (x - y) = (\nabla f(x) - \nabla f(y)) \cdot (x - y) - \mu\,|x - y|^2 \leq (L - \mu)\,|x - y|^2\,.$$

It follows from Proposition 6.6 that ϕ is convex and $(L - \mu)$-smooth. By Proposition 6.7, we have

$$(\nabla\phi(x) - \nabla\phi(y)) \cdot (x - y) \geq \frac{1}{L - \mu}\,\left|\nabla\phi(x) - \nabla\phi(y)\right|^2\,,$$

which is precisely (6.23). □

Remark 6.1. We make the following observation regarding (6.23) in Corollary 6.2. By definition, if a function f is μ-strongly convex, then it is also $\hat{\mu}$-strongly convex for any $\hat{\mu} \in [0, \mu]$. Similarly, if f is L-smooth, then it is also $\hat{L}$-smooth for all $\hat{L} \geq L$. Hence, (6.23) should hold for all $\hat{L} \geq L$ in place of L and all $\hat{\mu} \in [0, \mu]$ in place of μ. Taking $L \to \infty$ in (6.23), we recover (6.10). Taking $\mu = 0$, (6.23) reduces to (6.18).

6.3 Exercises

6.1. Show that a linear function $f(x) = a^\top x + b$, where $x \in \mathbb{R}^n$, $a \in \mathbb{R}^n$, and $b \in \mathbb{R}$, is convex.

6.2. Show that a quadratic function $f(x) = \frac{1}{2}x^\top Q x + a^\top x + b$, where $x \in \mathbb{R}^n$, $Q \in \mathbb{R}^{n\times n}$, $a \in \mathbb{R}^n$, and $b \in \mathbb{R}$, satisfies the following properties:

(1) f is convex if and only if Q is positive semi-definite.
(2) f is μ-strongly convex with $\mu > 0$ if and only if Q is positive definite and $\lambda_{\min}(Q) \geq \mu$.
(3) f is L-smooth for any $L \geq \|Q\|$.

6.3. Show that if $f : \mathbb{R}^n \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R}$ are both convex, then $f + g$ is also convex.

6.4. Let $f : \mathbb{R}^n \to \mathbb{R}$ be μ-strongly convex. Show that $g(x) = f(x) + a^\top x + b$, where $a \in \mathbb{R}^n$ and $b \in \mathbb{R}$, is also μ-strongly convex.

6.5. Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable on $\mathbb{R}^n$. Show that, if f is L-smooth, then

$$|(\nabla f(y) - \nabla f(x)) \cdot (y - x)| \leq L\,|y - x|^2 \tag{6.24}$$

holds for all $x, y \in \mathbb{R}^n$. Does this imply that f is L-smooth? What if we know that f is twice continuously differentiable on $\mathbb{R}^n$?

6.6. Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is twice continuously differentiable on $\mathbb{R}^n$. Show that f is μ-strongly convex if and only if $\lambda_{\min}(\nabla^2 f(x)) \geq \mu$ for all $x \in \mathbb{R}^n$.

6.7. Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is L-smooth and μ-strongly convex. Let $x^* \in \mathbb{R}^n$ be such that $\nabla f(x^*) = 0$. Write $f^* = f(x^*)$. Prove the following inequalities:

(1) $\frac{1}{2L}\,|\nabla f(x)|^2 \leq f(x) - f^* \leq \frac{L}{2}\,|x - x^*|^2$ for all $x \in \mathbb{R}^n$.
(2) $\frac{\mu}{2}\,|x - x^*|^2 \leq f(x) - f^* \leq \frac{1}{2\mu}\,|\nabla f(x)|^2$ for all $x \in \mathbb{R}^n$.

6.8. Suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is continuously differentiable on $\mathbb{R}^n$. Show that, if f is μ-strongly convex, then

$$|\nabla f(y) - \nabla f(x)| \geq \mu\,|y - x| \tag{6.25}$$

holds for all $x, y \in \mathbb{R}^n$. Does (6.25) imply that f is μ-strongly convex? What if we know that f is twice continuously differentiable on $\mathbb{R}^n$?

6.9. Prove Proposition 6.6.

Chapter 7

Gradient Descent

In this chapter, we introduce gradient descent for optimizing a smooth objective function. The gradient descent method can be traced back to Cauchy et al. (1847) and was originally developed for root finding. The basic idea is that if we keep moving in the opposite direction of the gradient of a given function, with a sufficiently small step size, the function value should keep decreasing until either the function reaches a local minimum or the gradient vanishes. In this chapter, we formalize this idea and analyze the convergence of the basic gradient descent algorithm under various regularity assumptions.

7.1 Gradient Descent Algorithm

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a differentiable function. Consider the following iteration:

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k), \quad k = 0, 1, \ldots, \tag{7.1}$$

where $x_0 \in \mathbb{R}^n$ is the initial value and $\alpha_k > 0$ is the step size.

This is the basic form of the gradient descent algorithm, often referred to as standard (or vanilla) gradient descent. Intuitively, starting with an initial value x_0, the algorithm (7.1) iteratively updates the value in the opposite direction of the gradient of the function at its current value.

We need some basic regularity assumptions on f to analyze the performance of algorithm (7.1). The minimal assumption we take in this chapter is that f is L-smooth. Recall that for an L-smooth function f, we have the following inequality (Proposition 6.4):

$$f(y) \leq f(x) - \nabla f(x) \cdot (y - x) + \frac{L}{2} |y - x|^2 . \tag{7.2}$$

Applying this inequality to a single iteration of (7.1), with $y = x_{k+1}$ and $x = x_k$, gives

$$\begin{aligned} f(x_{k+1}) &\leq f(x_k) - \nabla f(x_k) \cdot (x_{k+1} - x_k) + \frac{L}{2} |x_{k+1} - x_k|^2 \\ &= f(x_k) - \alpha_k |\nabla f(x_k)|^2 + \frac{L\alpha_k^2}{2} |\nabla f(x_k)|^2 . \end{aligned} \tag{7.3}$$

If the gradient $\nabla f(x_k)$ does not vanish at x_k, then the right-hand side of (7.3) is a quadratic function of α_k, whose leading coefficient is $\frac{L}{2} |\nabla f(x_k)|^2 > 0$. We can choose $\alpha_k = \frac{1}{L}$ to minimize the right-hand side (assuming we aim to derive a tighter upper bound on $f(x_{k+1})$) and obtain

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{2L} |\nabla f(x_k)|^2 . \tag{7.4}$$

From (7.4), we see that as long as $\nabla f(x_k) \neq 0$, we expect $f(x_{k+1}) < f(x_k)$. This is where the name *gradient descent* comes from.

Why is Gradient Descent Favored in Deep Learning?

Optimization methods can be broadly categorized into zero-order, first-order, and second-order methods. Zero-order methods rely solely on function evaluations without using derivative information, making them inefficient for high-dimensional problems. First-order methods, such as gradient descent, utilize first derivatives (gradients) to guide optimization, offering a balance between computational efficiency and convergence speed. Second-order methods, such as Newton's method, leverage second derivatives (Hessian matrices) to achieve faster convergence but at a significantly higher computational cost.

In deep learning, first-order methods, particularly stochastic gradient descent (SGD) and its variants, are preferred due to their scalability and efficiency. Computing

exact second-order information is often infeasible for deep neural networks, as it requires storing and inverting large Hessian matrices. In contrast, first-order methods leverage backpropagation to efficiently compute gradients with respect to model parameters. This significantly reduces computational cost while maintaining effective convergence. As a result, first-order methods provide a practical trade-off between computational efficiency and convergence speed, making them well-suited for optimizing deep models with millions, billions, or even trillions of parameters.

7.2 Convergence Analysis

We formalize the following convergence result for gradient descent with L-smooth objective functions.

Proposition 7.1. *Assume that f is an L-smooth function and that there exists some $f^* \in \mathbb{R}$ such that $f(x) \geq f^*$ for all $x \in \mathbb{R}^n$. Then, we have*

$$\lim_{k\to\infty} \nabla f(x_k) = 0$$

for the sequence $\{x_k\}$ obtained using the standard gradient descent algorithm (7.1) with $\alpha_k = \frac{1}{L}$. We also have

$$\min_{0\leq k\leq N} |\nabla f(x_k)|^2 \leq \frac{2L(f(x_0) - f^*)}{N+1}. \tag{7.5}$$

Proof. Rearranging (7.4), we have

$$\frac{1}{2L} |\nabla f(x_k)|^2 \leq f(x_k) - f(x_{k+1}). \tag{7.6}$$

Adding this from $k = 0$ to $k = N$ gives

$$\frac{1}{2L} \sum_{k=0}^{N} |\nabla f(x_k)|^2 \leq \sum_{k=0}^{N} (f(x_k) - f(x_{k+1})) = f(x_0) - f(x_{N+1}) \leq f(x_0) - f^*. \tag{7.7}$$

It follows that $\sum_{k=0}^{\infty} |\nabla f(x_k)|^2$ is a convergent series, which implies $|\nabla f(x_k)| \to 0$ as $k \to \infty$. From (7.7), we can also obtain (7.5). □

Example 7.1. Consider $f(x) = x$. Then, f is L-smooth. Running (7.1) on f with a

constant step size gives $x_{k+1} = x_k - \alpha$. Clearly, this does not lead to the convergence of x_k or $f(x_k)$. Furthermore, $\nabla f(x_k) \equiv 1$, which does not satisfy the conclusion of Proposition 7.1. This is because f does not satisfy the assumption of Proposition 7.1 that it is lower bounded by some f^*.

Proposition 7.1 shows that for a non-convex (not necessarily convex) function, we can only establish that $\nabla f(x_k) \to 0$ for iterates of the gradient descent algorithm. Without further assumptions, we cannot conclude convergence of x_k or $\{f(x_k)\}$ (see Example 7.1). Even if $\{x_k\}$ converges, we do not know whether it converges to a local minimum (see Exercise 7.2).

Remark 7.1 (Avoidance of saddle points). A saddle point is a stationary point that is neither a local minimum nor a local maximum. A stationary point x^* of f is said to be a strict saddle point if $\lambda_{\min}(\nabla^2 f(x^*)) < 0$. Lee et al. (2016) shows that gradient descent on L-smooth functions with a step size $\alpha < \frac{1}{L}$ almost surely converges to a local minimum, in the sense that a random initial condition has zero probability of lying in the local stable center manifold of a strict saddle point.

The next result assumes that the objective function is convex and L-smooth. It is reasonable to assume that f has at least one stationary point $x^* \in \mathbb{R}^n$, i.e., $\nabla f(x^*) = 0$. Due to convexity, x^* is also a global minimum, meaning $f^* = f(x^*) = \min_{x \in \mathbb{R}^n} f(x)$. We adopt this assumption for the following result.

Proposition 7.2. *Assume that f is an L-smooth and convex function. Let $x^* \in \mathbb{R}^n$ be a global minimum of f and $f^* = f(x^*)$. Then, we have*

$$f(x_k) - f^* \leq \frac{L|x_0 - x^*|^2}{2(k+1)} \tag{7.8}$$

for the sequence $\{x_k\}$ obtained using the standard gradient descent algorithm (7.1) with $\alpha_k = \frac{1}{L}$.

Proof. With $\alpha_k = \frac{1}{L}$, we have

$$\begin{aligned}|x_{k+1} - x^*|^2 &= |x_k - x^* - \frac{1}{L}\nabla f(x_k)|^2 \\ &= |x_k - x^*|^2 - \frac{2}{L}(x_k - x^*) \cdot \nabla f(x_k) + \frac{1}{L^2}|\nabla f(x_k)|^2 \\ &\leq |x_k - x^*|^2 - \frac{2}{L}(f(x_k) - f^*), \end{aligned} \tag{7.9}$$

where we used the fact that (Proposition 6.7(4))

$$f^* \geq f(x_k) + f(x_k) \cdot (x^* - x_k) + \frac{1}{2L}|\nabla f(x_k)|^2.$$

Adding the above inequality from 0 to k and using the fact that $f(x_k)$ is nonincreasing (see (7.4)) yield

$$(1+k)\frac{2}{L}(f(x_k) - f^*) \leq \sum_{i=0}^{k} \frac{2}{L}(f(x_i) - f^*) \leq |x_0 - x^*|^2 - |x_{k+1} - x^*|^2 \leq |x_0 - x^*|^2,$$

which is (7.8). □

It is natural to ask whether the sequence of iterates $\{x_k\}$ generated by the gradient descent algorithm actually converges to a minimizer of f. The following result answers this question.

Proposition 7.3. *Assume that f is an L-smooth and convex function. Denote the set of global minima for f by X and assume it is non-empty. Let $\{x_k\}$ be the sequence obtained using the standard gradient descent algorithm (7.1) with $\alpha_k = \frac{1}{L}$. We have*

$$\lim_{k\to\infty} x_k = x^*$$

for some $x^* \in X$.

Proof. Let $x^\star$ be any minimizer of f. From (7.9), we can see that $|x_k - x^\star|$ is monotonically decreasing (and bounded from below by 0). Hence it converges, which also implies that $\{x_k\}$ is bounded and has a convergent subsequence $\{x_{n_k}\}$.

Suppose that $x_{n_k} \to x^*$. By Proposition 7.1 and continuity of ∇f, we have $\nabla f(x^*) =$

$\lim_{k\to\infty} \nabla f(x_{n_k}) = 0$. This shows that x^* is a global minimizer, i.e., $x^* \in X$. By (7.9) again, we have $|x_k - x^*|$ converges. The limit must be zero because $\left|x_{n_k} - x^*\right| \to 0$ as $k \to \infty$. □

The next convergence result assumes that the objective function is simultaneously μ-strongly convex and L-smooth.

Proposition 7.4. *Assume that f is an L-smooth and μ-strongly convex function. Let $x^* \in \mathbb{R}^n$ be a global minimum of f and $f^* = f(x^*)$. Then, we have*

$$|x_k - x^*|^2 \le \left(\frac{L-\mu}{L+\mu}\right)^{2k} |x_0 - x^*|^2 \tag{7.10}$$

for the sequence $\{x_k\}$ obtained using the standard gradient descent algorithm (7.1) with $\alpha_k = \frac{2}{L+\mu}$.

Proof. With a step size $\alpha > 0$, we have

$$\begin{aligned}
|x_{k+1} - x^*|^2 &= |x_k - x^* - \alpha \nabla f(x_k)|^2 \\
&= |x_k - x^*|^2 - 2\alpha(x_k - x^*) \cdot \nabla f(x_k) + \alpha^2 |\nabla f(x_k)|^2 \\
&\le |x_k - x^*|^2 - \frac{2\alpha\mu L}{\mu + L}|x_k - x^*|^2 + (\alpha^2 - \frac{2\alpha}{\mu + L})|\nabla f(x_k)|^2,
\end{aligned}$$

where we used the fact that (Corollary 6.2)

$$(x - x^*) \cdot \nabla f(x) \ge \frac{\mu L}{\mu + L}|x - x^*|^2 + \frac{1}{\mu + L}|\nabla f(x)|^2.$$

Setting the step size $\alpha = \frac{2}{\mu+L}$ will lead to

$$|x_{k+1} - x^*|^2 \le \left(1 - \frac{4\mu L}{(\mu + L)^2}\right)|x_k - x^*|^2,$$

which implies (7.10). □

Remark 7.2. As shown in Proposition 7.4, for L-smooth and μ-strongly convex functions, the choice of step size $\alpha_k = \frac{2}{L+\mu}$ is better than the standard choice $\alpha_k = \frac{1}{L}$. In

fact, with $\alpha_k = \frac{1}{L}$, we have, similar to the proof of Proposition 7.2,

$$\begin{aligned}
|x_{k+1} - x^*|^2 &= |x_k - x^* - \frac{1}{L}\nabla f(x_k)|^2 \\
&= |x_k - x^*|^2 - \frac{2}{L}(x_k - x^*) \cdot \nabla f(x_k) + \frac{1}{L^2}|\nabla f(x_k)|^2 \\
&\leq |x_k - x^*|^2 - \frac{2}{L}(f(x_k) - f^*) \\
&\leq |x_k - x^*|^2 - \frac{\mu}{L}|x_k - x^*|^2,
\end{aligned}$$

where for the first inequality we used Proposition 6.7(4) for L-smooth convex functions (cf. proof of Proposition 7.2) and for the second inequality we used the definition of strong-convexity. This leads to

$$|x_k - x^*|^2 \leq \left(1 - \frac{\mu}{L}\right)^k |x_0 - x^*|^2. \tag{7.11}$$

One can easily see that this convergence rate is slightly worse than (7.10). Furthermore, getting a convergence rate for $f(x_k) - f^*$ from (7.10) is straightforward by L-smoothness of f. We have

$$f(x_k) - f(x^*) \leq \frac{L}{2}|x_k - x^*|^2 \leq \left(\frac{L-\mu}{L+\mu}\right)^{2k} \frac{L|x_0 - x^*|^2}{2}. \tag{7.12}$$

Remark 7.3. For convex and L-smooth functions, we have proved a convergence rate $O(1/k)$ for the standard gradient descent. For μ-strongly convex and L-smooth functions, a linear convergence rate $\left(\frac{L-\mu}{L+\mu}\right)^{2k}$ is established. We shall see in the next chapter that these are not the optimal rate that can be achieved by first-order methods.

7.3 Exercises

7.1. Consider the vanilla gradient descent algorithm (7.1). Under the same assumption as in Proposition 7.1, does the limit $\lim_{k\to\infty} f(x_k)$ exist? Assume that there exists $x^* \in \mathbb{R}^n$ such that $f^* = f(x^*)$, i.e., x^* is a global minimum. Do we have $f(x_k) \to f^*$? What about $x_k \to x^*$?

7.2. Consider $f(x, y) = x^2 + y^4 - 2y^2$. Find all stationary points of f. Consider the gradient descent algorithm (7.1) with an initial condition of the form $(x_0, 0)$ for some $x_0 \in \mathbb{R}$ and a step size $\alpha > 0$. What does $\{(x_k, y_k)\}$ converge to?

7.3. Assume that $f : \mathbb{R}^n \to \mathbb{R}$ is an L-smooth and μ-strongly convex function. Furthermore, assume that it is twice continuously differentiable. Consider the gradient descent algorithm (7.1) with a constant step size $\alpha > 0$.
(1) Show that, by Taylor's theorem, we have

$$|\nabla f(x_{k+1})| \leq \sup_{z \in \mathbb{R}^n} \left\| I_n - \alpha \nabla^2 f(z) \right\| |\nabla f(x_k)| .$$

(2) Show that

$$\sup_{z \in \mathbb{R}^n} \left\| I_n - \alpha \nabla^2 f(z) \right\| \leq \max(|1 - \alpha L|, |1 - \alpha \mu|).$$

From this, show that the optimal choice of step size (in terms of minimizing the right-hand side of the above) is $\alpha = \frac{2}{L+\mu}$. Show that this leads to

$$|\nabla f(x_k)|^2 \leq \left(\frac{L - \mu}{L + \mu} \right)^{2k} |\nabla f(x_0)|^2 .$$

(3) Let $x^* \in \mathbb{R}^n$ be a stationary point of f. From part (2), derive convergence rates for $f(x_k)$ and x_k.

7.4. Assume that $f : \mathbb{R}^n \to \mathbb{R}$ is an L-smooth and μ-strongly convex function. Furthermore, assume that it is twice continuously differentiable. Consider the gradient descent algorithm (7.1) with a constant step size $\alpha > 0$. Consider the map $g : \mathbb{R}^n \to \mathbb{R}^n$ defined by

$$g(x) = x - \alpha \nabla f(x).$$

(1) Let $x^* \in \mathbb{R}^n$ be a stationary point of f. Show that x^* is a fixed point of f, i.e., $g(x^*) = x^*$.
(2) Using Taylor's theorem, show that

$$|g(x) - g(y)| \leq \sup_{z \in \mathbb{R}^n} \left\| I_n - \alpha \nabla^2 f(z) \right\| |x - y| \leq \max(|1 - \alpha L|, |1 - \alpha \mu|) |x - y| .$$

(3) Conclude from part (2) that, with $\alpha = \frac{2}{L+\mu}$, we have

$$|x_k - x^*|^2 \leq \left(\frac{L-\mu}{L+\mu}\right)^{2k} |x_0 - x^*|^2. \tag{7.13}$$

7.5. Let $f : \mathbb{R}^n \to \mathbb{R}$ be an L-smooth and μ-strongly convex function. Consider the gradient descent algorithm (7.1) with a constant step size $\alpha = \frac{1}{L}$. Let $x^* \in \mathbb{R}^n$ be a stationary point of f and $f^* = f(x^*)$. From (7.3) and using the definition of strong convexity, prove that

$$f(x_k) - f^* \leq \left(1 - \frac{\mu}{L}\right)^k (f(x_0) - f^*).$$

Furthermore, show that from this you can also obtain convergence rates for x_k and $\nabla f(x_k)$. Compare the convergence rate with that from Proposition 7.4. Which one is better?

Chapter 8

Gradient Descent with Momentum

In the previous chapter, we introduced the vanilla gradient descent algorithm and analyzed its convergence. It turns out that first-order methods can outperform vanilla gradient descent by incorporating *momentum*, an additional term in gradient descent that carries information from past gradient updates to accelerate convergence. In this chapter, we introduce two important variants: Nesterov's accelerated gradient (NAG) method (Nesterov, 1983) and Polyak's heavy-ball (HB) method (Polyak, 1964). Both methods leverage momentum to accelerate convergence, but they do so in different ways. We will present their formulations and analyze their convergence rates.

8.1 Gradient Descent with Momentum

We first introduce two of the most popular momentum methods in their simplest forms.

The iterations of *Polyak's heavy-ball* (HB) method are given by

$$x_{k+1} = x_k - \alpha \nabla f(x_k) + \beta(x_k - x_{k-1}), \tag{8.1}$$

where $\alpha > 0$ and $\beta > 0$. Clearly, if $\beta = 0$, HB reduces to standard gradient descent.

The iterations of *Nesterov's accelerated gradient* (NAG) are given by

$$\begin{aligned} y_{k+1} &= x_k - \alpha \nabla f(x_k), \\ x_{k+1} &= y_{k+1} + \beta(y_{k+1} - y_k), \end{aligned} \tag{8.2}$$

where $\alpha > 0$ and $\beta > 0$. Clearly, if $\beta = 0$, NAG also reduces to standard gradient descent.

8.2 Convergence Analysis for Quadratic Functions

We first analyze convergence of NAG and HB methods on convex quadratic functions. Consider

$$f(x) = \frac{1}{2} x^\top Q x + b^\top x + a, \tag{8.3}$$

where $Q \in \mathbb{R}^{n \times n}$ is a positive definite symmetric matrix, $b \in \mathbb{R}^n$, and $a \in \mathbb{R}$. Let $\mu = \lambda_{\min}(Q)$ and $L = \lambda_{\max}(Q)$. Clearly, f is μ-strongly convex and L-smooth. We have $\nabla f(x) = Qx + b$. The global minimum of f is $x^* = -Q^{-1}b$.

Since $\nabla f(x) = Qx + b = Q(x - x^*)$, iterations of HB in (8.1) reduce to

$$x_{k+1} = x_k - \alpha Q(x_k - x^*) + \beta(x_k - x_{k-1}). \tag{8.4}$$

Introduce

$$\xi_k = \begin{bmatrix} x_k - x^* \\ x_{k-1} - x^* \end{bmatrix}.$$

We can rewrite (8.4) as a linear system of difference equations

$$\xi_{k+1} = \underbrace{\begin{bmatrix} (1+\beta)I - \alpha Q & -\beta I \\ I & 0 \end{bmatrix}}_{:=A_1} \xi_k = A_1 \xi_k, \tag{8.5}$$

where I is the n-dimensional identity matrix.

Similarly, for NAG, we can introduce

$$\xi_k = \begin{bmatrix} y_k - x^* \\ y_{k-1} - x^* \end{bmatrix}$$

and rewrite (8.2) as

$$\xi_{k+1} = \underbrace{\begin{bmatrix} (1+\beta)(I-\alpha Q) & -\beta(I-\alpha Q) \\ I & 0 \end{bmatrix}}_{:=A_2} \xi_k = A_2 \xi_k. \tag{8.6}$$

Therefore, the convergence analysis of NAG and HB methods reduces to the stability analysis of the linear systems (8.5) and (8.6). We perform this analysis by examining the spectral radius of the system matrix defining these linear systems.

Definition 8.1 (Spectral radius)**.** The *spectral radius* of a matrix $A \in \mathbb{R}^{n\times n}$ is defined as

$$\rho(A) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } A\}.$$

Consider the discrete-time linear system

$$z_{k+1} = Az_k. \tag{8.7}$$

We have the following conclusions that connect exponential stability of (8.7) with the spectral radius of A.

Proposition 8.1. *Let $A \in \mathbb{R}^{n\times n}$. We have $\lim_{k\to\infty} A^k = 0$ if and only if $\rho(A) < 1$.*

For a proof, see (Horn and Johnson, 2013, Theorem 5.6.12). The statement above essentially says that the linear system (8.7) is asymptotically stable if and only if $\rho(A) < 1$. The spectral radius also characterizes the exponential convergence rate for solutions of (8.7) as indicated by the following result, which can be found in (Horn and Johnson, 2013, Corollary 5.6.14).

Proposition 8.2 (Gelfand formula)**.** *Let $\|\cdot\|$ denote any matrix norm for matrices in $\mathbb{R}^{n\times n}$ and $A \in \mathbb{R}^{n\times n}$. We have $\rho(A) = \lim_{k\to\infty} \|A^k\|^{1/k}$.*

Now let $\|\cdot\|$ denote the matrix 2-norm. It is implied by Proposition 8.2 that, given any $A \in \mathbb{R}^{n\times n}$ and any $\varepsilon > 0$, there exists a constant $C = C(A, \varepsilon)$ such that

$$|z_k| = \left\|A^k z_0\right\| \le \left\|A^k\right\| |z_0| \le C(\rho(A)+\varepsilon)^k |z_0|$$

for all $k = 1, 2, \ldots$. Hence $\rho(A)$ characterizes the asymptotic convergence rate of solutions of (8.7), provided that $\rho(A) < 1$.

Returning to the convergence analysis of SNAG and HB methods, we just need to compute the spectral radii of the system matrices of (8.5) and (8.6). This is summarized in the following theorem.

Proposition 8.3. *Let $\kappa = \frac{L}{\mu}$. We have the following results on the spectral radii of the system matrices of (8.5) and (8.6).*

(1) For HB method, with $\alpha = \frac{4}{(\sqrt{L}+\sqrt{\mu})^2}$ and $\beta = \left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^2$, we have $\rho(A_1) = \frac{\sqrt{\kappa}+1}{\sqrt{\kappa}-1}$.
(2) For NAG method, with $\alpha = \frac{1}{L}$ and $\beta = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}$, we have $\rho(A_2) = 1 - \frac{1}{\sqrt{\kappa}}$.

Proof. (1) We first show that we can reduce the analysis to spectral radii of the following 2×2 matrices

$$T_i = \begin{bmatrix} (1+\beta) - \alpha\lambda_i & -\beta \\ 1 & 0 \end{bmatrix}, \quad i = 1, 2, \ldots, n, \tag{8.8}$$

where $\lambda_i \in [\mu, L]$, $i = 1, 2, \ldots, n$, are eigenvalues of Q. To see this, consider an eigenvalue decomposition of Q as $Q = S\Lambda S^\top$, where S is an orthogonal matrix and $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$. Then

$$A_1 = \begin{bmatrix} S & 0 \\ 0 & S \end{bmatrix} \begin{bmatrix} (1+\beta)I - \alpha\Lambda & -\beta I \\ I & 0 \end{bmatrix} \begin{bmatrix} S & 0 \\ 0 & S \end{bmatrix}^\top,$$

which, by similarity, shows that the eigenvalues of A_1 are precisely eigenvalues of all the matrices T_i, $i = 1, 2, \ldots, n$, defined above in (8.8). The characteristic polynomial of T_i is

$$\nu^2 + (\alpha\lambda_i - \beta - 1)\nu + \beta = 0, \tag{8.9}$$

where $\lambda_i \in [\mu, L]$. Solving this gives

$$\nu_1, \nu_2 = \frac{1 + \beta - \alpha\lambda_i \pm \sqrt{(\alpha\lambda_i - \beta - 1)^2 - 4\beta}}{2}.$$

For fixed α and β, it follows that

$$\rho(A_1) \leq \max_{\lambda\in[\mu,L]} \max(|\nu_1|, |\nu_2|) = \max_{\lambda\in[\mu,L]} \begin{cases} \frac{1}{2}|1+\beta-\alpha\lambda| + \frac{1}{2}\sqrt{\Delta}, \text{ if } \Delta \geq 0, \\ \sqrt{\beta}, \text{ if } \Delta < 0, \end{cases}$$

where $\Delta = (\alpha\lambda - \beta - 1)^2 - 4\beta$. With $\alpha = \frac{4}{(\sqrt{L}+\sqrt{\mu})^2}$, $\beta = \left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^2 = \left(\frac{\sqrt{L}-\sqrt{\mu}}{\sqrt{L}+\sqrt{\mu}}\right)^2$, and $\lambda \in [\mu, L]$, we have

$$\begin{aligned} \Delta &= \left[\frac{4\lambda}{(\sqrt{L}+\sqrt{\mu})^2} - \left(\frac{\sqrt{L}-\sqrt{\mu}}{\sqrt{L}+\sqrt{\mu}}\right)^2 - 1\right]^2 - 4\beta \\ &= \left[\frac{4\lambda - 2(L+\mu)}{(\sqrt{L}+\sqrt{\mu})^2}\right]^2 - 4\left(\frac{\sqrt{L}-\sqrt{\mu}}{\sqrt{L}+\sqrt{\mu}}\right)^2 \\ &\leq \left[\frac{2(L-\mu)}{(\sqrt{L}+\sqrt{\mu})^2}\right]^2 - \frac{4(L-\mu)^2}{(\sqrt{L}+\sqrt{\mu})^4} = 0. \end{aligned}$$

Hence

$$\rho(A_1) \leq \sqrt{\beta} = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}.$$

(2) The proof is very similar to part (1). We only need to investigate spectral radii of the following 2×2 matrices

$$T_i = \begin{bmatrix} (1+\beta)(1-\alpha\lambda_i) & -\beta(1-\alpha\lambda_i) \\ 1 & 0 \end{bmatrix}, \quad i = 1, 2, \ldots, n. \tag{8.10}$$

The characteristic polynomial of T_i is

$$\nu^2 - (1+\beta)(1-\alpha\lambda_i)\nu + \beta(1-\alpha\lambda_i) = 0, \tag{8.11}$$

where $\lambda_i \in [\mu, L]$. It follows that

$$\rho(A_2) \leq \max_{\lambda\in[\mu,L]} \max(|\nu_1|, |\nu_2|) = \max_{\lambda\in[\mu,L]} \begin{cases} \frac{1}{2}|(1+\beta)(1-\alpha\lambda)| + \frac{1}{2}\sqrt{\Delta}, \text{ if } \Delta \geq 0, \\ \sqrt{\beta(1-\alpha\lambda)}, \text{ if } \Delta < 0, \end{cases}$$

where $\Delta = (1+\beta)^2(1-\alpha\lambda)^2 - 4\beta(1-\alpha\lambda)$. With $\alpha = \frac{1}{L}$, $\beta = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1} = \frac{\sqrt{L}-\sqrt{\mu}}{\sqrt{L}+\sqrt{\mu}}$, and $\lambda \in [\mu, L]$, we have

$$\begin{aligned}\Delta &= \left(1+\frac{\sqrt{L}-\sqrt{\mu}}{\sqrt{L}+\sqrt{\mu}}\right)^2\left(1-\frac{\lambda}{L}\right)^2 - 4\frac{\sqrt{L}-\sqrt{\mu}}{\sqrt{L}+\sqrt{\mu}}\left(1-\frac{\lambda}{L}\right)\\ &= \left(\frac{4(L-\lambda)-4(L-\mu)}{(\sqrt{L}+\sqrt{\mu})^2}\right)\left(1-\frac{\lambda}{L}\right) \le 0.\end{aligned}$$

Hence

$$\rho(A_2) \le \max_{\lambda\in[\mu,L]} \sqrt{\beta(1-\alpha\lambda)} \le \sqrt{\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\left(1-\frac{1}{\kappa}\right)} = 1-\frac{1}{\sqrt{\kappa}}.$$

□

Remark 8.1. The proof of Proposition 8.3 is based on elementary linear algebra (see Polyak (1964) and (Lessard et al., 2016, Appendix A)). It is shown in (Lessard et al., 2016, Proposition 1) that the choice of parameters $\alpha = \frac{4}{(\sqrt{L}+\sqrt{\mu})^2}$ and $\beta = \left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^2$ for HB is optimal in the sense that they achieve the best worst-case convergence rates for quadratic functions that are μ-strongly convex and L-smooth. It is also stated in (Lessard et al., 2016, Proposition 1) that the optimal parameters for NAG should be $\alpha = \frac{4}{3L+\mu}$ and $\beta = \frac{\sqrt{3\kappa+1}-2}{\sqrt{3\kappa+1}+2}$, although a proof is omitted. Interested readers can work out the details by following the argument in (Lessard et al., 2016, Appendix A) or the proof of Proposition 8.3 above to directly show that this choice of α and β leads to $\rho(A_2) \le 1 - \frac{2}{\sqrt{3\kappa+1}}$, which slightly improves the rate for NAG in Proposition 8.3 but remains worse than HB (Exercise 8.1). Furthermore, (Lessard et al., 2016, Appendix B) shows that the HB method fails to converge in general, even for strongly convex functions. This contrasts with the NAG method, which can be shown to achieve acceleration for general convex and strongly convex functions. We present the analysis in the next section. Finally, we note that the convergence analysis of the heavy-ball method for general convex and strongly convex functions has been investigated in Ghadimi et al. (2015) under various assumptions on the parameter choices α and β, although it does not demonstrate acceleration compared to standard gradient descent in such settings.

8.3 Convergence Analysis for Convex Functions

In this section, we analyze convergence rates of Nesterov's accelerated gradient for both general convex and strongly convex functions.

Consider Nesterov's accelerated gradient (NAG) method:

$$\begin{aligned} x_{k+1} &= y_k - \alpha_k \nabla f(y_k), \\ y_k &= x_k + \beta_k (x_k - x_{k-1}), \quad k \geq 0, \end{aligned} \tag{8.12}$$

where $x_0 = x_{-1}$ are given as initial conditions and α_k and β_k are parameters to be chosen.

We first analyze the convergence of NAG (8.12) for general convex functions.

Theorem 8.1. *Let $f : \mathbb{R}^n \to \mathbb{R}$ be L-smooth and convex. Let x^* be a minimum of f and $f^* = f(x^*)$. Then the NAG algorithm (8.12) with $\alpha_k = \alpha \in (0, \frac{1}{L}]$ and $\beta_k = \frac{k-1}{k+r-1}$, $r \geq 3$, leads to*

$$f(x_k) - f^* \leq \frac{(r-1)^2 |x_0 - x^*|^2}{2\alpha(k+r-2)^2}, \quad k \geq 0. \tag{8.13}$$

Proof. Define $a_0 = 0$ and $a_k = \frac{k+r-2}{r-1}$, $k = 1, 2, \ldots$. We can rewrite $\beta_k = \frac{\frac{k+r-2}{r-1} - 1}{\frac{k+r-1}{r-1}} = \frac{a_k - 1}{a_{k+1}}$, $k \geq 1$. Introduce

$$p_k := (a_k - 1)(x_k - x_{k-1}) \overset{(8.12)}{=} a_{k+1}(y_k - x_k), \quad k \geq 1. \tag{8.14}$$

Let $p_0 = 0$. Equation (8.14) also holds for $k = 0$ by the initial condition $x_0 = x_{-1}$ and $y_0 = x_0$. By (8.14) and (8.12), it is straightforward to verify that

$$\begin{aligned} p_{k+1} + x_{k+1} &= (a_{k+1} - 1)(x_{k+1} - x_k) + x_{k+1} \\ &= a_{k+1} x_{k+1} - a_{k+1} x_k + x_k \\ &= a_{k+1}(y_k - \alpha \nabla f(y_k)) - a_{k+1} x_k + x_k \\ &= a_{k+1}(y_k - x_k) + x_k - a_{k+1} \alpha \nabla f(y_k) \\ &= p_k + x_k - a_{k+1} \alpha \nabla f(y_k), \quad k \geq 0. \end{aligned} \tag{8.15}$$

It follows that

$$
\begin{aligned}
&|p_{k+1} + x_{k+1} - x^*|^2 \\
&= |p_k + x_k - x^*|^2 - 2\alpha a_{k+1}(p_k + x_k - x^*) \cdot \nabla f(y_k) \\
&+ a_{k+1}^2 \alpha^2 |\nabla f(y_k)|^2 \\
&= |p_k + x_k - x^*|^2 - 2\alpha a_{k+1}(a_{k+1} - 1)(y_k - x_k) \cdot \nabla f(y_k) \\
&- 2\alpha a_{k+1}(y_k - x^*) \cdot \nabla f(y_k) + a_{k+1}^2 \alpha^2 |\nabla f(y_k)|^2 .
\end{aligned} \tag{8.16}
$$

By (7.3), we have

$$
f(x_{k+1}) \leq f(y_k) + (-\alpha + \frac{L\alpha^2}{2}) |\nabla f(y_k)|^2 . \tag{8.17}
$$

By convexity of f, we have

$$
(y_k - x_k) \cdot \nabla f(y_k) \geq f(y_k) - f(x_k). \tag{8.18}
$$

and

$$
(y_k - x^*) \cdot \nabla f(y_k) \geq f(y_k) - f^*. \tag{8.19}
$$

Substituting (8.18) and (8.19) into (8.16) gives

$$
\begin{aligned}
&|p_{k+1} + x_{k+1} - x^*|^2 \\
&\leq |p_k + x_k - x^*|^2 - 2\alpha a_{k+1}(a_{k+1} - 1)(f(y_k) - f(x_k)) \\
&\quad - 2\alpha a_{k+1}(f(y_k) - f^*) + a_{k+1}^2 \alpha^2 |\nabla f(y_k)|^2 \\
&\leq |p_k + x_k - x^*|^2 - 2\alpha a_{k+1}^2 (f(y_k) - f^*) \\
&\quad + 2\alpha a_k^2 (f(x_k) - f^*) + a_{k+1}^2 \alpha^2 |\nabla f(y_k)|^2 \\
&\leq |p_k + x_k - x^*|^2 - 2\alpha a_{k+1}^2 (f(x_{k+1}) - f^*) \\
&\quad + 2\alpha a_k^2 (f(x_k) - f^*) \\
&\quad + a_{k+1}^2 \alpha^2 (-1 + L\alpha) |\nabla f(y_k)|^2 \\
&\leq |p_k + x_k - x^*|^2 - 2\alpha a_{k+1}^2 (f(x_{k+1}) - f^*) \\
&\quad + 2\alpha a_k^2 (f(x_k) - f^*),
\end{aligned} \tag{8.20}
$$

where the first inequality follows from substituting (8.18) and (8.19) into (8.16), the second inequality used $a_{k+1}(a_{k+1} - 1) \leq a_k^2$ for all $k \geq 0$ (which can be verified

directly), the third inequality follows from (8.17), and the fourth inequality holds because $\alpha \leq \frac{1}{L}$.

Inequality (8.20), by rearrangement, shows that the function

$$V_k = |p_k + x_k - x^*|^2 + 2\alpha a_k^2 (f(x_k) - f^*) \tag{8.21}$$

is a *Lyapunov function* in the sense that it is nonincreasing w.r.t. k. It follows that

$$2\alpha a_k^2 (f(x_k) - f^*) \leq V_k \leq V_0 = |x_0 - x^*|^2 ,$$

which gives (8.13). □

Remark 8.2. The proof of Theorem 8.1 essentially follows the argument in the original proof in Nesterov (1983). We made the choice of step size more explicit, following Su et al. (2016). Additionally, we reformulated the more general step size choice, β_k, considered in Su et al. (2016), to align with the format of the original proof in Nesterov (1983).

Corollary 8.1. *The special case of Theorem 8.1 with $\alpha = \frac{1}{L}$ and $\beta_k = \frac{k-1}{k+2}$ gives*

$$f(x_k) - f^* \leq \frac{2L\,|x_0 - x^*|^2}{(k+1)^2}, \quad k \geq 0. \tag{8.22}$$

Remark 8.3. Here, acceleration refers to the $1/k^2$ convergence rate, compared to the $1/k$ convergence rate achieved by standard gradient descent (Proposition 7.2).

Similar to what we did for vanilla gradient descent in Proposition 7.3, we prove that the sequences $\{x_k\}$ and $\{y_k\}$ generated by the NAG algorithm (8.12) actually converge to a minimizer. The proof is largely based on Jang and Ryu (2025), where the authors presented an AI-assisted proof of what had been described as an open question, with some simplifications to make the argument self-contained.

Proposition 8.4. *Assume that f is an L-smooth and convex function. Denote the set of global minima for f by X and assume it is non-empty. Let $\{x_k\}$ and $\{y_k\}$ be the sequences obtained using the NAG algorithm (8.12) under the same step size choices as in Theorem*

8.1. We have

$$\lim_{k\to\infty} x_k = x^*$$

for some $x^* \in X$.

Proof. By the conclusion of Theorem 8.1, the sequence V_k is nonincreasing and bounded from below. Hence it converges. This implies that $|p_k + x_k - x^*|^2$ is bounded. As a result, $\{p_k + x_k\}$ is also bounded. By (8.15), (8.14), and (8.12), we have

$$\begin{aligned} p_{k+1} + x_{k+1} &= p_k + x_k - a_{k+1}\alpha\nabla f(y_k) \\ &= a_{k+1}(y_k - x_k) + x_k + a_{k+1}(x_{k+1} - y_k) \\ &= a_{k+1}x_{k+1} + (1 - a_{k+1})x_k, \quad k \geq 0. \end{aligned} \tag{8.23}$$

This can be rewritten as

$$x_{k+1} = \left(1 - \frac{1}{a_{k+1}}\right) x_k + \frac{1}{a_{k+1}}(p_{k+1} + x_{k+1}), \quad k \geq 0. \tag{8.24}$$

Note that $1/a_{k+1} \in (0, 1]$ for all $k \geq 0$. Let M be such that $\sup_{k\geq 0} |p_k + x_k| \leq M$. We claim that

$$|x_k| \leq \max(|x_0|, M), \quad \forall k \geq 0. \tag{8.25}$$

It clearly holds for $k = 0$. Suppose that it holds for some $k \geq 0$. Then by (8.24), we have

$$\begin{aligned} |x_{k+1}| &\leq \left(1 - \frac{1}{a_{k+1}}\right) |x_k| + \frac{1}{a_{k+1}} |p_{k+1} + x_{k+1}| \\ &\leq \left(1 - \frac{1}{a_{k+1}}\right) \max(|x_0|, M) + \frac{1}{a_{k+1}} \max(|x_0|, M) = \max(|x_0|, M), \end{aligned}$$

and (8.25) holds by induction. Consequently, $\{x_k\}$ is bounded.

To show that $\{x_k\}$ is convergent, we only need to show that it cannot have two convergent subsequences with different limits (i.e., two different cluster points). Let $\{x_{m_k}\}$ and $\{x_{n_k}\}$ be two subsequences of $\{x_k\}$ such that $\lim_{k\to\infty} x_{m_k} = x_1^*$ and $\lim_{k\to\infty} x_{n_k} = x_2^*$. By Theorem 8.1 and continuity of f, we have $f(x_1^*) = f(x_2^*) = f^*$. Hence both x_1^* and x_2^* are minima of f.

By the proof of Theorem 8.1, we can define two Lyapunov functions V_k^1 and V_k^2 as in (8.21) with x^* replaced by x_1^* and x_2^*, respectively. Both V_k^1 and V_k^2 converge as $k \to \infty$. Consider

$$V_k^1 - V_k^2 = -2(p_{k+1} + x_{k+1}) \cdot (x_1^* - x_2^*) + \left|x_1^*\right|^2 - \left|x_2^*\right|^2 \tag{8.26}$$

By (8.24), this implies

$$V_k^1 - V_k^2 = -2a_{k+1}x_{k+1} \cdot (x_1^* - x_2^*) - 2(1 - a_{k+1})x_k \cdot (x_1^* - x_2^*) + \left|x_1^*\right|^2 - \left|x_2^*\right|^2 \tag{8.27}$$

Define $\theta_k = -2x_k \cdot (x_1^* - x_2^*)$ and $v_k = V_k^1 - V_k^2 - \left|x_1^*\right|^2 + \left|x_2^*\right|^2$. The recursion above can be rewritten as

$$\theta_{k+1} = \left(1 - \frac{1}{a_{k+1}}\right)\theta_k + \frac{1}{a_{k+1}}v_k, \quad k \geq 0. \tag{8.28}$$

Since both $\left\{V_k^1\right\}$ and $\left\{V_k^1\right\}$ converge, we have $v_k \to v^*$ for some $v^* \in \mathbb{R}$.
Claim: $\theta_k \to v^*$ as $k \to \infty$.
Proof of the claim: We can subtract v^* from both sides of (8.28) and obtain

$$\theta_{k+1} - v^* = \left(1 - \frac{1}{a_{k+1}}\right)(\theta_k - v^*) + \frac{1}{a_{k+1}}(v_k - v^*), \quad k \geq 0. \tag{8.29}$$

Choose any $\varepsilon > 0$. There exists $K > 0$ such that $|v_k - v^*| \leq \varepsilon$ for all $k \geq K$. We consider two cases:
(1) If $|\theta_k - v^*| \leq \varepsilon$ for some $k \geq K$, then

$$|\theta_{k+1} - v^*| \leq \left(1 - \frac{1}{a_{k+1}}\right)|\theta_k - v^*| + \frac{1}{a_{k+1}}\varepsilon \leq \left(1 - \frac{1}{a_{k+1}}\right)\varepsilon + \frac{1}{a_{k+1}}\varepsilon = \varepsilon.$$

By induction, this shows that $|\theta_k - v^*| \leq \varepsilon$ for all subsequent k as well.
(2) Assume that case (1) does not happens, i.e., $|\theta_k - v^*| > \varepsilon$ for all $k \geq K$. By $|v_k - v^*| \leq \varepsilon$ for $k \geq K$ and (8.29), we can obtain

$$|\theta_{k+1} - v^*| \leq \left(1 - \frac{1}{a_{k+1}}\right)|\theta_k - v^*| + \frac{1}{a_{k+1}}\varepsilon, \quad k \geq K, \tag{8.30}$$

which can be rewritten as

$$|\theta_{k+1} - v^*| - \varepsilon \le \left(1 - \frac{1}{a_{k+1}}\right)(|\theta_k - v^*)| - \varepsilon), \quad k \ge K. \tag{8.31}$$

It follows that

$$|\theta_N - v^*| - \varepsilon \le (|\theta_K - v^*)| - \varepsilon) \prod_{k=K}^{N} \left(1 - \frac{1}{a_{k+1}}\right), \quad N \ge K. \tag{8.32}$$

Note that $\sum_{k=K}^{\infty} \frac{1}{a_{k+1}} = \infty$, and

$$\log \prod_{k=K}^{N} \left(1 - \frac{1}{a_{k+1}}\right) = \sum_{k=K}^{N} \log\left(1 - \frac{1}{a_{k+1}}\right) \le -\sum_{k=K}^{N} \frac{1}{a_{k+1}} \to -\infty, \quad \text{as} \quad N \to \infty,$$

where we used the elementary inequality $\log(1-x) \le -x$ for $x \in (0,1)$. It follows that

$$\prod_{k=K}^{N} \left(1 - \frac{1}{a_{k+1}}\right) \to 0, \quad \text{as} \quad N \to \infty.$$

and $|\theta_k - v^*| - \varepsilon \to 0$ as $k \to \infty$ in view of (8.32). Hence there exists some $K' > 0$ such that $|\theta_k - v^*| \le 2\varepsilon$ for all $k \ge K'$.

In either of the two cases, we have shown that for all sufficiently large k, we have $|\theta_k - v^*| \le 2\varepsilon$. Since ε is arbitrary, we must have $\theta_k \to v^*$ as $k \to \infty$. ■

Now passing to the limit with $\{x_{m_k}\} \to x_1^*$ and $\{x_{n_k}\} \to x_2^*$ in $\theta_k = -2x_k \cdot (x_1^* - x_2^*)$, we obtain

$$-2x_1^* \cdot (x_1^* - x_2^*) = v^* = -2x_2^* \cdot (x_1^* - x_2^*),$$

which implies $2\left|x_1^* - x_2^*\right|^2 = 0$ and $x_1^* = x_2^*$. Hence $\{x_k\}$ converges to a single minimizer x^*.

We now show that $\{y_k\}$ also converges to x^*. From (8.14), we have

$$y_k = \frac{p_k}{a_{k+1}} + x_k = \frac{p_k + x_k}{a_{k+1}} + \left(1 - \frac{1}{a_{k+1}}\right) x_k \to x^*,$$

because $a_k \to \infty$ and $p_k + x_k$ is bounded. □

We now consider strongly convex functions.

Theorem 8.2. *Let $f : \mathbb{R}^n \to \mathbb{R}$ be L-smooth and μ-strongly convex. Let x^* be a minimum of f and $f^* = f(x^*)$. Then the NAG algorithm (8.12) with $\alpha_k = \frac{1}{L}$ and $\beta_k = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}$, where $\kappa = \frac{L}{\mu}$, leads to*

$$f(x_k) - f^* \leq \left(1 - \frac{1}{\sqrt{\kappa}}\right)^k \left(f(x_0) - f^* + \frac{\mu}{2}|x_0 - x^*|^2\right)$$

for all $k \geq 0$.

Proof. Introduce

$$v_k = (\sqrt{\kappa} + 1)y_k - \sqrt{\kappa}x_k, \quad k \geq 0. \tag{8.33}$$

By the initial condition $x_0 = x_{-1}$ and (8.12), we have $v_0 = x_0 = y_0$. By (8.33) and (8.12), it is also straightforward to verify that

$$v_{k+1} = v_k + \frac{1}{\sqrt{\kappa}}(y_k - v_k) - \frac{1}{\mu\sqrt{\kappa}}\nabla f(y_k). \tag{8.34}$$

Define

$$V_k = f(x_k) - f^* + \frac{\mu}{2}|v_k - x^*|^2.$$

It follows that

$$\begin{aligned}
V_{k+1} &= f(x_{k+1}) - f^* + \frac{\mu}{2}|v_{k+1} - x^*|^2 \\
&\overset{(7.4)}{\leq} f(y_k) - \frac{1}{2L}|\nabla f(y_k)|^2 - f^* + \frac{\mu}{2}|v_{k+1} - x^*|^2 \\
&\overset{(8.34)}{=} f(y_k) - \frac{1}{2L}|\nabla f(y_k)|^2 - f^* \\
&\quad + \frac{\mu}{2}|v_k - x^*|^2 + \frac{\mu}{2\kappa}|y_k - v_k|^2 + \frac{1}{2\mu\kappa}|\nabla f(y_k)|^2 \\
&\quad - \frac{1}{\kappa}(y_k - v_k)\cdot\nabla f(y_k) + \frac{\mu}{\sqrt{\kappa}}(v_k - x^*)\cdot(y_k - v_k) \\
&\quad - \frac{1}{\sqrt{\kappa}}(v_k - x^*)\cdot\nabla f(y_k)
\end{aligned} \tag{8.35}$$

Rewrite[1]

$$
\begin{aligned}
&(v_k - x^*) \cdot (y_k - v_k) \\
&\quad = \frac{1}{2}(|y_k - x^*|^2 - |v_k - x^*|^2 - |y_k - v_k|^2),
\end{aligned} \tag{8.36}
$$

and

$$
\begin{aligned}
&-(v_k - x^*) \cdot \nabla f(y_k) \\
&\quad = (y_k - v_k) \cdot \nabla f(y_k) + (x^* - y_k) \cdot \nabla f(y_k).
\end{aligned} \tag{8.37}
$$

By μ-strong convexity of f, we have

$$
(x^* - y_k) \cdot \nabla f(y_k) \leq f^* - f(y_k) - \frac{\mu}{2} |y_k - x^*|^2 . \tag{8.38}
$$

Substituting (8.38) into (8.37), then substituting (8.37) and (8.36) into (8.35), and canceling out terms, we obtain

$$
\begin{aligned}
V_{k+1} \leq\ & (1 - \frac{1}{\sqrt{\kappa}})(f(y_k) - f^*) + (1 - \frac{1}{\sqrt{\kappa}})\frac{\mu}{2} |v_k - x^*|^2 \\
& + (\frac{\mu}{2\kappa} - \frac{\mu}{2\sqrt{\kappa}}) |y_k - v_k|^2 \\
& - (\frac{1}{\kappa} - \frac{1}{\sqrt{\kappa}})(y_k - v_k) \cdot \nabla f(y_k).
\end{aligned} \tag{8.39}
$$

Applying convexity of f, we have

$$
f(y_k) \leq f(x_k) - (x_k - y_k) \cdot \nabla f(y_k). \tag{8.40}
$$

[1] This elementary equality

$$
(a - b) \cdot (c - a) = \frac{1}{2}\left(|c - b|^2 - |c - a|^2 - |a - b|^2\right)
$$

is sometimes called the three-point identity. To see an easy proof, write $c - b = (a - b) + (c - a)$ and compute $|c - b|^2$.

Putting (8.40) into (8.39) and noticing that $\frac{\mu}{2\kappa} - \frac{\mu}{2\sqrt{\kappa}} \leq 0$ because $\kappa \geq 1$, we have

$$V_{k+1} \leq (1 - \frac{1}{\sqrt{\kappa}})(f(x_k) - f^*) + (1 - \frac{1}{\sqrt{\kappa}})\frac{\mu}{2}|v_k - x^*|^2$$
$$-\left[(\frac{1}{\kappa} - \frac{1}{\sqrt{\kappa}})(y_k - v_k) + (1 - \frac{1}{\sqrt{\kappa}})(x_k - y_k)\right] \cdot \nabla f(y_k),$$

where the last term vanishes because by (8.33) we have $y_k - v_k = \sqrt{\kappa}(x_k - y_k)$. We have proved that $V_{k+1} \leq (1 - \frac{1}{\sqrt{\kappa}})V_k$ for $k \geq 0$. The conclusion follows immediately. □

Remark 8.4. For strongly convex functions, acceleration of convergence by Nesterov's method (8.12) refers to improving the convergence rates of the standard gradient descent method, $(1 - 1/\kappa)^k$ (with step size $\alpha = 1/L$; see Remark 7.2) or $((\kappa - 1)/(\kappa + 1))^{2k} = (1 - 2/(\kappa + 1))^{2k}$ (with step size $\alpha = 2/(L + \mu)$; Proposition 7.4), to $(1 - 1/\sqrt{\kappa})^k$, as shown in Theorem 8.2, which is a significant improvement for large κ.

Remark 8.5. The proof is inspired by the estimating sequence argument in Nesterov (2013), which treats both general (weakly) and strongly convex functions in a unified fashion. While estimating sequences are powerful, their construction is inductive, and the analysis of the algorithm becomes entangled with its design. Lyapunov analysis is well known for effectively analyzing the stability of dynamical systems Khalil (2002). Since optimization algorithms can be viewed as discrete-time dynamical systems, it is not surprising that Lyapunov functions can be used to analyze their convergence Wilson et al. (2021). In particular, Wilson et al. (2021) provides a comprehensive study of Lyapunov analysis for accelerated methods in optimization as discretizations of continuous-time dynamical systems. Despite this comprehensive work, we believe a concise and direct Lyapunov analysis can be beneficial in revealing the convergence behavior of algorithm (8.12). We encourage interested readers to contrast the analysis in Wilson et al. (2021), particularly Proposition 13, as well as the proof using estimating sequences in (Nesterov, 2013, Section 2.2) with the proof of Theorem 8.2 presented here.

8.4 Exercises

8.1. Let $L \geq \mu$ and define $\kappa = \frac{L}{\mu}$. Show that the NAG method (8.2) with $\alpha = \frac{4}{3L+\mu}$ and $\beta = \frac{\sqrt{3\kappa+1}-2}{\sqrt{3\kappa+1}+2}$ can achieve convergence rate close to $1 - \frac{2}{\sqrt{3\kappa+1}}$ on quadratic functions that are μ-strongly convex and L-smooth by proving that the spectral radius of A_2 defined in (8.6) satisfies $\rho(A_2) \leq 1 - \frac{2}{\sqrt{3\kappa+1}}$.

8.2. If $\beta = 0$, both the HB algorithm (8.1) and NAG algorithm (8.2) reduce to the standard gradient descent (GD) $x_{k+1} = x_k - \alpha \nabla f(x_k)$. Consider a quadratic objective function

$$f(x) = \frac{1}{2} x^\top Q x + b^\top x + a.$$

Let $x^* = -Q^{-1} b$ and define $\xi_k = x_k - x^*$.

(a) Write down the linear system $\xi_{k+1} = A \xi_k$ for the iterations of GD.

(b) Use the same argument as in Section 8.2 and the proof of Proposition 8.3 to show that the spectral radius of A is minimized and equal to $(\kappa-1)/(\kappa+1)$ when the step size $\alpha = 2/(L+\mu)$. Compare this with the conclusion of Proposition 7.4.

8.3. Define $a_0 = 0$ and $a_k = \frac{k+r-2}{r-1}$, $k = 1, 2, \ldots$, where $r \geq 3$. Verify that $a_{k+1}(a_{k+1} - 1) \leq a_k^2$ for all $k \geq 0$ (used in the proof of Theorem 8.1).

Chapter 9

Stochastic Gradient Descent

In this chapter, we introduce one of the cornerstones of deep learning—stochastic gradient descent (SGD)—and analyze its convergence.

9.1 Why Stochastic Gradient?

When training neural networks with big datasets, computing the exact gradient of the loss across the entire dataset for each update step can be computationally expensive. Stochastic gradient descent (SGD) addresses this challenge by using random subsets of the data (mini-batches) to approximate the gradient, dramatically reducing the computation needed per update. This efficiency has made SGD a cornerstone of deep learning optimization, enabling practitioners to train modern deep networks on massive datasets that would otherwise be impractical to handle.

9.1.1 SGD for Empirical Risk Minimization

We use empirical risk minimization as a motivation for SGD. More formally, we consider a parameterized hypothesis class $\mathscr{H} = \{h_w \mid w \in \mathscr{W}\}$, where w represents the learnable parameters. To learn the optimal hypothesis h_w, we aim to minimize the empirical risk:

$$\hat{R}_S(h) = \frac{1}{m} \sum_{i=1}^{m} L(h_w, z_i),$$

where $S = \{z_1, z_2, \ldots, z_m\}$ represents the training dataset of m samples drawn from the underlying distribution $\mathcal{D}$. A natural approach is to apply gradient descent, which updates the model parameters w iteratively using the full gradient of the empirical risk:

$$w^{(t+1)} = w^{(t)} - \eta_t \nabla_w \hat{R}_S(h_{w^{(t)}}).$$

However, computing the full gradient

$$\nabla_w \hat{R}_S(h_w) = \frac{1}{m} \sum_{i=1}^{m} \nabla_w L(h_w, z_i)$$

can be prohibitively expensive for large datasets. Instead, SGD approximates this gradient by using a randomly selected mini-batch $S_b \subseteq S$ of size b, leading to the update rule:

$$w^{(t+1)} = w^{(t)} - \eta_t \frac{1}{b} \sum_{z \in S_b} \nabla_w L(h_{w^{(t)}}, z).$$

By adjusting the batch size b, we can balance the trade-off between computational efficiency and the variance of gradient updates.

One usual assumption is that the stochastic gradient is an unbiased estimator of the full gradient. This holds for mini-batch stochastic gradient descent, as described above for empirical risk minimization, if the mini-batch samples are drawn uniformly at random from the dataset. Formally, this is verified by

$$\begin{aligned}\mathbb{E}_{S_b}\left[\frac{1}{b}\sum_{z\in S_b}\nabla_w L(h_w, z)\right] &= \frac{1}{b}\sum_{z\in S} P(z \in S_b)\nabla_w L(h_w, z) = \frac{1}{b}\sum_{z\in S}\frac{b}{m}\nabla_w L(h_w, z)\\ &= \sum_{z\in S}\frac{1}{m}\nabla_w L(h_w, z) = \nabla_w \hat{R}_S(h_w).\end{aligned}$$

9.2 Convergence Analysis

The iteration of SGD is given by

$$x_{t+1} = x_t - \alpha_t g_t, \quad t \geq 1, \tag{9.1}$$

where $g_t := \nabla f(x_t; \xi_t)$ is the stochastic gradient at x_t (with randomness ξ_t) and α_t is the step size.

We make the following assumption on the objective function f and the stochastic gradient $\nabla f(x; \xi)$, where ξ is the source of randomness. For simplicity of presentation, we shall only focus L-smooth and strongly convex functions.

Assumption 9.1 (L-smoothness)**.** *The continuously differentiable function $f : \mathbb{R}^d \to \mathbb{R}$ is bounded from below by $f^* := \inf_{x \in \mathbb{R}^d} f(x) \in \mathbb{R}$ and its gradient ∇f is L-Lipschitz, i.e., $\|\nabla f(x) - \nabla f(y)\| \le L \|x - y\|$ for all $x, y \in \mathbb{R}^d$.*

Assumption 9.2. *The function $f : \mathbb{R}^n \to \mathbb{R}$ is L-smooth and μ-strongly convex. The function f has a minimizer $x^* \in \mathbb{R}$ and $f^* = f(x^*)$.*

Assumption 9.3. *The stochastic gradient is unbiased in the sense that $\mathbb{E}[\nabla f(x; \xi)] = f(x)$ for all $x \in \mathbb{R}^n$. There exist nonnegative constants A, B, and C such that*

$$\mathbb{E}[|\nabla f(x; \xi)|^2] \le A(f(x) - f^*) + B\,|\nabla f(x)|^2 + C \tag{9.2}$$

for all $x \in \mathbb{R}^n$.

Remark 9.1. Assumption 9.3 was proposed in Khaled and Richtárik (2020) as "the weakest assumption" for analysis of SGD in the non-convex setting. When f is convex, it is in fact equivalent to seemingly weaker forms (see Exercise 9.1).

We will investigate convergence rates both in the almost sure sense and in expectation. We need the following classical result on supermartingale convergence.

Proposition 9.1 (Robbins and Siegmund (1971))**.** *Let $\{X_t\}$, $\{Y_t\}$, and $\{Z_t\}$ be three sequences of random variables that are adapted to a filtration $\{\mathcal{F}_t\}$. Let $\{\gamma_t\}$ be a sequence of nonnegative real numbers such that $\Pi_{t=1}^{\infty}(1 + \gamma_t) < \infty$. Suppose that the following conditions hold:*

(1) X_t, Y_t, and Z_t are nonnegative for all $t \ge 1$.
(2) $\mathbb{E}[Y_{t+1} \,|\, \mathcal{F}_t] \le (1 + \gamma_t)Y_t - X_t + Z_t$ for all $t \ge 1$.
(3) $\sum_{t=1}^{\infty} Z_t < \infty$ holds almost surely.

Then $\sum_{t=1}^{\infty} X_t < \infty$ almost surely and Y_t converges almost surely.

We will not prove this result. Interested readers may refer to Robbins and Siegmund

(1971) (see also Liu (2021)). Here, almost sure convergence means that the sequence of random variables converges to a limit with probability one. In other words, except for a set of outcomes with probability zero, the sequence eventually stays arbitrarily close to the limit.

A special case of Proposition 9.1 occurs when all sequences $\{X_t\}$, $\{Y_t\}$, and $\{Z_t\}$ are deterministic. In this case, we can still conclude that $\sum_{t=1}^{\infty} X_t < \infty$ and that Y_t converges (see Exercise 9.2).

We use the following notation for asymptotic comparisons of positive real-valued sequences $\{a_t\}$ and $\{b_t\}$ indexed by t.

- The little-o notation, $a_t = o(b_t)$, indicates that $a_t/b_t \to 0$ as $t \to \infty$.
- The big-O notation, $a_t = O(b_t)$, indicates the existence of a constant $C > 0$ such that $a_t \leq Cb_t$ for all sufficiently large t.
- Lastly, the big-Θ notation, $a_t = \Theta(b_t)$, means that both $a_t = O(b_t)$ and $b_t = O(a_t)$, implying that a_t and b_t grow at the same asymptotic rate.

The following lemma, as a corollary of Proposition 9.1, provides concrete estimates of almost sure convergence rates for sequences of random variables satisfying a supermartingale property.

Lemma 9.1 (Liu and Yuan (2022, 2024))**.** *If $\{Y_t\}$ is a sequence of nonnegative random variables satisfying*

$$\mathbb{E}[Y_{t+1} \mid \mathcal{F}_t] \leq (1 - c_1\alpha_t)Y_t + c_2\alpha_t^2, \tag{9.3}$$

for all $t \geq 1$, where $\{\alpha_t\}$ is sequence of positive real numbers such that $\alpha_t = \Theta\left(\frac{1}{t^{1-\theta}}\right)$ for some $\theta \in (0, \frac{1}{2})$, and c_1 and c_2 are positive constants. Then, for any $\varepsilon \in (2\theta, 1)$,

$$Y_t = o\left(\frac{1}{t^{1-\varepsilon}}\right), \quad \textit{almost surely.}$$

Proof. By the choice of $\alpha_t = \Theta(\frac{1}{t^{1-\theta}})$, where $\theta \in (0, \frac{1}{2})$, there exists some $\eta > 0$ such that $c_1\alpha_t \geq \frac{\eta}{t^{1-\theta}}$ for all $t \geq 1$. We shall make use of the elementary inequality

$$(t+1)^{1-\varepsilon} \leq t^{1-\varepsilon} + (1-\varepsilon)t^{-\varepsilon}, \quad t > 0, \quad \varepsilon \in (0, 1), \tag{9.4}$$

which can be proved, for instance, as follows. Let $g(x) = x^{1-\varepsilon}$ for $x > 0$. Then

$g'(x) = (1-\varepsilon)x^{-\varepsilon}$ is decreasing on $(0,\infty)$. By the mean value theorem,

$$(t+1)^{1-\varepsilon} - t^{1-\varepsilon} = g'(\xi) \le g'(t) = (1-\varepsilon)t^{-\varepsilon},$$

where $\xi \in (t, t+1)$, which implies inequality (9.4). For the convenience of the readers, recall (9.3) as

$$\mathbb{E}[Y_{t+1} \,|\, \mathcal{F}_t] \le (1 - c_1\alpha_t)Y_t + c_2\alpha_t^2,$$

Multiplying both sides with $(t+1)^{1-\varepsilon}$ and applying inequality (9.4) and $c_1\alpha_t \ge \frac{\eta}{t^{1-\theta}}$ lead to

$$\begin{aligned}
\mathbb{E}[(t+1)^{1-\varepsilon}Y_{t+1} \,|\, \mathcal{F}_t] &\le (t+1)^{1-\varepsilon}(1 - c_1\alpha_t)Y_t + c_2(t+1)^{1-\varepsilon}\alpha_t^2 \\
&\le \left[t^{1-\varepsilon} + (1-\varepsilon)t^{-\varepsilon}\right]\left(1 - \frac{\eta}{t^{1-\theta}}\right)Y_t + c_2(t+1)^{1-\varepsilon}\alpha_t^2 \\
&= \left(1 + \frac{1-\varepsilon}{t}\right)\left(1 - \frac{\eta}{t^{1-\theta}}\right)t^{1-\varepsilon}Y_t + c_2(t+1)^{1-\varepsilon}\alpha_t^2 \\
&= \left[1 + \frac{1-\varepsilon}{t} - \frac{\eta}{t^{1-\theta}} - \frac{\eta(1-\varepsilon)}{t^{2-\theta}}\right]t^{1-\varepsilon}Y_t + c_2(t+1)^{1-\varepsilon}\alpha_t^2,
\end{aligned}$$

where the last two equations are straightforward algebraic rearrangements. Clearly, as $t \to \infty$, the term $\frac{1-\varepsilon}{t}$ is dominated by $\frac{\eta}{2t^{1-\theta}}$. Hence, there exists some $T > 1$ sufficiently large such that, for all $t \ge T$, $\frac{1-\varepsilon}{t} - \frac{\eta}{t^{1-\theta}} - \frac{\eta(1-\varepsilon)}{t^{2-\theta}} \le \frac{\eta}{2t^{1-\theta}} - \frac{\eta}{t^{1-\theta}} = -\frac{\eta}{2t^{1-\theta}}$. It follows that

$$\mathbb{E}[(t+1)^{1-\varepsilon}Y_{t+1} \,|\, \mathcal{F}_t] \le t^{1-\varepsilon}Y_t - \frac{\eta}{2t^{1-\theta}}t^{1-\varepsilon}Y_t + c_2(t+1)^{1-\varepsilon}\alpha_t^2, \quad t \ge T.$$

With $\hat{Y}_t = t^{1-\varepsilon}Y_t$, $X_t = \frac{\eta}{2t^{1-\theta}}t^{1-\varepsilon}Y_t$, $Z_t = c_2(t+1)^{1-\varepsilon}\alpha_t^2 = \Theta\left(\frac{1}{t^{1+\varepsilon-2\theta}}\right)$, and $\gamma_t = 0$, the conditions of Proposition 9.1 are met for all $t \ge$T with $\hat{Y}_t$ in place of Y_t. By Proposition 9.1, we have $t^{1-\varepsilon}Y_t$ converges and $\sum_{t=T}^{\infty} X_t < \infty$ almost surely. We must have $t^{1-\varepsilon}Y_t \to 0$ almost surely, since $\sum_{t=T}^{\infty} \frac{\eta}{2t^{1-\theta}} = \infty$ and $\sum_{t=T}^{\infty} X_t = \sum_{t=T}^{\infty} \frac{\eta}{2t^{1-\theta}}t^{1-\varepsilon}Y_t < \infty$ almost surely. Otherwise, a contradiction would arise by the limit comparison test for convergence of infinite series. The conclusion follows. □

The following lemma, when used together with Proposition 9.1, is useful for almost sure convergence rate analysis in a slightly different setting than Lemma 9.1.

Lemma 9.2. *Let $\{X_t\}$ be a sequence of nonnegative real numbers and $\{\alpha_t\}$ be a decreasing sequence of positive real numbers such that the following holds:*

$$\sum_{t=1}^{\infty} \alpha_t X_t < \infty \quad \textit{and} \quad \sum_{t=2}^{\infty} \frac{\alpha_t}{\sum_{i=1}^{t-1} \alpha_i} = \infty.$$

Define $w_t = \frac{2\alpha_t}{\sum_{i=1}^{t} \alpha_i}$, $Y_1 = X_1$, and

$$Y_{t+1} = (1 - w_t)Y_t + w_t X_t, \quad t \geq 1. \tag{9.5}$$

Then

$$Y_t = o\left(\frac{1}{\sum_{i=1}^{t-1} \alpha_i}\right) \quad \textit{and} \quad \min_{1 \leq i \leq t-1} X_i = o\left(\frac{1}{\sum_{i=1}^{t-1} \alpha_i}\right). \tag{9.6}$$

Proof. Note that $w_1 = 2$ and $Y_2 = Y_1$. Since α_t is monotonically decreasing, $w_t \in [0, 1]$ for $t \geq 2$. It follows that, for each $t \geq 2$, Y_t is a weighted average of all numbers in $\{X_1, \cdots, X_{t-1}\}$. Furthermore, by (9.5) we have

$$Y_{t+1} \sum_{i=1}^{t} \alpha_i = Y_t \sum_{i=1}^{t-1} \alpha_i - \alpha_t Y_t + 2\alpha_t X_t, \quad t \geq 1. \tag{9.7}$$

Let $\hat{Y}_t = Y_t \sum_{i=1}^{t-1} \alpha_i$. Then conditions of Proposition 9.1 are met with $\hat{Y}_t$ in place of Y_t, $\alpha_t Y_t$ in place of X_t, and $2\alpha_t X_t$ in place of Z_t. It follows from Proposition 9.1 that $Y_{t+1} \sum_{i=1}^{t} \alpha_i$ converges[1] and $\sum_{t=1}^{\infty} \alpha_t Y_t < \infty$. Since $\sum_{t=2}^{\infty} \frac{\alpha_t}{\sum_{i=1}^{t-1} \alpha_i} = \infty$, $\sum_{t=1}^{\infty} \alpha_t Y_t < \infty$, and $\lim_{t\to\infty} \frac{\alpha_t Y_t}{\frac{\alpha_t}{\sum_{i=1}^{t-1} \alpha_i}} = \lim_{t\to\infty} Y_t \sum_{i=1}^{t-1} \alpha_i$ exists, we must this limit equal 0 by the limit comparison test for series. Hence $Y_t = o\left(\frac{1}{\sum_{i=1}^{t-1} \alpha_i}\right)$. The other part of the conclusion follows by noting $\min_{1\leq i\leq t-1} X_i \leq Y_t$, because Y_t is a weighted average of $\{X_1, \cdots, X_{t-1}\}$. □

Remark 9.2. A concrete convergence rate $o\left(\frac{1}{t^{\frac{1}{2}-\varepsilon}}\right)$ results from (9.6) if we choose $\alpha_t = \frac{\alpha}{t^{\frac{1}{2}+\varepsilon}}$ for some $\alpha > 0$ and $\varepsilon \in (0, \frac{1}{2})$, because then we have $\sum_{i=1}^{t-1} \alpha_i = \Theta(t^{\frac{1}{2}-\varepsilon})$,

[1] While no random sequences are involved here, Proposition 9.1 is still applicable with almost sure convergence replaced by convergence. A direct proof is possible using the monotone convergence theorem for real numbers.

$\frac{\alpha_t}{\sum_{i=1}^{t-1}\alpha_i} = \Theta(\frac{1}{t})$, and $\sum_t \frac{\alpha_t}{\sum_{i=1}^{t-1}\alpha_i} = \infty$. Note that (9.6) is an asymptotic statement (by the definition of the little-o notation), and the summation $\sum_{i=1}^{t-1}\alpha_i$ is non-empty for $t \geq 2$.

We recall the following fact for strongly convex functions. By Proposition 6.8 and the fact that x^* is a stationary point, we have

$$\frac{1}{2\mu}\|\nabla f(x)\|^2 \geq f(x) - f^*, \quad \forall x \in \mathbb{R}^d. \tag{9.8}$$

Theorem 9.1. *Consider the iterates of SGD (9.1).*

(1) If Assumptions 9.2 and 9.3 hold and $\alpha_t = \Theta\left(\frac{1}{t^{1-\theta}}\right)$ for some $\theta \in (0, \frac{1}{2})$, then

$$f(x_t) - f^* = o\left(\frac{1}{t^{1-\varepsilon}}\right), \quad \forall \varepsilon \in (2\theta, 1), \text{ almost surely,}$$

and

$$\mathbb{E}[f(x_t) - f^*] = o\left(\frac{1}{t^{1-\varepsilon}}\right), \quad \forall \varepsilon \in (2\theta, 1),$$

(2) If Assumptions 9.1 and 9.3 hold and $\{\alpha_t\}$ is a decreasing sequence of positive real numbers satisfying $\sum_{t=1}^{\infty}\alpha_t^2 < \infty$ and $\sum_{t=2}^{\infty}\frac{\alpha_t}{\sum_{i=1}^{t-1}\alpha_i} = \infty$, then almost surely

$$\min_{1\leq i\leq t-1}\|\nabla f(x_i)\|^2 = o\left(\frac{1}{\sum_{i=1}^{t-1}\alpha_i}\right). \tag{9.9}$$

In particular, if we choose $\alpha_t = \frac{\alpha}{t^{\frac{1}{2}+\varepsilon}}$ with $\alpha > 0$ and $\varepsilon \in (0, \frac{1}{2})$, then almost surely

$$\min_{1\leq i\leq t-1}\|\nabla f(x_i)\|^2 = o\left(\frac{1}{t^{\frac{1}{2}-\varepsilon}}\right). \tag{9.10}$$

Proof. 1. We first consider the strongly convex case. By L-smoothness of f and (7.3), we have

$$f(x_{t+1}) \leq f(x_t) - \alpha_t \nabla f(x_t)\cdot g_t + \frac{L\alpha_t^2}{2}\|g_t\|^2.$$

Taking conditional expectation w.r.t. x_t, denoted by $\mathbb{E}_t[\cdot] := \mathbb{E}[\cdot|x_t]$, and using (9.8)

lead to

$$
\begin{aligned}
\mathbb{E}_t\left[f(x_{t+1})-f^*\right] &\overset{(9.2)}{\leq} f(x_t)-f^*-\alpha_t\left\|\nabla f(x_t)\right\|^2+\frac{L\alpha_t^2}{2}\left[A(f(x_t)-f^*)+B\left\|\nabla f(x_t)\right\|^2+C\right] \\
&= (1+\frac{LA\alpha_t^2}{2})(f(x_t)-f^*)-(\alpha_t-\frac{LB\alpha_t^2}{2})\left\|\nabla f(x_t)\right\|^2+\frac{LC\alpha_t^2}{2} \\
&\leq (1+\frac{LA\alpha_t^2}{2})(f(x_t)-f^*)-2\mu(\alpha_t-\frac{LB\alpha_t^2}{2})(f(x_t)-f^*)+\frac{LC\alpha_t^2}{2} \\
&= (1-2\mu\alpha_t+(LA/2+LB\mu)\alpha_t^2)(f(x_t)-f^*)+\frac{LC\alpha_t^2}{2} \\
&\leq (1-\mu\alpha_t)(f(x_t)-f^*)+\frac{LC\alpha_t^2}{2}, \qquad (9.11)
\end{aligned}
$$

provided that $(LA/2+LB\mu)\alpha_t \leq \mu$. The conclusion follows from Lemma 9.1.

To show convergence in expectation, taking expectation from both sides of (9.11) gives

$$
\mathbb{E}[f(x_{t+1})]-f^* \leq (1-\mu\alpha_t)(\mathbb{E}[f(x_t)]-f^*)+\frac{LC\alpha_t^2}{2}. \qquad (9.12)
$$

Applying Lemma 9.1 to the deterministic sequence $\{\mathbb{E}[f(x_t)]\}$ implies the conclusion.

2. For the non-convex case, by L-smoothness and as in (9.11), we obtain

$$
\begin{aligned}
\mathbb{E}_t\left[f(x_{t+1})-f^*\right] &\leq f(x_t)-f^*-\alpha_t\left\|\nabla f(x_t)\right\|^2+\frac{L\alpha_t^2}{2}\left[A(f(x_t)-f^*)+B\left\|\nabla f(x_t)\right\|^2+C\right] \\
&\leq (1+\frac{LA\alpha_t^2}{2})(f(x_t)-f^*)-\left(\alpha_t-\frac{LB\alpha_t^2}{2}\right)\left\|\nabla f(x_t)\right\|^2+\frac{LC\alpha_t^2}{2} \\
&\leq (1+\frac{LA\alpha_t^2}{2})(f(x_t)-f^*)-\frac{1}{2}\alpha_t\left\|\nabla f(x_t)\right\|^2+\frac{LC\alpha_t^2}{2}, \qquad (9.13)
\end{aligned}
$$

provided that $LB\alpha_t \leq 1$. By Proposition 9.1, $\sum_{t=1}^{\infty}\alpha_t\left\|\nabla f(x_t)\right\|^2<\infty$ almost surely. The conclusions follow from Lemma 9.2 and Remark 9.2. Convergence in expectation follows by taking one more expectation over both sides of (9.13) and applying Lemma 9.2 to the deterministic sequence $\left\{\mathbb{E}[\|\nabla f(x_t)\|^2]\right\}$. □

Remark 9.3. While we only focused on the case of L-smooth and strongly convex objective functions, interested readers can refer to Liu and Yuan (2022, 2024) for

an analysis of almost sure convergence rates of SGD and SGD with momentum, including stochastic Nesterov's accelerated gradient descent and stochastic heavy-ball methods, in both nonconvex and general (weakly) convex settings. The work in Liu and Yuan (2022, 2024) provides almost sure convergence guarantees, in contrast to most of the literature, which focuses on convergence in expectation (Bottou et al., 2018).

9.3 Exercises

9.1. Suppose that f is L-smooth and μ-strongly convex ($\mu > 0$). Assume that f has a minimizer x^* and $f^* = f(x^*)$. Consider the following assumption on stochastic gradient of f. There exist nonnegative constants B and C such that

$$\mathbb{E}[|\nabla f(x;\xi)|^2] \leq B\,|\nabla f(x)|^2 + C \tag{9.14}$$

for all $x \in \mathbb{R}^n$. Show that (9.14) is equivalent to (9.2).

9.2. Prove the following deterministic version of Proposition 9.1. Let $\{X_t\}$, $\{Y_t\}$, and $\{Z_t\}$ be three sequences of real numbers. Let $\{\gamma_t\}$ be a sequence of nonnegative real numbers such that $\Pi_{t=1}^{\infty}(1+\gamma_t) < \infty$. Suppose that the following conditions hold:

(1) X_t, Y_t, and Z_t are nonnegative for all $t \geq 1$.
(2) $Y_{t+1} \leq (1+\gamma_t)Y_t - X_t + Z_t$ for all $t \geq 1$.
(3) $\sum_{t=1}^{\infty} Z_t < \infty$.

Then $\sum_{t=1}^{\infty} X_t < \infty$ and Y_t converges.

Part III

Generalization

Chapter 10

PAC Learning

In this chapter, we introduce the framework of probably approximately correct (PAC) learning theory (Valiant, 1984), which provides a formal approach to understanding when a learning algorithm can generalize from finite data. It ensures that, with high probability, the true risk of the learned hypothesis is close to the empirical risk.

10.1 A Formal Learning Model

We introduce the following basic ingredients of learning theory.

- **Input set**: X.
- **Label/output set**: Y. For example, $Y = \{0, 1\}$ or $Y = [0, 1]$.
- **Samples (training data)**: a finite set of samples $S = \{(x_i, y_i)\}_{i=1}^m \in (X \times Y)^m$.
- **Data generating distribution:** each sample is drawn from a distribution $\mathcal{D}$ over $X \times Y$.
- **Loss function**: $L : Y \times Y \to \mathbb{R}_{\geq 0}$.
- **Hypothesis (function) class**: a set of functions $\mathcal{H}$, where each $h \in \mathcal{H}$ maps X into Y.
- **Learning algorithm (learner)**: a family of maps $\mathcal{A} = \{A_m\}_{m=1}^\infty$, where $A_m : (X \times Y)^m \to \mathcal{H}$.
- **Empirical risk**: The empirical risk for some $h \in \mathcal{H}$ and sample set $S = \{(x_i, y_i)\}_{i=1}^m$

is defined as

$$\hat{R}_S(h) = \frac{1}{m}\sum_{i=1}^{m} L(h(x_i), y_i). \tag{10.1}$$

- **Expected risk**: The expected risk for some $h \in \mathscr{H}$ is defined as

$$R(h) = \mathbb{E}_{(x,y)\sim\mathscr{D}}[L(h(x), y)]. \tag{10.2}$$

If the distribution $\mathscr{D}$ is emphasized, we write the expected risk as $R_{\mathscr{D}}(h)$.

Example 10.1 (Concept learning (binary classification))**.** In this context, $Y = \{0, 1\}$ and, for each finite set of samples $S = \{(x_i, y_i)\}_{i=1}^m \in (X \times Y)^m$, the learner outputs a concept $h_m : X \to \{0, 1\}$ using $h_m = A_m(S)$. The loss can be defined as

$$L(y, y') = |y - y'| = \begin{cases} 1, & y \neq y', \\ 0, & y = y'. \end{cases}$$

Such a loss is often called the 0-1 loss. We can also assume that there is a target concept $c : X \to \{0, 1\}$ to be learned. Data generation in this context often assumes that samples $(x_1, \ldots, x_m)$ are independently and identically distributed (i.i.d.) according to some fixed but unknown distribution on X, with labels obtained using $y_i = c(x_i)$.

Example 10.2 (Function learning (regression))**.** If the target function $f : X \to Y$ to be learned has a bounded range, we can assume without loss of generality that $Y = [0, 1]$. One common choice of loss is $L(y, y') = |y - y'|^2$, which corresponds to the use of mean squared error (MSE) for training and empirical error and L^2 distance when measuring generalization error in the function space.

10.2 PAC Learning

Definition 10.1. *A hypothesis class $\mathscr{H}$ is said to be* ***probably approximately correct (PAC)-learnable*** *if there exists an algorithm $\{A_m\}_{m=1}^{\infty}$ and a function $m_{\mathscr{H}} : (0, 1)^2 \to \mathbb{N}$ such that, for every $\varepsilon \in (0, 1)$, every $\delta \in (0, 1)$, and every distribution $\mathscr{D}$ on $X \times Y$, if $m \geq m(\varepsilon, \delta)$, then $h_S = A_m(S)$ returned by the algorithm, with $S = \{(x_i, y_i)\}_{i=1}^m \in (X \times Y)^m$ being i.i.d. samples generated from the distribution $\mathscr{D}$, i.e., $S \sim \mathscr{D}^m$, satisfies*

$$\mathbb{P}_{S\sim\mathscr{D}^m}[R(h_S) > \inf_{h\in\mathscr{H}} R(h) + \varepsilon] \leq \delta. \tag{10.3}$$

When such an algorithm $\mathcal{A}$ exists, it is called a PAC-learning algorithm for $\mathcal{H}$.

In other words, PAC learning guarantees that, with probability at least $1 - \delta$ (over the distribution of generating m samples), the gap between expected risk $R(h_S)$ achieved by the learned hypothesis h_S and the optimally achievable expected risk $\inf_{h \in \mathcal{H}} R(h)$ is smaller than ε. The notion of PAC learnability introduced above is known as *agnostic* PAC learning, where it is not assumed that data generation follows a realizable target function in $\mathcal{H}$.

The notion of PAC learnability implies a uniform convergence bound that does not depend on the distribution. Furthermore, it deals learnability for the entire hypothesis $\mathcal{H}$ and not a particular function in $\mathcal{H}$. In other words, we assume that $\mathcal{H}$ is known to the algorithm, but a particular target function is not. This is a natural assumption to make.

The function $m_{\mathcal{H}}$ describes (an upper bound of) the *sample complexity* of learning $\mathcal{H}$. When the number of samples $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$, learning to ε accuracy with probability at least $1 - \delta$ is guaranteed.

10.3 Learnability of Finite Hypothesis Class

We prove that if $\mathcal{H}$ is finite, then $\mathcal{H}$ is learnable.

Theorem 10.1. *Suppose that $\mathcal{H}$ is finite with cardinality $|\mathcal{H}|$. Assume that the loss function L takes values in $[0, 1]$. Then $\mathcal{H}$ is PAC learnable with sample complexity bounded by*

$$m_{\mathcal{H}}(\varepsilon, \delta) \leq \left\lceil \frac{2 \log(2|\mathcal{H}|/\delta)}{\varepsilon^2} \right\rceil, \tag{10.4}$$

where $\lceil \cdot \rceil$ denotes the ceiling function, which rounds a real number up to the nearest integer.

The following lemma plays a crucial role in bounding the probability that the sum of bounded independent random variables deviates from its expected value by more than a certain amount. Such inequalities are commonly referred to as concentration inequalities (see Appendix C).

Lemma 10.1 (Hoeffding's inequality)**.** *Let X_i, $i = 1, \ldots, m$, be independent random*

variables taking values in $[a_i, b_i]$*. Then, for any* $t > 0$*,*

$$\mathbb{P}\left(S_m - \mathbb{E}\left[S_m\right] \geq t\right) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^m (b_i - a_i)^2}\right), \tag{10.5}$$

$$\mathbb{P}\left(S_m - \mathbb{E}\left[S_m\right] \leq -t\right) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^m (b_i - a_i)^2}\right), \tag{10.6}$$

$$\mathbb{P}\left(\left|S_m - \mathbb{E}\left[S_m\right]\right| \geq t\right) \leq 2\exp\left(-\frac{2t^2}{\sum_{i=1}^m (b_i - a_i)^2}\right), \tag{10.7}$$

where $S_m = \sum_{i=1}^m X_i$ *and* $\mathbb{E}[S_m]$ *is the expectation of* S_m*.*

Proof of Theorem 10.1. Let the samples be $S = \{(x_i, y_i)\}_{i=1}^m$ and define $X_i = L(h(x_i), y_i)$. Then $\{X_i\}_{i=1}^m$ are independently and identically distributed. Fix any $\varepsilon \in (0, 1)$. Applying Hoeffding's inequality (10.7) (Lemma 10.1) to $X_1, \ldots, X_m$, with $t = m\varepsilon$ and $[a_i, b_i] = [0, 1]$, we have

$$\begin{aligned}
\mathbb{P}(\left|\hat{R}_S(h) - R(h)\right| \geq \varepsilon) &= \mathbb{P}\left(\left|\frac{1}{m}\sum_{i=1}^m X_i - \mathbb{E}[X_1]\right| \geq \varepsilon\right) \\
&= \mathbb{P}\left(\left|\sum_{i=1}^m X_i - \mathbb{E}[\sum_{i=1}^m X_i]\right| \geq m\varepsilon\right) \\
&\leq 2\exp\left(-2m\varepsilon^2\right)
\end{aligned}$$

It follows from the union bound (see Appendix C) that

$$\begin{aligned}
\mathbb{P}(\left\{\exists h \in \mathscr{H} \text{ s. t. } \left|\hat{R}_S(h) - R(h)\right| \geq \varepsilon\right\}) &= \mathbb{P}(\bigcup_{h\in\mathscr{H}} \left\{\left|\hat{R}_S(h) - R(h)\right| \geq \varepsilon\right\}) \\
&\leq \sum_{h\in\mathscr{H}} \mathbb{P}(\left|\hat{R}_S(h) - R(h)\right| \geq \varepsilon) \\
&\leq 2\left|\mathscr{H}\right|\exp\left(-2m\varepsilon^2\right).
\end{aligned}$$

For any $\delta \in (0, 1)$, if $m \geq \frac{\log(2|\mathscr{H}|/\delta)}{2\varepsilon^2}$, we have

$$\mathbb{P}(\left\{\exists h \in \mathscr{H} \text{ s. t. } \left|\hat{R}_S(h) - R(h)\right| \geq \varepsilon\right\}) \leq 2\left|\mathscr{H}\right|\exp\left(-2m\varepsilon^2\right) \leq \delta,$$

which implies that, with probability at least $1-\delta$,

$$\left|\hat{R}_S(h) - R(h)\right| \leq \varepsilon \tag{10.8}$$

holds for all $h \in \mathcal{H}$, provided that $m \geq \frac{\log(2|\mathcal{H}|/\delta)}{2\varepsilon^2}$.

Let h_S be learned by empirical risk minimization (ERM), i.e.,

$$h_S = \arg\min_{h\in\mathcal{H}} \hat{R}_S(h). \tag{10.9}$$

By (10.8), if $m \geq \frac{\log(2|\mathcal{H}|/\delta)}{2(\varepsilon/2)^2} = \frac{2\log(2|\mathcal{H}|/\delta)}{\varepsilon^2}$, then we have, with probability at least $1-\delta$, for any $h \in \mathcal{H}$,

$$R(h_S) \leq \hat{R}_S(h_S) + \frac{\varepsilon}{2} \leq \hat{R}_S(h) + \frac{\varepsilon}{2} \leq R(h) + \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = R(h) + \varepsilon.$$

Taking the minimum of the right-hand side over all $h \in \mathcal{H}$ provides the PAC learning guarantee $\mathbb{P}(R(h_S) \leq \min_{h\in\mathcal{H}} R(h) + \varepsilon) \geq 1-\delta$. □

We emphasize that the learnability result above is based on uniform convergence of the empirical risk to expected risk for all hypotheses in the hypothesis class. With $\varepsilon = \sqrt{\frac{\log(2|\mathcal{H}|/\delta)}{2m}}$, (10.8) gives the convergence rate

$$\left|\hat{R}_S(h) - R(h)\right| \leq \sqrt{\frac{\log(2\,|\mathcal{H}|\,/\delta)}{2m}}, \quad \forall h \in \mathcal{H}, \tag{10.10}$$

with probability at least $1-\delta$. Note that this $O(1/\sqrt{m})$ convergence rate is also independent of the underlying data distribution.

10.4 Fast Rate under Realizability Assumption for 0-1 Loss

We consider a special case of Theorem 10.1, where the uniform convergence rate can be improved. The *realizability* assumption is satisfied for a hypothesis class $\mathcal{H}$ if there exists a hypothesis $h^* \in \mathcal{H}$ such that $R(h^*) = 0$. Since the loss function is non-negative, the realizability assumption implies $\mathbb{P}(L(h^*(x), y) = 0) = 1$. Consequently, with probability 1, $\hat{R}_S(h^*) = 0$ on any set of samples S drawn i.i.d. from the data distribution. Thus, if we use the ERM hypothesis $h_S = \arg\min_{h\in\mathcal{H}} \hat{R}_S(h)$, we have

$\hat{R}_S(h_S) = 0$ with probability 1.

Theorem 10.2. *Suppose that $\mathcal{H}$ is finite with cardinality $|\mathcal{H}|$. Assume that the realizability assumption holds and the loss function L takes values in $\{0, 1\}$. Then $\mathcal{H}$ is PAC learnable with sample complexity bounded by*

$$m_{\mathcal{H}}(\varepsilon, \delta) \leq \frac{\log(|\mathcal{H}|/\delta)}{\varepsilon}. \tag{10.11}$$

Proof. Fix any $\varepsilon \in (0, 1)$. Define $\mathcal{H}_\varepsilon = \{h \in \mathcal{H} : R(h) \geq \varepsilon\}$. Since we have a 0-1 loss,

$$R(h) = \mathbb{E}[L(h(x), y)] = \mathbb{E}[\mathbf{1}_{\{L(h(x),y)=1\}}] = P(L(h(x), y) = 1).$$

If $h \in \mathcal{H}_\varepsilon$, then $\mathbb{P}(L(h(x), y) = 0) \leq 1 - \varepsilon$. It follows that, for any sample set S of size m drawn i.i.d. from the data distribution, we have

$$\mathbb{P}(\hat{R}_S(h) = 0) \leq (1 - \varepsilon)^m \leq e^{-m\varepsilon}, \tag{10.12}$$

where we used the inequality that $1 - x \leq e^{-x}$ for $x \geq 0$, which implied $(1 - x)^m \leq (e^{-x})^m = e^{-mx}$ for $x \in [0, 1]$. Let h_S be an ERM hypothesis. Then $\hat{R}_S(h_S) = 0$ with probability 1. We have

$$\begin{aligned}
\mathbb{P}(\{R(h_S) \geq \varepsilon\}) &= \mathbb{P}(\{R(h_S) \geq \varepsilon,\ \hat{R}_S(h_S) = 0\}) \\
&\leq \mathbb{P}\left(\bigcup_{h \in \mathcal{H}_\varepsilon} \{\hat{R}_S(h) = 0\}\right) \\
&\leq \sum_{h \in \mathcal{H}_\varepsilon} \mathbb{P}(\{\hat{R}_S(h) = 0\}) \\
&\leq |\mathcal{H}|\, e^{-m\varepsilon},
\end{aligned}$$

where we used the union bound for probability and (10.12). If $m \geq \frac{\log(|\mathcal{H}|/\delta)}{\varepsilon}$, then $\mathbb{P}(\{S : R(h_S) \geq \varepsilon\}) \leq \delta$. The conclusion follows. □

The conclusion of the theorem can be equivalently stated as follows: with probability at least $1 - \delta$,

$$R(h_S) \leq \frac{\log(|\mathcal{H}|/\delta)}{m}, \tag{10.13}$$

where h_S is an ERM hypothesis given by $h_S = \arg\min_{h \in \mathcal{H}} \hat{R}_S(h)$. Note that this

convergence rate, $O(1/m)$, improves upon the $O(1/\sqrt{m})$ rate in (10.10). Equivalently, the sample complexity in the realizable case is $O(1/\epsilon)$, which improves upon the $O(1/\epsilon^2)$ rate without the realizability assumption. Note that the convergence rate (10.13) applies to any learning algorithm that consistently outputs a hypothesis with zero empirical risk on the training data. Such an algorithm is called consistent.

It is clear from the proof of Theorem 10.2 that the improvement in sample complexity and convergence rate comes from replacing the use of Hoeffding's inequality with a simpler counting argument when the realizability assumption is satisfied and the loss is 0-1.

10.4.1 An Application to Neural Networks

Remark 10.1 (How relevant is a finite hypothesis class?). While the finite hypothesis class assumption appears to be quite restrictive, we can still use it to obtain rough estimates of sample complexity in practical situations. Suppose that the hypothesis class $\mathscr{H}$ is a neural network parameterized by W real numbers (weights and biases). In practice, each real number can be represented as a 64-bit floating-point number. Then, the hypothesis class effectively has size 2^{64W}. Applying Theorem 10.1, the sample complexity of learning this hypothesis class is bounded by

$$\frac{128W\log(2) + 2\log(2/\delta)}{\epsilon^2}.$$

In the following chapters, we will rigorously discuss the learnability and sample complexity of infinite hypothesis classes.

10.5 No-Free-Lunch Theorem

In this section, we present a result showing that there is no "universal learner"—one that performs well across all data distributions and all tasks. This result is commonly known as the no-free-lunch theorem. We focus on binary classification and follow the presentation in Shalev-Shwartz and Ben-David (2014).

Theorem 10.3. *Let $\mathscr{A} = \{A_m\}_{m=1}^{\infty}$ be any learning algorithm for binary classification with 0-1 loss over an input domain X. Suppose that the size of training samples m is smaller that $\frac{|X|}{2}$. Then there exists a distribution $\mathscr{D}$ on $X \times \{0,1\}$ and a function $h : X \to \{0,1\}$*

such that

(1) $R_{\mathcal{D}}(h) = 0$;
(2) $\mathbb{P}_{S\sim\mathcal{D}^m}[R_{\mathcal{D}}(h_S) \geq \frac{1}{8}] \geq \frac{1}{7}$, *where* $h_S = A_m(S)$.

This result essentially states that if the sample size does not exceed half of the input domain size, then we cannot achieve arbitrarily good performance with arbitrarily high probability across all distributions and all tasks (target functions).

Proof. Let C denote any subset of X of size $2m$, i.e., $|C| = 2m$. There are $N := 2^{2m}$ possible functions from C to $\{0,1\}$. Denote these functions as $h_1, h_2, \ldots, h_N$. For each h_i, let $\mathcal{D}_i$ be a distribution (probability measure) over $X \times \{0,1\}$ defined by

$$\mathcal{D}_i(\{(x,y)\}) = \begin{cases} \dfrac{1}{2m}, & \text{if } y = h_i(x),\ x \in C, \\ 0, & \text{otherwise.} \end{cases}$$

Given any set of samples of size m from $X \times \{0,1\}$, denote $h_S = A_m(S)$. We show that there exists $i \in \{1, \ldots, N\}$ such that

$$\mathbb{E}_{S\sim\mathcal{D}_i^m}[R_{\mathcal{D}_i}(h_S)] \geq \frac{1}{4}. \tag{10.14}$$

Claim: (10.14) implies that

$$\mathbb{P}_{S\sim\mathcal{D}_i^m}[R_{\mathcal{D}_i}(h_S) \geq \frac{1}{8}] \geq \frac{1}{7}. \tag{10.15}$$

Since

$$R_{\mathcal{D}_i}(h_i) = \frac{1}{2m} \sum_{\substack{x\in C,\\ y=h_i(x)}} |h_i(x) - y| = 0,$$

the conclusion of the theorem follows from (10.15) with $\mathcal{D} = D_i$ and $h = h_i$. We can make h defined on X by arbitrarily extending it from C to X.
Proof of the claim: Denote the random variable $\theta := R_{\mathcal{D}_i}(h_S)$. Then $\theta \in [0,1]$ (because

of 0-1 loss). We have $\mathbb{E}[\theta] = \frac{1}{4}$. Write $p = \mathbb{P}[\theta \geq \frac{1}{8}]$. Then (10.14) implies that

$$\begin{aligned}\frac{1}{4} \leq \mathbb{E}[\theta] &= \int_{\{\theta \geq \frac{1}{8}\}} \theta d\mathbb{P} + \int_{\{\theta < \frac{1}{8}\}} \theta d\mathbb{P} \leq \int_{\{\theta \geq \frac{1}{8}\}} d\mathbb{P} + \frac{1}{8} \int_{\{\theta < \frac{1}{8}\}} d\mathbb{P} \\ &= p + \frac{1}{8}(1-p),\end{aligned}$$

which implies $p \geq \frac{1}{7}$.

We are left with the proof of (10.14). There are $K := (2m)^m$ possible sequences of m inputs drawn from C. Denote them by $X_1, X_2, \ldots, X_K$. Let S_j^i denote the set of m samples from $X \times \{0,1\}$, where the inputs are X_j and the outputs corresponding to labels given by h_i. Clearly, if the distribution is $\mathcal{D}_i$, then the only possible training sets of size m are $S_1^i, S_2^i, \ldots, S_K^i$, and they have the same probability of being drawn. It follows that

$$\mathbb{E}_{S \sim \mathcal{D}_i^m}[R_{\mathcal{D}_i}(h_S)] = \frac{1}{K} \sum_{j=1}^{K} R_{\mathcal{D}_i}(h_{S_j^i}).$$

Consequently,

$$\begin{aligned}\max_{1 \leq i \leq N} \mathbb{E}_{S \sim \mathcal{D}_i^m}[R_{\mathcal{D}_i}(h_S)] &\geq \frac{1}{N} \sum_{i=1}^{N} \frac{1}{K} \sum_{j=1}^{K} R_{\mathcal{D}_i}(h_{S_j^i}) \\ &= \frac{1}{K} \sum_{j=1}^{K} \frac{1}{N} \sum_{i=1}^{N} \frac{1}{2m} \sum_{\substack{x \in C, \\ y = h_i(x)}} \left| h_{S_j^i}(x) - y \right| \qquad (10.16)\end{aligned}$$

Fix any $j \in \{1, \ldots, K\}$. Let $v_1, v_2, \ldots, v_l$ denote the input samples in C that are not seen in $X_j = (x_1, x_2, \ldots, x_m)$. Then $l \geq m$. It follows that

$$\begin{aligned}\frac{1}{N} \sum_{i=1}^{N} \frac{1}{2m} \sum_{\substack{x \in C, \\ y = h_i(x)}} \left| h_{S_j^i}(x) - y \right| &\geq \frac{1}{N} \sum_{i=1}^{N} \frac{1}{2m} \sum_{r=1}^{l} \left| h_{S_j^i}(v_r) - h_i(v_r) \right| \\ &\geq \frac{1}{N} \sum_{i=1}^{N} \frac{1}{2l} \sum_{r=1}^{l} \left| h_{S_j^i}(v_r) - h_i(v_r) \right| \\ &= \frac{1}{2l} \sum_{r=1}^{l} \frac{1}{N} \sum_{i=1}^{N} \left| h_{S_j^i}(v_r) - h_i(v_r) \right|. \qquad (10.17)\end{aligned}$$

Fix any $r \in \{1, \ldots, l\}$. For each $i \in \{1, \ldots, N\}$, let $h_{i'}$ be the function in $\{h_1, h_2 \ldots, h_N\}$ that only differs from h_i by their evaluations at v_r. Then $S_j^i = S_j^{i'}$ because evaluations of h_i' and h_i coincide on X_j. It follows that

$$\left|h_{S_j^i}(v_r) - h_i(v_r)\right| + \left|h_{S_j^{i'}}(v_r) - h_{i'}(v_r)\right| = 1,$$

which, together with (10.17) and (10.16), implies

$$\begin{aligned}\max_{1\le i\le N} \mathbb{E}_{S\sim\mathcal{D}_i^m}[R_{\mathcal{D}_i}(h_S)] &\ge \frac{1}{K}\sum_{j=1}^{K}\frac{1}{2l}\sum_{r=1}^{l}\frac{1}{N}\sum_{i=1}^{N}\left|h_{S_j^i}(v_r) - h_i(v_r)\right| \\ &\ge \frac{1}{K}\sum_{j=1}^{K}\frac{1}{2l}\sum_{r=1}^{l}\frac{1}{2} = \frac{1}{4}, \end{aligned} \tag{10.18}$$

which is precisely (10.14). □

10.6 Summary

A finite hypothesis class of real-valued functions is (agnostically) PAC learnable. The sample complexity of learning, $O(1/\epsilon^2)$, is established by proving a uniform convergence property with a rate of $O(1/\sqrt{m})$ between empirical risk and expected risk, with high probability, for all hypotheses in the class. For the realizable (consistent) case with 0-1 loss, we demonstrate that the sample complexity can be improved to $O(1/\epsilon)$, and the uniform convergence rate to $O(1/m)$.

10.7 Exercises

10.1. The sample complexity bound for learning a finite hypothesis class is

$$m_{\mathcal{H}}(\varepsilon, \delta) \le \left\lceil \frac{2\log(2|\mathcal{H}|/\delta)}{\varepsilon^2} \right\rceil.$$

Investigate how the parameters $|\mathcal{H}|$, δ, and ε affect the required number of samples.

(1) Suppose $|\mathcal{H}| = 1000$, $\varepsilon = 0.1$, and $\delta = 0.05$. Compute the sample complexity bound.

(2) Keeping the other parameters fixed, recompute the bound in each of the following cases:
   (1) $|\mathscr{H}|$ increases to 10 000;
   (2) δ decreases to 0.005;
   (3) ε decreases to 0.05.
(3) Compare the results obtained above. Which parameter has the largest effect on the required number of samples?
(4) Explain your observation by examining how each parameter appears in the sample complexity formula.

Chapter 11

VC-Dimension

In Chapter 10, we saw that finite hypothesis classes are learnable, and the sample complexity of learning is upper-bounded by the logarithm of their size. In this chapter, we introduce a characterization of learnability for infinite hypothesis classes using the notion of Vapnik–Chervonenkis (VC) dimension (Vapnik and Chervonenkis, 1971).

11.1 Definitions of Growth Function and VC-Dimension

The cardinality of a finite set A, i.e., the number of elements in A, is denoted by $|A|$.

Definition 11.1 (Restriction of function class on finite set)**.** *Let $\mathscr{H}$ be a class of functions from Z to $\{0,1\}$. Let $S \subseteq Z$. Given $h \in \mathscr{H}$, let $h\big|_S : S \to \{0,1\}$ denote the restriction of h on S. Denote $\mathscr{H}\big|_S = \left\{h\big|_S : h \in \mathscr{H}\right\}$, i.e., the set of restrictions to S of all functions in $\mathscr{H}$.*

Definition 11.2 (Shattering)**.** *A finite set $S \subseteq Z$ is said to be shattered by a class of functions $\mathscr{H}$ from Z to $\{0,1\}$, if $H\big|_S$ has cardinality $2^{|S|}$, i.e., it contains every possible function $h : S \to \{0,1\}$.*

Definition 11.3 (Growth function)**.** *Given a class of functions $\mathscr{H}$ from Z to $\{0,1\}$, the*

growth function of $\mathcal{H}$, $\Pi_{\mathcal{H}} : \mathbb{N} \to \mathbb{N}$, is defined by

$$\Pi_{\mathcal{H}}(m) = \max_{\substack{S \subseteq Z \\ |S|=m}} \left|\mathcal{H}\big|_S\right|.$$

Intuitively, the growth function of $\mathcal{H}$ represents the largest number of patterns (i.e., assignments of 0-1 labels) that can be realized by $\mathcal{H}$ on a given number of samples.

Clearly, $\Pi_{\mathcal{H}}(m) \leq 2^m$ for all m, because there are at most 2^m functions from m samples to $\{0,1\}$. If $\mathcal{H}$ is finite, then $\Pi_{\mathcal{H}}(m) \leq |\mathcal{H}|$ for all m and $\Pi_{\mathcal{H}}(m) = |\mathcal{H}|$ for m sufficiently large. We shall see that the growth function precisely plays the role of $|\mathcal{H}|$ in characterizing the learnability and sample complexity of infinite classes of functions.

Definition 11.4 (VC-dimension)**.** *Given a class of functions $\mathcal{H}$ from Z to $\{0,1\}$, the VC-dimension of $\mathcal{H}$, denoted by VC-dim($\mathcal{H}$), is defined to the size of the largest subset of Z shattered by $\mathcal{H}$. If $\mathcal{H}$ can shatter sets of arbitrarily large size, then VC-dim($\mathcal{H}$) $= \infty$.*

If VC-dim($\mathcal{H}$) $= d$, then for any $m \leq d$, we have $\Pi_{\mathcal{H}}(m) = 2^m$. This is because if $\mathcal{H}$ shatters a set, it also shatters all its subsets (Exercise 11.1).

Example 11.1 (VC-dimension of threshold classifiers)**.** A threshold classifier on the real line is a function of the form

$$h_c(x) = \mathbf{1}_{[c,\infty)}(x),$$

where $\mathbf{1}_A$ is the indicator function of a set A, defined by

$$\mathbf{1}_A(x) = \begin{cases} 1, & \text{if } x \in A, \\ 0, & \text{if } x \notin A. \end{cases}$$

Consider $\mathcal{H} = \{h_c : c \in \mathbb{R}\}$. Clearly, for a singleton set $C = \{c_1\}$, $h_c\big|_C$ maps C to 0 if $c > c_1$ and maps C to 1 if $c \leq c_1$. Hence, $\mathcal{H}$ shatters C. For any set $C = \{c_1, c_2\}$ where we assume $c_1 < c_2$ without loss of generality, we can easily check that there is no threshold classifier h_c that classifies c_1 as 1 and c_2 as 0, because if $c_1 \geq c$, then we also have $c_2 \geq c$. It follows that VC-dim($\mathcal{H}$) $= 1$.

Example 11.2 (VC-dimension of intervals)**.** Let $\mathcal{H} = \{\mathbf{1}_{[a,b]} : a, b \in \mathbb{R}, a < b\}$. For any set $C = \{c_1, c_2\}$ where $c_1 < c_2$, we can easily show that $\mathcal{H}$ shatters C (see Exercise 11.2). For an arbitrary set $C = \{c_1, c_2, c_3\}$, where we assume $c_1 < c_2 < c_3$ without loss of generality, it can be verified that $\mathcal{H}$ cannot achieve the label $(1, 0, 1)$ on input (c_1, c_2, c_3), because an interval containing both c_1 and c_3 must also contain c_2.

Example 11.3 (VC-dimension of a finite hypothesis class)**.** Let $\mathcal{H}$ be an arbitrary finite hypothesis class of cardinality $|\mathcal{H}|$ from X to $\{0, 1\}$. Let C be a subset of X. Since there are a total of $2^{|C|}$ functions from C to $\{0, 1\}$ and $\left|\mathcal{H}\big|_C\right| \leq |\mathcal{H}|$, $\mathcal{H}$ cannot shatter C if $|\mathcal{H}| < 2^{|C|}$. In other words, the largest subset of X shattered by $\mathcal{H}$ must have size no larger than $\log_2(|\mathcal{H}|)$. This implies VC-dim$(\mathcal{H}) \leq \log_2(|\mathcal{H}|)$. We can show that the gap between VC-dim$(\mathcal{H})$ and $\log_2(\mathcal{H})$, however, can be arbitrarily large (Exercise 11.3).

11.2 Fundamental Theorem of Statistical Learning for Binary Classification

The following fundamental theorem of statistical learning states that a hypothesis class from X to $\{0, 1\}$ is PAC learnable if and only if it has a finite VC-dimension.

Theorem 11.1. *Let $\mathcal{H}$ be a hypothesis class from X to $\{0, 1\}$ and the loss be 0-1. Then $\mathcal{H}$ is PAC learnable if and only if VC-dim$(H) < \infty$. Furthermore, the sample complexity of learning $\mathcal{H}$ is bounded by*

$$m_{\mathcal{H}}(\varepsilon, \delta) \leq C \frac{d \log(1/\varepsilon) + \log(1/\delta)}{\varepsilon^2}, \tag{11.1}$$

where $C > 0$ is an absolute constant and $d =$ VC-dim(H), and this can be achieved by the ERM learner.

The proof of necessity is based on the no-free-lunch theorem (Theorem 10.3). In particular, we state the follow corollary of Theorem 10.3.

Corollary 11.1. *Let $\mathcal{H}$ be a hypothesis class from X to $\{0, 1\}$ and the loss be 0-1. Let m be the size of training samples. If there exists a set C of size $2m$ such that $\mathcal{H}$ shatters C, then, for any learning algorithm $\mathcal{A} = \{A_m\}_{m=1}^{\infty}$, there exists a distribution $\mathcal{D}$ over $X \times \{0, 1\}$ and a function $h \in \mathcal{H}$ such that $R_{\mathcal{D}}(h) = 0$ and $\mathbb{P}_{S \sim \mathcal{D}^m}[R_{\mathcal{D}}(h_S) \geq \frac{1}{8}] \geq \frac{1}{7}$, where $h_S = A_m(S)$.*

The corollary immediately follows from Theorem 10.3 and its proof, because if H shatters C, then h constructed in the proof of Theorem 10.3 indeed belongs to $\mathscr{H}$.

To prove the sufficiency part of Theorem 11.1, we rely on the following result on uniform convergence of empirical risk based on the growth function.

Theorem 11.2 (Uniform convergence from growth function)**.** *Let $\mathscr{H}$ be a hypothesis class from X to $\{0,1\}$ and the loss be 0-1. Let $\mathscr{D}$ be a distribution over $X \times \{0,1\}$. For any integer $m \geq 1$ and $\varepsilon \in (0,1)$, we have*

$$\mathbb{P}_{S\sim\mathscr{D}^m}(\exists h \in \mathscr{H} \text{ s.t. } \left|R_{\mathscr{D}}(h) - \hat{R}_S(h)\right| \geq \varepsilon) \leq 4\Pi_{\mathscr{H}}(2m)\exp\left(-\frac{m\varepsilon^2}{8}\right). \tag{11.2}$$

We shall not prove this result directly. For a direct proof, see Theorem 4.3 of Anthony and Bartlett (2009). It relies on a standard but lengthy argument involving techniques such as symmetrization, permutations, and reduction to a finite hypothesis class. Instead, we will prove a sharper result in Chapter 12 as a corollary of generalization bounds using Rademacher complexity.

We also need the following lemma, which bounds the growth function using the VC-dimension, known as Sauer's lemma (or Sauer-Shelah Lemma). For a proof, see (Mohri et al., 2018, Theorem 3.17 and Corollary 3.18), (Shalev-Shwartz and Ben-David, 2014, Lemma 6.10), (Anthony and Bartlett, 2009, Theorems 3.6 and 3.7), or Ngo (2010).

Lemma 11.1 (Sauer's lemma)**.** *Let $\mathscr{H}$ be a hypothesis class from X to $\{0,1\}$. Suppose that VC-dim$(\mathscr{H}) = d < \infty$. Then, for $m \geq d$, we have*

$$\Pi_{\mathscr{H}}(m) \leq \sum_{i=0}^{d}\binom{m}{i} \leq \left(\frac{em}{d}\right)^d = O(m^d). \tag{11.3}$$

We are now ready to prove Theorem 11.1.

Proof of Theorem 11.1. The necessary part follows from Corollary 11.1, because for any sample size m, $\mathscr{H}$ can shatter a set of size $2m$ due to VC-dim$(\mathscr{H}) = \infty$. The conclusion follows from Corollary 11.1.

Fix any distribution $\mathcal{D}$ over $X \times \{0,1\}$ and a sample size $m \geq 1$. By the definition of infimum, for any $\alpha > 0$, there exists $h^* \in \mathcal{H}$ such that $R_{\mathcal{D}}(h^*) \leq \inf_{h\in\mathcal{H}} R_{\mathcal{D}}(h) + \alpha$. Let S denote a sample set of size m and h_S be an ERM hypothesis given by $h_S = \arg\min_{h\in\mathcal{H}} \hat{R}_S(h)$. It follows that

$$\begin{aligned}
R_{\mathcal{D}}(h_S) - \inf_{h\in\mathcal{H}} R_{\mathcal{D}}(h) &\leq R_{\mathcal{D}}(h_S) - R_{\mathcal{D}}(h^*) + \alpha \\
&= R_{\mathcal{D}}(h_S) - \hat{R}_S(h_S) + \underbrace{\hat{R}_S(h_S) - \hat{R}_S(h^*)}_{\leq 0} + \hat{R}_S(h^*) - R_{\mathcal{D}}(h^*) + \alpha \\
&\leq R_{\mathcal{D}}(h_S) - \hat{R}_S(h_S) + \hat{R}_S(h^*) - R_{\mathcal{D}}(h^*) + \alpha.
\end{aligned}$$

Using (11.2), we obtain that, with probability $1-\delta$,

$$R_{\mathcal{D}}(h_S) - \inf_{h\in\mathcal{H}} R_{\mathcal{D}}(h) \leq 2\varepsilon + \alpha.$$

provided that

$$4\Pi_{\mathcal{H}}(2m)\exp\left(-\frac{m\varepsilon^2}{8}\right) \leq \delta. \tag{11.4}$$

Since $\alpha > 0$ is arbitrary, by continuity of probability, we have

$$R_{\mathcal{D}}(h_S) - \inf_{h\in\mathcal{H}} R_{\mathcal{D}}(h) \leq 2\varepsilon$$

with probability $1-\delta$, provided that δ satisfies (11.4). To satisfies the condition of PAC learning (10.3), we need δ to satisfy (11.4) with $\varepsilon/2$ in place of ε; that is,

$$4\Pi_{\mathcal{H}}(2m)\exp\left(-\frac{m\varepsilon^2}{32}\right) \leq \delta. \tag{11.5}$$

By Sauer's lemma (Lemma 11.1), (11.5) holds if

$$4\left(\frac{2em}{d}\right)^d \exp\left(-\frac{m\varepsilon^2}{32}\right) \leq \delta, \tag{11.6}$$

or equivalently,

$$m \geq \frac{32}{\varepsilon^2}(d\log m + d\log(2e/d) + \log(4/\delta)). \tag{11.7}$$

Using the elementary inequality $\log x \leq ax - \log a - 1$ for all $a, x > 0$ (Exercise 11.5),

we have

$$\log m \leq \frac{\varepsilon^2}{64d} m - \log(\frac{\varepsilon^2}{64d}) - 1.$$

Hence, to satisfy (11.7), it suffices to have

$$m \geq \frac{m}{2} + \frac{32}{\varepsilon^2}(d \log(128/\varepsilon^2) + \log(4/\delta)), \tag{11.8}$$

or

$$m \geq \frac{64}{\varepsilon^2}(2d \log(8\sqrt{2}/\varepsilon) + \log(4/\delta)), \tag{11.9}$$

which clearly implies (11.1). □

Remark 11.1 (Improved generalization bounds for the realizable case). Similar to Theorem 10.2, under the realizability assumption, i.e., there exists h^* such that $R_{\mathcal{D}}(h^*) = 0$, the sample complexity bound (11.1) in Theorem 11.1 can be further improved (see (Anthony and Bartlett, 2009, Theorem 4.8)) to

$$m_{\mathcal{H}}(\varepsilon, \delta) \leq C \frac{d \log(1/\varepsilon) + \log(1/\delta)}{\varepsilon}. \tag{11.10}$$

The main idea, again, is to replace the uniform convergence bound (11.2), obtained using Hoeffding's inequality, with a simple counting argument.

11.2.1 Applications to Neural Networks

Neural networks produce real-valued outputs. To determine the VC-dimension of a real-valued neural network, we typically consider its ability to classify points by introducing a threshold function. That is, we define a binary function of the form:

$$h(x) = \mathbf{1}_{\{f(x) > \theta\}},$$

where $f(x)$ is the network's real-valued output and θ is a fixed threshold. The VC-dimension of the neural network is then defined as the VC dimension of this thresholded function class.

For ReLU networks, we summarize the known upper and lower bounds on their VC dimension as follows (DeVore et al., 2021, Lemma 3.11, Theorem 3.12). We only focus on networks with a single output.

Proposition 11.1 (VC-dimension of shallow ReLU networks)**.** *Let $\mathcal{M}_W(ReLU; d)$ denote the class of one-hidden-layer ReLU networks with width W and d input variables. We have*

(1) VC-dim$(\mathcal{M}_W(ReLU; 1)) = W + 1$.
(2) For $d \geq 2$, VC-dim$(\mathcal{M}_W(ReLU; d)) \leq C_0 W \log_2(W)$, where C_0 depends only on d.
(3) For $d \geq 4$, VC-dim$(\mathcal{M}_W(ReLU; d)) \geq c_0 W \log_2(W)$, where c_0 depends only on d.
(4) For $d = 2, 3$, VC-dim$(\mathcal{M}_W(ReLU; d)) \geq W + 1$.

Proposition 11.2 (VC-dimension of deep ReLU networks)**.** *Let $\mathcal{M}^L_{W_0}(ReLU; d)$ denote the class of ReLU networks with width W_0, depth L, and d input variables. Let W_0 be fixed but sufficiently large depending only on d. There are fixed constants c_1 and C_1, depending only on d, such that*

$$c_1 L^2 \leq VC\text{-}dim(\mathcal{M}^L_{W_0}(ReLU; d)) \leq C_1 L^2.$$

For proofs of the results above and further references, see (DeVore et al., 2021, Lemma 3.11). It seems that for the cases $d = 2, 3$, there is a logarithmic factor of discrepancy between the lower and upper bounds. We also remark that upper bounds on the VC-dimension of a hypothesis class can be used to lower bound the degree of approximation provided by functions in this class in the infinity norm. For more details, see DeVore et al. (2021) as well as Yarotsky (2017); Lu et al. (2021).

Remark 11.2 (Practical usefulness of VC-dimension bounds for neural networks). The bounds above describe the asymptotic scaling of the VC-dimension with respect to the network size. The constants c_0, C_0, c_1, C_1 are typically not explicit and depend on the input dimension d. Consequently, when these bounds are combined with the generalization result in Theorem 11.1, the resulting sample-complexity estimates are often extremely large for modern neural networks. For this reason, VC-dimension bounds are mainly useful for establishing theoretical learnability and understanding how model capacity scales with architecture size, rather than for computing practical numerical generalization guarantees.

11.3 Exercises

11.1. Show that, if $\mathcal{H}$ shatters a set S, then it also shatters all the subsets of S.

11.2. Let $\mathscr{H} = \{\mathbf{1}_{[a,b]} : a, b \in \mathbb{R},\ a < b\}$. For an arbitrary set $C = \{c_1, c_2\}$ where $c_1 < c_2$, show that $\mathscr{H}$ shatters C. This implies VC-dim$(\mathscr{H}) \geq 2$. Do we really need to show that $\mathscr{H}$ shatters an *arbitrary* two-point set C, or is it sufficient to show that it shatters a *specific* two-point set to conclude VC-dim$(\mathscr{H}) \geq 2$?

11.3. Let $\mathscr{H}$ be an arbitrary finite hypothesis class. Show that VC-dim$(\mathscr{H}) \leq \log_2(\mathscr{H})$. Furthermore, show that the gap between VC-dim$(\mathscr{H})$ and $\log_2(\mathscr{H})$ can be arbitrarily large.

11.4. Determine the VC-dimension of $\mathscr{M}_1^L(\text{ReLU}; 1)$, the class of ReLU networks with width 1, depth L, and one input. (Hint: show that all functions in this class are monotone.)

11.5. Prove that $\ln x \leq ax - \ln a - 1$ for all $a, x > 0$. (Hint: prove that $e^y \geq 1 + y$ holds for all $y \in \mathbb{R}$ and take $y = ax - 1$ with $a, x > 0$. Alternatively, prove that $\log y \leq y - 1$ for $y > 0$.)

Chapter 12

Rademacher Complexity

There are different notions of complexity for hypothesis classes that can be used to reduce the analysis of infinite hypothesis classes to that of finite ones. To this end, we have introduced growth functions and VC-dimension in Chapter 11. In this chapter, we introduce another such notion, Rademacher complexity, and demonstrate its application to generalization analysis for both classification and regression tasks.

12.1 Rademacher Complexity

Definition 12.1 (Empirical Rademacher complexity)**.** Let $\mathcal{G}$ be a set of functions from Z (an arbitrary input domain) to some bounded interval $[a, b] \subseteq \mathbb{R}$. Let $S = (z_1, z_2, \ldots, z_m)$ be a sample set of size m from Z. The *empirical Rademacher complexity* of $\mathcal{G}$ with respect to S is defined as

$$\widehat{\mathrm{Rad}}_S(\mathcal{G}) = \mathbb{E}_\sigma \left[\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i g(z_i) \right], \tag{12.1}$$

where $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_m)$, and each σ_i, called a Rademacher variable, is an independent uniform random variable taking value in $\{-1, 1\}$.

Definition 12.2 (Rademacher complexity)**.** Let $\mathcal{D}$ be a distribution over Z. The

Rademacher complexity of $\mathcal{G}$ with respect to $\mathcal{D}$ for sample size m is defined as

$$\mathrm{Rad}_m(\mathcal{G}) = \mathbb{E}_{S\sim\mathcal{D}^m}[\widehat{\mathrm{Rad}}_S(\mathcal{G})]. \tag{12.2}$$

Intuitively, Rademacher complexity measures how well a function class can fit random noise. Specifically, it quantifies the ability of a hypothesis class to correlate with random ±1 labels, reflecting its capacity to fit arbitrary patterns in data.

12.2 Generalization Bounds from Rademacher Complexity

We first state the following generalization bound based on Rademacher complexity. The presentation follows (Mohri et al., 2018, Theorem 3.3). We make it slightly more general to allow the function bounds to be an arbitrary interval instead of being restricted to $[0, 1]$, clarify a few kep steps, and added a few more conclusions.

Theorem 12.1. *Let $\mathcal{G}$ be a family of functions from Z to $[a, b]$. Let $S = (z_1, z_2, \ldots, z_m)$ denote m samples drawn i.i.d. from Z. For any $\delta \in (0, 1)$, each of the following holds with probability at least $1 - \delta$ with respect to the draw of S:*

$$\mathbb{E}_{z\sim\mathcal{D}}[g(z)] \le \frac{1}{m}\sum_{i=1}^{m} g(z_i) + 2\,\mathrm{Rad}_m(\mathcal{G}) + (b-a)\sqrt{\frac{\log\frac{1}{\delta}}{2m}}, \tag{12.3}$$

$$\mathbb{E}_{z\sim\mathcal{D}}[g(z)] \le \frac{1}{m}\sum_{i=1}^{m} g(z_i) + 2\widehat{\mathrm{Rad}}_S(\mathcal{G}) + 3(b-a)\sqrt{\frac{\log\frac{2}{\delta}}{2m}}. \tag{12.4}$$

$$\frac{1}{m}\sum_{i=1}^{m} g(z_i) \le \mathbb{E}_{z\sim\mathcal{D}}[g(z)] + 2\,\mathrm{Rad}_m(\mathcal{G}) + (b-a)\sqrt{\frac{\log\frac{1}{\delta}}{2m}}, \tag{12.5}$$

$$\frac{1}{m}\sum_{i=1}^{m} g(z_i) \le \mathbb{E}_{z\sim\mathcal{D}}[g(z)] + 2\widehat{\mathrm{Rad}}_S(\mathcal{G}) + 3(b-a)\sqrt{\frac{\log\frac{2}{\delta}}{2m}}, \tag{12.6}$$

$$\left|\mathbb{E}_{z\sim\mathcal{D}}[g(z)] - \frac{1}{m}\sum_{i=1}^{m} g(z_i)\right| \le 2\,\mathrm{Rad}_m(\mathcal{G}) + (b-a)\sqrt{\frac{\log\frac{2}{\delta}}{2m}}, \tag{12.7}$$

$$\left|\mathbb{E}_{z\sim\mathcal{D}}[g(z)] - \frac{1}{m}\sum_{i=1}^{m} g(z_i)\right| \le 2\widehat{\mathrm{Rad}}_S(\mathcal{G}) + 3(b-a)\sqrt{\frac{\log\frac{4}{\delta}}{2m}}, \tag{12.8}$$

The proof relies on the following lemma, known as McDiarmid's inequality (see Appendix C).

Lemma 12.1 (McDiarmid's inequality)**.** *Let $X_1, \ldots, X_n$ be independent random variables taking values in some space $\mathcal{X}$, and let $f : \mathcal{X}^n \to \mathbb{R}$ be a function satisfying the bounded differences property: for some constants c_i,*

$$\sup_{x_1,\ldots,x_n,x_i'} \left| f(x_1, \ldots, x_i, \ldots, x_n) - f(x_1, \ldots, x_i', \ldots, x_n) \right| \leq c_i, \quad \forall i \in \{1, \ldots, n\}.$$

Define the expectation $\mu = \mathbb{E}[f(X_1, \ldots, X_n)]$. Then, for any $\epsilon > 0$,

$$\mathbb{P}\left(f(X_1, \ldots, X_n) - \mu \geq \epsilon\right) \leq \exp\left(\frac{-2\epsilon^2}{\sum_{i=1}^n c_i^2}\right),$$

$$\mathbb{P}\left(f(X_1, \ldots, X_n) - \mu \leq -\epsilon\right) \leq \exp\left(\frac{-2\epsilon^2}{\sum_{i=1}^n c_i^2}\right),$$

$$\mathbb{P}\left(|f(X_1, \ldots, X_n) - \mu| \geq \epsilon\right) \leq 2\exp\left(\frac{-2\epsilon^2}{\sum_{i=1}^n c_i^2}\right).$$

Note that Hoeffding's inequality (Lemma 10.1) is a special case of McDiarmid's inequality when $f(X_1, \ldots, X_n) = \sum_{i=1}^n X_i$.

Proof of Theorem 12.1. Let $S = (z_1, z_2, \ldots, z_m)$ denote any m samples drawn i.i.d. from Z. Given $g \in \mathcal{G}$, define the empirical average of g over S by

$$\hat{\mathbb{E}}_S[g] = \frac{1}{m}\sum_{i=1}^m g(z_i),$$

and denote the true expectation of g over $\mathcal{D}$ by $\mathbb{E}[g] = \mathbb{E}_{z\sim D}[g(z)]$. Define

$$\Delta(S) = \sup_{g\in\mathcal{G}} \left(\mathbb{E}[g] - \hat{\mathbb{E}}_S[g]\right).$$

Let S' be another set of m samples but only differ from S by one sample. We have

$$\Delta(S') - \Delta(S) \leq \sup_{g \in \mathcal{G}} (\hat{\mathbb{E}}_S[g] - \hat{\mathbb{E}}_{S'}[g]) = \sup_{g \in \mathcal{G}} \frac{g(z_i) - g(z_i')}{m} \leq \frac{b-a}{m},$$

where z_i and z_i' are the only pair of different samples in S and S'. By symmetry, we can also show that $\Delta(S') - \Delta(S) \leq \frac{b-a}{m}$, and therefore $|\Delta(S') - \Delta(S)| \leq \frac{b-a}{m}$. Applying McDiarmid's inequality (Lemma 12.1), we obtain that

$$\Delta(S) \leq \mathbb{E}_{S \sim \mathcal{D}^m}[\Delta(S)] + (b-a)\sqrt{\frac{\log \frac{1}{\delta}}{2m}} \tag{12.9}$$

holds with probability at least $1 - \delta$. The term $\mathbb{E}_{S \sim \mathcal{D}^m}[\Delta(S)]$ can be bounded as

$$\begin{aligned}
\mathbb{E}_{S \sim \mathcal{D}^m}[\Delta(S)] &= \mathbb{E}_{S \sim \mathcal{D}^m} \sup_{g \in \mathcal{G}} \left(\mathbb{E}[g] - \hat{\mathbb{E}}_S[g] \right) \\
&= \mathbb{E}_{S \sim \mathcal{D}^m} \sup_{g \in \mathcal{G}} \mathbb{E}_{S' \sim \mathcal{D}^m} \left[\hat{\mathbb{E}}_{S'}[g] - \hat{\mathbb{E}}_S[g] \right] \\
&\leq \mathbb{E}_{S \sim \mathcal{D}^m} \mathbb{E}_{S' \sim \mathcal{D}^m} \sup_{g \in \mathcal{G}} \left(\hat{\mathbb{E}}_{S'}[g] - \hat{\mathbb{E}}_S[g] \right) \\
&= \mathbb{E}_{S \sim \mathcal{D}^m} \mathbb{E}_{S' \sim \mathcal{D}^m} \sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{m} (g(z_i') - g(z_i)) \\
&= \mathbb{E}_{S \sim \mathcal{D}^m} \mathbb{E}_{S' \sim \mathcal{D}^m} \sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i (g(z_i') - g(z_i)), \\
&= \mathbb{E}_{\sigma} \mathbb{E}_{S \sim \mathcal{D}^m} \mathbb{E}_{S' \sim \mathcal{D}^m} \sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i (g(z_i') - g(z_i)),
\end{aligned} \tag{12.10}$$

where the second equality uses the fact that samples in S' are drawn i.i.d. from $\mathcal{D}$, the inequality follows from the monotonicity of integration and the definition of supremum. In the second last equality, we introduce the Rademacher variables $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_m)$, where each σ_i is independent and uniformly drawn from $\{-1, 1\}$. For each fixed σ, $\sigma_i = 1$ means the term being summed is not changed and $\sigma_i = -1$ means the term being summed swapped to $g(z_i) - g(z_i')$. Since S and S' are drawn i.i.d. from $\mathcal{D}$, swapping each pair (z_i, z_i') does not change their joint distribution. Hence introducing independent Rademacher variables σ_i, which encode such swaps, does not change the value of the overall expectation, and the

second last equality follows. This means the term remains constant over all choices of σ_i, and the final equality follows.

Now, again assuming that we can change the order of integration, we can further derive

$$\begin{aligned}
&\mathbb{E}_\sigma \mathbb{E}_{S\sim\mathcal{D}^m} \mathbb{E}_{S'\sim\mathcal{D}^m} \sup_{g\in\mathcal{G}} \frac{1}{m}\sum_{i=1}^m \sigma_i(g(z_i') - g(z_i)) \\
&\le \mathbb{E}_\sigma \mathbb{E}_{S'\sim\mathcal{D}^m} \sup_{g\in\mathcal{G}} \frac{1}{m}\sum_{i=1}^m \sigma_i g(z_i') + \mathbb{E}_\sigma \mathbb{E}_{S\sim\mathcal{D}^m} \sup_{g\in\mathcal{G}} \frac{1}{m}\sum_{i=1}^m -\sigma_i g(z_i), \\
&\le \mathbb{E}_{S'\sim\mathcal{D}^m} \mathbb{E}_\sigma \sup_{g\in\mathcal{G}} \frac{1}{m}\sum_{i=1}^m \sigma_i g(z_i') + \mathbb{E}_{S\sim\mathcal{D}^m} \mathbb{E}_\sigma \sup_{g\in\mathcal{G}} \frac{1}{m}\sum_{i=1}^m -\sigma_i g(z_i) \\
&= 2\,\mathrm{Rad}_m(\mathcal{G}).
\end{aligned} \tag{12.11}$$

Substituting (12.11) into (12.10), and then into (12.9), we obtain that

$$\Delta(S) \le 2\,\mathrm{Rad}_m(\mathcal{G}) + (b-a)\sqrt{\frac{\log\frac{1}{\delta}}{2m}} \tag{12.12}$$

holds with probability at least $1-\delta$, which is (12.3).

Next, we bound the gap between $\mathrm{Rad}_m(\mathcal{G})$ and $\widehat{\mathrm{Rad}}_S(\mathcal{G})$. Note that changing one sample in S can at most change $\widehat{\mathrm{Rad}}_S(\mathcal{G})$ by $(b-a)/m$. We can apply McDiarmid's inequality and obtain that, with probability $1-\delta/2$ over the draw of S,

$$\mathrm{Rad}_m(\mathcal{G}) \le \widehat{\mathrm{Rad}}_S(\mathcal{G}) + (b-a)\sqrt{\frac{\log\frac{2}{\delta}}{2m}}.$$

Restating (12.12) with $\delta/2$ in place of δ and using the union bound (Exercise 12.2), we can obtain (12.4).

We can define

$$\Delta(S) = \sup_{g\in\mathcal{G}} \left(\hat{\mathbb{E}}_S[g] - \mathbb{E}[g]\right).$$

and run the same argument as above to prove (12.5) and (12.6) (Exercise 12.3).

From the one-sided bounds (12.3) and (12.5), we can easily use the union bound to obtain (12.7). Similarly, we can obtain (12.8) from (12.4) and (12.6). We leave these as an exercise (Exercise 12.4). □

The generalization bounds in Theorem 12.1 based on Rademacher complexity are stated for generic functions. We can easily link it generalization bounds that quantify the gaps between the expected risk and the empirical risk of a learning problem.

Corollary 12.1. *Let X denote the input domain and Y denote the output domain. Let $L : Y \times Y \to [0, c]$ be a loss function. Let $\mathcal{H}$ be a hypothesis class from X to Y. Define*

$$\mathcal{G} = \{(x, y) \mapsto L(h(x), y) : h \in \mathcal{H}\}.$$

For every $\delta \in (0, 1)$ and integer $m \geq 1$, the following bounds hold for all $h \in \mathcal{H}$ with probability at least $1 - \delta$, over the draw of m samples S i.i.d. from a data distribution $\mathcal{D}$ over $X \times Y$,

$$\left|R_{\mathcal{D}}(h) - \hat{R}_S(h)\right| \leq 2\,\mathrm{Rad}_m(\mathcal{G}) + c\sqrt{\frac{\log \frac{2}{\delta}}{2m}}, \tag{12.13}$$

$$\left|R_{\mathcal{D}}(h) - \hat{R}_S(h)\right| \leq 2\widehat{\mathrm{Rad}}_S(\mathcal{G}) + 3c\sqrt{\frac{\log \frac{4}{\delta}}{2m}}, \tag{12.14}$$

Proof. We only focused on the two-sided bounds here. These bounds immediately follow from the bounds (12.7) and (12.8). □

In both Theorem 12.1 and Corollary 12.1, the bounds that involve $\widehat{\mathrm{Rad}}_S(\mathcal{G})$ are data-dependent in the sense that it explicitly depends on the sample set S. While such bounds can potentially be less conservative, computing the empirical Rademacher complexity can still be computationally challenging. By definition, it is given by

$$\widehat{\mathrm{Rad}}_S(\mathcal{G}) = \mathbb{E}_\sigma\left[\sup_{g \in \mathcal{G}} \frac{1}{m}\sum_{i=1}^{m} \sigma_i g(z_i)\right].$$

In fact, for a fixed σ, this is as difficult as maximizing the objective function $\frac{1}{m}\sum_{i=1}^{m} \sigma_i g(z_i)$. In the next section, we show how to relate Rademacher complexity to classical notions of complexity in the context of binary classification.

12.3 An Application to Binary Classification

We now connect the generalization bounds in Theorem 12.1 and Corollary 12.1 to learning guarantees for binary classification.

We first establish the following lemmas.

Lemma 12.2 (Massart's lemma)**.** *Let $A \subseteq \mathbb{R}^m$ be a finite set. Then*

$$\mathbb{E}_\sigma \left[\frac{1}{m} \sup_{a \in A} \sum_{i=1}^{m} \sigma_i a_i \right] \leq r(A) \frac{\sqrt{2 \log |A|}}{m}, \tag{12.15}$$

where $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_m)$ with each σ_i being an independent uniform random variable taking value in $\{-1, 1\}$, $a = (a_1, a_2, \ldots, a_m)$, $|A|$ is the cardinality of A, and $r(A)$ is the radius of A defined as $r(A) = \frac{1}{2} \max_{a,b \in A} |a - b|$.

Usually, Massart's lemma is stated as

$$\mathbb{E}_\sigma \left[\frac{1}{m} \sup_{a \in A} \sum_{i=1}^{m} \sigma_i a_i \right] \leq r \frac{\sqrt{2 \log |A|}}{m}, \tag{12.16}$$

where $r = \max_{a \in A} |a|$. For a proof, see (Mohri et al., 2018, Theorem 3.7). Here, we stated it with a bound involving the radius $r(A)$ instead. By the definition of radius, there exists a point $a^* \in \mathbb{R}^m$ such that $|a - a^*| \leq r(A)$ for all $a \in A$. It follows from (12.16) that

$$\begin{aligned} \mathbb{E}_\sigma \left[\frac{1}{m} \sup_{a \in A} \sum_{i=1}^{m} \sigma_i a_i \right] &= \mathbb{E}_\sigma \left[\frac{1}{m} \sup_{a \in A} \sum_{i=1}^{m} \sigma_i (a_i - a^*) \right] \\ &\leq \max_{a \in A} |a - a^*| \frac{\sqrt{2 \log |A|}}{m} \leq r(A) \frac{\sqrt{2 \log |A|}}{m}, \end{aligned}$$

which is (12.15).

Lemma 12.3 (Bounding Rademacher complexity with growth function)**.** *Let $\mathscr{H}$ be a hypothesis class from X to $\{0, 1\}$. Then, for each integer $m \geq 1$,*

$$\mathrm{Rad}_m(\mathscr{H}) \leq \frac{1}{2} \sqrt{\frac{2 \log \Pi_{\mathscr{H}}(m)}{m}}, \tag{12.17}$$

where $\Pi_{\mathcal{H}}$ is the growth function of $\mathcal{H}$.

Proof. Let $S = (x_1, x_2, \ldots, x_m)$ be a set of m samples drawn from X. Define

$$A = \{(h(x_1), h(x_2), \ldots, h(x_m)) : \ h \in \mathcal{H}\}.$$

Then $|A| \leq \Pi_{\mathcal{H}}(m)$ by the definition of the growth function. Since each $h(x_i)$ takes value in $\{0, 1\}$, the radius of A is bounded by $\frac{\sqrt{m}}{2}$. By Massart's lemma (Lemma 12.2), we have

$$\mathrm{Rad}_m(\mathcal{H}) = \mathbb{E}_S \mathbb{E}_\sigma \left[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i h(x_i) \right] = \mathbb{E}_S \mathbb{E}_\sigma \left[\sup_{a \in A} \frac{1}{m} \sum_{i=1}^{m} \sigma_i a_i \right] \leq \frac{1}{2} \sqrt{\frac{2 \log \Pi_{\mathcal{H}}(m)}{m}}.$$

□

We can now prove the following generalization theorem for binary classification.

Theorem 12.2. *Let $\mathcal{H}$ be a hypothesis class from X to $\{0, 1\}$ and the loss be 0-1. Let S denote a set of m samples. Then the following holds with probability $1 - \delta$ over draw of S:*

$$\left| R_{\mathcal{D}}(h) - \hat{R}_S(h) \right| \leq 2\,\mathrm{Rad}_m(\mathcal{H}) + \sqrt{\frac{\log \frac{2}{\delta}}{2m}} \leq \sqrt{\frac{2 \log \Pi_{\mathcal{H}}(m)}{m}} + \sqrt{\frac{\log \frac{2}{\delta}}{2m}}. \tag{12.18}$$

Proof. We first connect the Rademacher complexity of $\mathcal{H}$ with that of

$$\mathcal{G} = \{(x, y) \mapsto L(h(x), y) : \ h \in \mathcal{H}\},$$

where L is a 0-1 loss given by $|h(x) - y|$. Since both $h(x), y \in \{0, 1\}$, we can rewrite

$$L(h(x), y) = \frac{1 - (2y - 1)(2h(x) - 1)}{2} = \begin{cases} 0, & h(x) = y, \\ 1, & h(x) \neq y. \end{cases}$$

Denote the set of samples by $S = \{(x_i, y_i)\}_{i=1}^m$. We have

$$\mathrm{Rad}_m(\mathcal{G}) = \mathbb{E}_S \mathbb{E}_\sigma \left[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i L(h(x_i), y_i) \right]$$

$$
\begin{aligned}
&= \mathbb{E}_S \mathbb{E}_\sigma \left[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i \frac{1 - (2y_i - 1)(2h(x_i) - 1)}{2} \right] \\
&= \mathbb{E}_S \mathbb{E}_\sigma \left[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i \frac{(2y_i - 1)(2h(x_i) - 1)}{2} \right] \\
&= \mathbb{E}_S \mathbb{E}_\sigma \left[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i (2h(x_i) - 1) 2 \right] \\
&= \mathbb{E}_S \mathbb{E}_\sigma \left[\sup_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i h(x_i) \right] \\
&= \mathrm{Rad}_m(\mathcal{H}).
\end{aligned}
$$

The remaining of the proof follows from Corollary 12.1 and Lemma 12.3. □

Using Theorem 12.2, we can obtain an improved bound on uniform convergence (and therefore generalization) for binary classification based on the growth function (and consequently the VC-dimension).

Corollary 12.2 (Improved uniform convergence from growth function)**.** . *Let $\mathcal{H}$ be a hypothesis class from X to $\{0,1\}$ and the loss be 0-1. Let $\mathcal{D}$ be a distribution over $X \times \{0,1\}$. For any integer $m \geq 1$ and $\varepsilon \in (0,1)$, we have*

$$
\mathbb{P}_{S \sim \mathcal{D}^m}(\exists h \in \mathcal{H} \text{ s.t. } \left| R_{\mathcal{D}}(h) - \hat{R}_S(h) \right| \geq \varepsilon) \leq 2\Pi_{\mathcal{H}}(m) \exp\left(-\frac{2m\varepsilon^2}{5} \right). \tag{12.19}
$$

Proof. Using the elementary inequality $\sqrt{2a} + \sqrt{\frac{b}{2}} \leq \sqrt{\frac{5}{2}(a+b)}$ for $a, b \geq 0$, we obtain from (12.18) that

$$
\begin{aligned}
\left| R_{\mathcal{D}}(h) - \hat{R}_S(h) \right| &\leq \sqrt{\frac{2 \log \Pi_{\mathcal{H}}(m)}{m}} + \sqrt{\frac{\log \frac{2}{\delta}}{2m}} \\
&\leq \sqrt{\frac{5}{2} \frac{\log \Pi_{\mathcal{H}}(m) + \log \frac{2}{\delta}}{m}} = \sqrt{\frac{5 \log \left(\Pi_{\mathcal{H}}(m) \frac{2}{\delta} \right)}{2m}}
\end{aligned}
$$

holds for all $h \in \mathscr{H}$ with probability at least $1 - \delta$. For any $\varepsilon \in (0, 1)$, taking

$$\varepsilon = \sqrt{\frac{5 \log \left(\Pi_{\mathscr{H}}(m) \frac{2}{\delta}\right)}{2m}}$$

or equivalently

$$\delta = 2\Pi_{\mathscr{H}}(m) \exp\left(-\frac{2m\varepsilon^2}{5}\right),$$

we obtain (12.19). □

To see that Corollary 12.2 improves Theorem 11.2, compare (12.19) with (11.2), restated below:

$$\mathbb{P}_{S \sim \mathscr{D}^m}(\exists h \in \mathscr{H} \text{ s.t. } \left|R_{\mathscr{D}}(h) - \hat{R}_S(h)\right| \geq \varepsilon) \leq 4\Pi_{\mathscr{H}}(2m) \exp\left(-\frac{m\varepsilon^2}{8}\right).$$

By Sauer's lemma (Lemma 11.1), which gives $\Pi_{\mathscr{H}}(m) \leq (em/d)^d$, we can also write the generalization bound in Theorem 12.2 in terms of the VC-dimension.

Corollary 12.3. *Let $\mathscr{H}$ be a hypothesis class from X to $\{0, 1\}$ and the loss be 0-1. Assume that $d = \text{VC-dim}(\mathscr{H}) < \infty$. Let S denote a set of m samples. Then the following holds with probability $1 - \delta$ over draw of S:*

$$\left|R_{\mathscr{D}}(h) - \hat{R}_S(h)\right| \leq \sqrt{\frac{2d \log \frac{em}{d}}{m}} + \sqrt{\frac{\log \frac{2}{\delta}}{2m}}. \tag{12.20}$$

12.4 An Application to Learning Real-Valued Functions

We now focus on generalization for learning real-valued functions. Note that both Theorem 12.1 and Corollary 12.1 are already applicable to hypothesis classes of real-valued functions. The only link we need to establish is between the Rademacher complexity of the underlying hypothesis class $\mathscr{H}$ and that of

$$\mathscr{G} = \{(x, y) \mapsto L(h(x), y) : h \in \mathscr{H}\},$$

which involves the loss function L. The following lemma, known as Talagrand's contraction lemma (Ledoux and Talagrand, 1991), facilitates this connection. The following formulation and proof follow that in (Mohri et al., 2018, Lemma 5.7).

Lemma 12.4 (Talagrand's lemma)**.** *Let $\mathscr{H}$ be a hypothesis class of real-valued functions from X to $[a, b]$. Let S denote a sample set of size m from X. Let $\phi_1, \ldots, \phi_m$ be l-Lipschitz functions from $\mathbb{R}$ to $\mathbb{R}$ and $\sigma_1, \ldots, \sigma_m$ be Rademacher random variables and write $\sigma = (\sigma_1, \ldots, \sigma_m)$. We have*

$$\mathbb{E}_\sigma \sup_{h \in \mathscr{H}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i \phi_i(h(x_i)) \leq l \mathbb{E}_\sigma \sup_{h \in \mathscr{H}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i h(x_i). \tag{12.21}$$

In particular, let $\phi : \mathbb{R} \to \mathbb{R}$ to an l-Lipschitz function and define

$$\phi \circ \mathscr{H} = \left\{ \phi \circ h : h \in \mathscr{H} \right\}.$$

We have

$$\widehat{\mathrm{Rad}}_S(\phi \circ \mathscr{H}) \leq l \widehat{\mathrm{Rad}}_S(\mathscr{H}), \quad \mathrm{Rad}_m(\phi \circ \mathscr{H}) \leq l \, \mathrm{Rad}_m(\mathscr{H}).$$

Proof. We only prove $\widehat{\mathrm{Rad}}_S(\phi \circ \mathscr{H}) \leq l \widehat{\mathrm{Rad}}_S(\mathscr{H})$, which would imply $\mathrm{Rad}_m(\phi \circ \mathscr{H}) \leq l \, \mathrm{Rad}_m(\mathscr{H})$ immediately.

Fix some $S = (x_1, x_2, \ldots, x_m)$. By definition, we have

$$\begin{aligned} \widehat{\mathrm{Rad}}_S(\phi \circ \mathscr{H}) &= \mathbb{E}_\sigma \sup_{h \in \mathscr{H}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i \phi(h(x_i)) \\ &= \frac{1}{m} \mathbb{E}_{(\sigma_1, \ldots, \sigma_{m-1})} \mathbb{E}_{\sigma_m} \sup_{h \in \mathscr{H}} \left[\sum_{i=1}^{m-1} \sigma_i \phi(h(x_i)) + \sigma_m \phi(h(x_m)) \right], \end{aligned}$$

where $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_m)$ with each σ_i being a Rademacher random variable.

By the definition of supremum, for every $\varepsilon > 0$, there exists $h_1, h_2 \in \mathscr{H}$ such that

$$\sum_{i=1}^{m-1} \sigma_i \phi(h_1(x_i)) + \phi(h_1(x_m)) \geq \sup_{h \in \mathscr{H}} \left[\sum_{i=1}^{m-1} \sigma_i \phi(h(x_i)) + \phi(h(x_m)) \right] - \varepsilon \tag{12.22}$$

and

$$\sum_{i=1}^{m-1} \sigma_i \phi(h_2(x_i)) - \phi(h_2(x_m)) \geq \sup_{h \in \mathcal{H}} \left[\sum_{i=1}^{m-1} \sigma_i \phi(h(x_i)) - \phi(h(x_m)) \right] - \varepsilon. \tag{12.23}$$

By the definition of $\mathbb{E}_{\sigma_m}$, we have

$$\begin{aligned}
&\mathbb{E}_{\sigma_m} \sup_{h \in \mathcal{H}} \left[\sum_{i=1}^{m-1} \sigma_i \phi(h(x_i)) + \sigma_m \phi(h(x_m)) \right] \\
&\quad = \frac{1}{2} \sup_{h \in \mathcal{H}} \left[\sum_{i=1}^{m-1} \sigma_i \phi(h(x_i)) + \phi(h(x_m)) \right] + \frac{1}{2} \sup_{h \in \mathcal{H}} \left[\sum_{i=1}^{m-1} \sigma_i \phi(h(x_i)) - \phi(h(x_m)) \right]
\end{aligned}$$

Let $s = \operatorname{sgn}(h_1(x_m) - h_2(x_m))$. It follows from (12.22) and (12.23) that

$$\begin{aligned}
&\mathbb{E}_{\sigma_m} \sup_{h \in \mathcal{H}} \left[\sum_{i=1}^{m-1} \sigma_i \phi(h(x_i)) + \sigma_m \phi(h(x_m)) \right] \\
&\quad \leq \frac{1}{2} \left[\sum_{i=1}^{m-1} \sigma_i \phi(h_1(x_i)) + \sum_{i=1}^{m-1} \sigma_i \phi(h_2(x_i)) + \phi(h_1(x_m)) - \phi(h_2(x_m)) \right] + \varepsilon \\
&\quad \leq \frac{1}{2} \left[\sum_{i=1}^{m-1} \sigma_i \phi(h_1(x_i)) + \sum_{i=1}^{m-1} \sigma_i \phi(h_2(x_i)) + ls(h_1(x_m) - h_2(x_m)) \right] + \varepsilon \\
&\quad = \frac{1}{2} \left[\sum_{i=1}^{m-1} \sigma_i \phi(h_1(x_i)) + lsh_1(x_m) + \sum_{i=1}^{m-1} \sigma_i \phi(h_2(x_i)) - lsh_2(x_m) \right] + \varepsilon \\
&\quad \leq \frac{1}{2} \left[\sup_{h \in \mathcal{H}} \sum_{i=1}^{m-1} \sigma_i \phi(h(x_i)) + lsh(x_m) + \sup_{h \in \mathcal{H}} \sum_{i=1}^{m-1} \sigma_i \phi(h(x_i)) - lsh(x_m) \right] \\
&\quad = \mathbb{E}_{\sigma_m} \sup_{h \in \mathcal{H}} \left[\sum_{i=1}^{m-1} \sigma_i \phi(h(x_i)) + \sigma_m lh(x_m) \right] + \varepsilon.
\end{aligned}$$

Since this holds for all $\varepsilon > 0$, we have proved

$$\mathbb{E}_{\sigma_m} \sup_{h \in \mathcal{H}} \left[\sum_{i=1}^{m-1} \sigma_i \phi(h(x_i)) + \sigma_m \phi(h(x_m)) \right] \leq \mathbb{E}_{\sigma_m} \sup_{h \in \mathcal{H}} \left[\sum_{i=1}^{m-1} \sigma_i \phi(h(x_i)) + \sigma_m lh(x_m) \right].$$

Repeating the same argument for each σ_i ($i \neq m$), we can show that

$$\widehat{\mathrm{Rad}}_S(\phi \circ \mathscr{H}) = \mathbb{E}_\sigma \sup_{h \in \mathscr{H}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i \phi(h(x_i)) \leq \mathbb{E}_\sigma \sup_{h \in \mathscr{H}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i l h(x_i) = l\widehat{\mathrm{Rad}}_S(\mathscr{H}).$$

□

We can now state the following generalization theorem for learning real-valued functions.

Theorem 12.3. *Let $\mathscr{H}$ be a hypothesis class of real-valued functions from X to $Y = [a, b]$. Let $L : Y \times Y \longrightarrow [0, c]$ be the loss function. Assume that $L(\cdot, y)$ is l-Lipschitz for each fixed $y \in Y$. For every $\delta \in (0, 1)$ and integer $m \geq 1$, the following bounds hold for all $h \in \mathscr{H}$ with probability at least $1 - \delta$, over the draw of m samples S i.i.d. from a data distribution $\mathscr{D}$ over $X \times Y$,*

$$\left|R_{\mathscr{D}}(h) - \hat{R}_S(h)\right| \leq 2l\,\mathrm{Rad}_m(\mathscr{H}) + c\sqrt{\frac{\log \frac{2}{\delta}}{2m}}, \tag{12.24}$$

$$\left|R_{\mathscr{D}}(h) - \hat{R}_S(h)\right| \leq 2l\widehat{\mathrm{Rad}}_S(\mathscr{H}) + 3c\sqrt{\frac{\log \frac{4}{\delta}}{2m}}, \tag{12.25}$$

Proof. Define

$$\mathscr{G} = \{(x, y) \mapsto L(h(x), y) : h \in \mathscr{H}\}.$$

We aim to prove that $\widehat{\mathrm{Rad}}_S(\mathscr{G}) \leq l\widehat{\mathrm{Rad}}_S(\mathscr{H})$ and $\mathrm{Rad}_m(\mathscr{G}) \leq l\,\mathrm{Rad}_m(\mathscr{H})$. Let $S = \{(x_i, y_i)\}_{i=1}^m$. For each y_i, $\phi_i(\cdot) := L(\cdot, y_i)$ is l-Lipschitz. By (12.21) from Talagrand's lemma (Lemma 12.4), we have

$$\widehat{\mathrm{Rad}}_S(\mathscr{G}) = \mathbb{E}_\sigma \sup_{h \in \mathscr{H}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i \phi_i(h(x_i)) \leq l\mathbb{E}_\sigma \sup_{h \in \mathscr{H}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i h(x_i) = l\widehat{\mathrm{Rad}}_S(\mathscr{H}).$$

The conclusion follows from that of Corollary 12.1. □

Similar to the Rademacher generalization bound for binary classification, the bound (12.25) for regression (learning real-valued functions) is data-dependent, making it potentially less conservative. However, estimating the empirical Rademacher complexity of a given hypothesis class can be computationally challenging. Just

as the Rademacher complexity can be linked to the growth function and then to the VC-dimension of a hypothesis class for binary classification, one can generalize the notions of the growth function and VC-dimension to real-valued functions. This leads to the concepts of the covering number and the pseudo-dimension of a hypothesis class of real-valued functions.

We introduce the pseudo-dimension below.

Definition 12.3. *Let $\mathscr{H}$ be a hypothesis class from Z to $\mathbb{R}$. A set of points $S = \{z_1, z_2, \ldots, z_m\} \subseteq Z$ is said to be P-shattered by $\mathscr{H}$ if there exist $t_1, t_2, \ldots, t_m \in \mathbb{R}$ such that the set*

$$\left\{(\mathbf{1}_{h(z_1)>t_1}, \ldots, \mathbf{1}_{h(z_m)>t_m}) : h \in \mathscr{H}\right\}$$

has cardinality 2^m, where

$$\mathbf{1}_{h(z_i)>t_i} = \begin{cases} 1, & \text{if } h(z_i) > t_i, \\ 0, & \text{otherwise.} \end{cases}$$

If this holds, we say that $t_1, \ldots, t_m$ witness the P-shattering. The size of the largest set that can be P-shattered by $\mathscr{H}$ is defined to be the pseudo-dimension of $\mathscr{H}$, denoted by P-dim(H).

Define

$$\mathscr{G} = \left\{(z, t) \mapsto \mathbf{1}_{h(z)>t} : h \in \mathscr{H}\right\}.$$

By definition, one can verify that P-dim$(\mathscr{H})$ = VC-dim(G) (Exercise 12.5). Furthermore, recall that we can define the VC-dimension of real-valued functions (such as neural networks) directly as

$$\text{VC-dim}(\mathscr{H}) = \text{VC-dim}\left\{z \mapsto \mathbf{1}_{h(z)>\theta} : h \in \mathscr{H}\right\}$$

for some threshold $\theta \in \mathbb{R}$. We can easily show that VC-dim$(\mathscr{H}) \leq$ P-dim$(\mathscr{H})$ (Exercise 12.6).

12.5 Exercises

12.1. Is the empirical Rademacher complexity $\widehat{\text{Rad}}_S(\mathscr{G})$ always non-negative?

12.2. Verify the last step of obtaining (12.4) in the proof of Theorem 12.1 using the

union bound (see Appendix C.1.2) of probability.

12.3. Prove (12.5) and (12.6) by repeating the same argument in the proof of Theorem 12.1 with

$$\Delta(S) = \sup_{g \in \mathcal{G}} \left(\hat{\mathbb{E}}_S[g] - \mathbb{E}[g] \right).$$

12.4. Verify the bounds (12.7) and (12.8) in the proof of Theorem 12.1.

12.5. Let $\mathcal{H}$ be a hypothesis class from Z to $\mathbb{R}$. Define

$$\mathcal{G} = \left\{ (z, t) \mapsto \mathbf{1}_{h(z) > t} : h \in \mathcal{H} \right\}.$$

Show that P-dim$(\mathcal{H})$ = VC-dim(G).

12.6. Let $\mathcal{H}$ be a hypothesis class from Z to $\mathbb{R}$. Define

$$\text{VC-dim}(\mathcal{H}) = \text{VC-dim} \left\{ z \mapsto \mathbf{1}_{h(z) > \theta} : h \in \mathcal{H} \right\}$$

where $\theta \in \mathbb{R}$ is fixed. Show that VC-dim$(\mathcal{H}) \leq$ P-dim$(\mathcal{H})$.

Chapter 13

Norm-Based Bounds

One of the main quests in theoretical deep learning is to understand the ability of deep neural networks to generalize. Classical notions of neural network complexity (e.g., VC-dimension, growth function, and pseudo-dimension) explicitly and strongly depend on the size of the networks. These results do not explain the generalization of modern neural networks well because their sizes are often much larger than the number of training examples. In this chapter, we introduce generalization bounds that can be made independent of network size (at least not so explicitly).

13.1 Problem Formulation

We consider feedforward neural networks of the form

$$x \mapsto W_L \rho(W_{L-1} \rho(\dots W_1 x)), \tag{13.1}$$

where $\rho : \mathbb{R} \to \mathbb{R}$ is a Lipschitz continuous activation function applied elementwise. Without loss of generality, we assume ρ has Lipschitz constant 1 (otherwise, we can scale ρ and absorb the scaling into the weight matrix). For each $l = 1, \dots, L$, $W_i \in \mathbb{R}^{d_l \times d_{l-1}}$. We assume that $d_0 = d$ (input dimension) and $d_L = 1$ (output dimension). We also assume that ρ is (positive) homogeneous, i.e., $\rho(\lambda t) = \lambda \rho(t)$ for all $t \in \mathbb{R}$ and $\lambda \geq 0$. These assumptions are satisfied by the ReLU activation (i.e.,

$\rho(t) = \max(0, t)$).

For each positive integer l, we let $\mathcal{F}_l$ denote the class of functions that are representable by a network in (13.1) with $L = l$.

In Chapter 12, we saw that the Rademacher complexity can be used to bound the gap between the expected value and the empirical average with high probability. In turn, it provides a sample complexity bound for learning real-valued functions (Theorem 12.1).

In this chapter, we aim to bound the Rademacher complexity of the function class $\mathcal{F}_L$ described in (13.1).

We recall that for a matrix $W \in \mathbb{R}^{n\times k}$, the induced matrix 2-norm and the Frobenius norm are defined, respectively, as

$$\|W\| := \sup_{|x|=1} |Wx|, \quad \|W\|_F := \left(\sum_{i=1}^{n}\sum_{j=1}^{k} |W_{ij}|^2\right)^{1/2}.$$

13.2 A Contraction Lemma with Frobenius Norm

We need the following lemme from Golowich et al. (2020).

Lemma 13.1. *Let $\rho : \mathbb{R} \to \mathbb{R}$ be a 1-Lipschitz and homogeneous activation function applied elementwise. Let $\mathcal{F}$ denote any class of vector-valued functions on X. Let $g : \mathbb{R} \to [0, \infty)$ be convex and monotonically increasing. Then*

$$\mathbb{E}_\sigma \sup_{\substack{f\in\mathcal{F} \\ \|W\|_F\le R}} g\left(\left|\sum_{i=1}^{m} \sigma_i \rho(Wf(x_i))\right|\right) \le 2\mathbb{E}_\sigma \sup_{f\in\mathcal{F}} g\left(R\left|\sum_{i=1}^{m} \sigma_i f(x_i)\right|\right), \tag{13.2}$$

where $\sigma = (\sigma_1, \ldots, \sigma_m)$, with each σ_i being an independent Rademacher random variable and $(x_1, \ldots, x_m)$ are m samples from X.

13.3 Properties of Sub-Gaussian Variables

We define sub-Gaussian random variables and provide an equivalent characterization using tail bounds.

Definition 13.1. Consider a random variable X with mean $\mathbb{E}[X] = \mu$. We say X is *sub-Gaussian* with variance proxy σ^2 if there exists some $\sigma > 0$ such that

$$\mathbb{E}[e^{\lambda(X-\mu)}] \leq e^{\frac{\sigma^2\lambda^2}{2}} \tag{13.3}$$

for all $\lambda \in \mathbb{R}$.

Proposition 13.1. *A random variable X is sub-Gaussian with variance proxy σ^2 if and only if*

$$\mathbb{P}[|X-\mu| \geq t] \leq 2e^{-\frac{t^2}{2\sigma^2}}, \quad \forall t \geq 0, \tag{13.4}$$

where $\mu = \mathbb{E}[X]$.

Proof. We first prove the "only if" part. For any $t \geq$. The technique we employ is commonly referred to as the *Chernoff bound*. By Markov's inequality (Proposition C.2 in Appendix C) and the sub-Gaussianity (13.9), we have, for any $\lambda > 0$,

$$\mathbb{P}(X-\mu \geq t) = \mathbb{P}(e^{\lambda(X-\mu)} \geq e^{\lambda t}) \leq \frac{\mathbb{E}[e^{\lambda(X-\mu)}]}{e^{\lambda t}} \leq e^{\frac{\sigma^2\lambda^2}{2}-\lambda t}.$$

Choosing $\lambda = \frac{t}{\sigma^2}$ minimizes the right-hand side and leads to the bound

$$\mathbb{P}(X-\mu \geq t) \leq e^{-\frac{t^2}{2\sigma^2}}.$$

Similarly, we can show that (Exercise 13.1)

$$\mathbb{P}(X-\mu \leq -t) \leq e^{-\frac{t^2}{2\sigma^2}}.$$

The conclusion follows from the union bound.

For the "if" part and additional equivalent characterizations of sub-Gaussianity, we refer the readers to (Wainwright, 2019, Theorem 2.6). □

13.4 Size-Independent Bounds for Rademacher Complexity

We make the following assumptions for functions in $\mathcal{F}_l$. The domain is restricted to X and we have $|x| \leq B$ for all $x \in X$. All the weight matrices satisfy

$$\|W_l\| \leq M_l, \quad \|W_l\|_F \leq F_l, \quad l = 1, 2, \ldots, L,$$

where $M_l \leq F_l$ for each l. Furthermore, denote

$$P_M(l) = \Pi_{i=1}^{l} M_i, \quad P_F(l) = \Pi_{i=1}^{l} F_i, \quad R(l) = P_M(l)/P_F(l).$$

It is clearly that $R(L) \leq R(L-1) \leq \cdots R(1) \leq R(0) := 1$. For notational consistency, we also let $P_M(0) = P_F(0) = 1$.

We prove the following theorem from Sellke (2024), which slightly improves the result in Golowich et al. (2020).

Theorem 13.1. *Consider the class of neural networks $\mathcal{F}_L$ defined by (13.1). For any set of m samples $S = (x_1, \ldots, x_m)$ from X, we have*

$$\widehat{\mathrm{Rad}}_S(\mathcal{F}_L) \leq \frac{2\sqrt{2\log 2} B P_F(L) \sqrt{\sum_{l=0}^{L-1} R(l)}}{\sqrt{m}}.$$

Since the right-hand side does not depend on S, the same bound holds for $\mathrm{Rad}_m(\mathcal{F}_L)$.

Proof. Without loss of generality, we can assume $B = 1$. Fix any set of m samples $S = (x_1, \ldots, x_m)$ from X. Choose any $0 = l_0 < l_1 < \ldots < l_k = L$. We first prove the following

$$\widehat{\mathrm{Rad}}_S(\mathcal{F}_L) \leq \frac{P_F(L)\sqrt{2\log 2} \sum_{j=1}^{k} R(l_{j-1})\sqrt{l_j - l_{j-1}}}{\sqrt{m}}. \tag{13.5}$$

Let

$$X_j = \frac{1}{m} \sup_{f \in \mathcal{F}_{l_j}} \sum_{i=1}^{m} \sigma_i f(x_i)$$

and $Y_j = X_j / P_F(l_j)$ for $j = 1, \ldots, k$. Note that both X_j and Y_j depend on the

Rademacher random variables in σ. Then (13.5) is equivalent to

$$\mathbb{E}_\sigma Y_k \leq \frac{\sqrt{2\log 2}\sum_{j=1}^{k} R(l_{j-1})\sqrt{l_j - l_{j-1}}}{\sqrt{m}}. \tag{13.6}$$

Let $\mathcal{F}_0$ denote the singleton set with the identity function. Then clearly $\mathbb{E}_\sigma Y_0 = 0$. We prove (13.6) by showing that

$$\mathbb{E}_\sigma Y_{j+1} \leq \mathbb{E}_\sigma Y_j + \frac{\sqrt{2\log 2}R(l_j)\sqrt{l_{j+1} - l_j}}{\sqrt{m}}. \tag{13.7}$$

holds for all $j = 0, \ldots k-1$.

For any positive λ_j, using Lemma 13.1, we can show that (Exercise 13.2)

$$\mathbb{E}_\sigma\left[\exp(\lambda_j Y_{j+1})\right] \leq 2^{l_{j+1}-l_j}\mathbb{E}_\sigma\left[\exp(\lambda_j Y_j)\right], \tag{13.8}$$

By the definition of P_M, we know that for all $|x| \leq 1$ and $f \in \mathcal{F}_{l_j}$, we have $|f(x)| \leq P_M(l_j)$. Changing one σ_i will change X_j by at most $2P_M(l_j)/m$. It follows from McDiarmid's inequality (Lemma 12.1) that, for all $t \geq 0$,

$$\mathbb{P}\left(\left|X_j - \mathbb{E}[X_j]\right| \geq t\right) \leq 2\exp\left(-\frac{mt^2}{2P_M(l_j)^2}\right),$$

which gives, for $Y_j = X_j/P_F(l_j)$,

$$\mathbb{P}\left(\left|Y_j - \mathbb{E}[Y_j]\right| \geq t\right) \leq 2\exp\left(-\frac{mP_F(l_j)^2t^2}{2P_M(l_j)^2}\right).$$

By the characterization of sub-Gaussian variables with variance proxy $P_M(l_j)^2/m$ (Proposition 13.1) that

$$\log\mathbb{E}\left[\exp\left(\lambda_j(Y_j - \mathbb{E}[Y_j])\right)\right] \leq \frac{P_M(l_j)^2\lambda_j^2}{2P_F(l_j)^2 m} \tag{13.9}$$

which gives

$$\log \mathbb{E}_\sigma \left[\exp\left(\lambda_j Y_j\right)\right] \leq \frac{P_M(l_j)^2 \lambda_j^2}{2P_F(l_j)^2 m} + \lambda_j \mathbb{E}_\sigma Y_j. \tag{13.10}$$

By Jensen's inequality (Proposition C.1 in Appendix C.1.4), (13.8), and (13.10), we have

$$\begin{aligned}
\mathbb{E}Y_{j+1} &\leq \frac{1}{\lambda_j} \log \mathbb{E}\left[\exp(\lambda_i Y_{j+1})\right] \\
&\leq \frac{1}{\lambda_j}\left(\log \mathbb{E}_\sigma\left[\exp(\lambda_j Y_j)\right] + (l_{j+1} - l_j)\log 2\right) \\
&\leq \frac{1}{\lambda_j}\left(\frac{P_M(l_j)^2\lambda_j^2}{2P_F(l_j)^2 m} + \lambda_j \mathbb{E}_\sigma Y_j + (l_{j+1} - l_j)\log 2\right) \\
&= \mathbb{E}_\sigma Y_j + \frac{P_M(l_j)^2 \lambda_j}{2P_F(l_j)^2 m} + \frac{(l_{j+1} - l_j)\log 2}{\lambda_j}.
\end{aligned}$$

Choosing $\lambda_j = \frac{P_F(l_j)\sqrt{2(l_{j+1}-l_j)\log 2}}{P_M(l_j)}$, we obtain

$$\mathbb{E}Y_{j+1} \leq \mathbb{E}_\sigma Y_j + \frac{P_F(l_j)\sqrt{2(l_{j+1} - l_j)\log 2}}{P_M(l_j)},$$

which proves (13.7).

Finally, we show that the bound in (13.5) can be optimized by strategically choosing l_j. We can take l_j to be the minimum l_j such that $R(d_j) \leq 2^{-j}$. This can be done at least for $j = 0$. If no such l_j can be found, we terminate with $l_k = L$. It follows from the definition of l_j and the Cauchy-Schwarz inequality that

$$\begin{aligned}
\sum_{j=1}^{k} R(l_{j-1})\sqrt{l_j - l_{j-1}} &\leq 2\sum_{j=1}^{k} \frac{\sqrt{l_j - l_{j-1}}}{2^j} \leq 2\sum_{j=1}^{k} \frac{\sqrt{l_j - l_{j-1}}}{\sqrt{2}^j}\frac{1}{\sqrt{2}^j} \leq 2\sqrt{\sum_{j=1}^{k}\frac{l_j - l_{j-1}}{2^j}\sum_{j=1}^{k}\frac{1}{2^j}} \\
&\leq 2\sqrt{\sum_{j=1}^{k}\frac{l_j - l_{j-1}}{2^j}} \leq 2\sqrt{\sum_{j=1}^{k}\sum_{l=l_{j-1}}^{l_j - 1} R(l)} \leq 2\sqrt{\sum_{l=0}^{L-1} R(l)}.
\end{aligned}$$

The proof is complete. □

Remark 13.1. While it is claimed that such bounds are "size-independent" (Golowich et al., 2020; Sellke, 2024), both the product of Frobenius norms and the term $\sqrt{\sum_{l=0}^{L-1} R(l)}$ can still depend on the depth L of the network. In particular, the product can potentially depend exponentially on L. It would be of interest to investigate under what conditions truly depth-independent bounds can be obtained. Interested readers can refer to Golowich et al. (2020) for further discussion.

13.5 Exercises

13.1. In the proof of Proposition 13.1, show that, if X is sub-Gaussian with variance proxy σ^2, then

$$\mathbb{P}(X - \mu \leq -t) \leq e^{-\frac{t^2}{2\sigma^2}}.$$

13.2. Prove (13.8) using Lemma 13.1.

Chapter 14

PAC-Bayes Bounds

All the generalization bounds we have seen so far apply uniformly to all hypotheses in a given class. PAC-Bayesian bounds offer a powerful alternative by analyzing the generalization of learning algorithms that output randomized predictors. Rather than bounding the error of each hypothesis uniformly, they control the expected error under a posterior distribution over hypotheses, relative to a prior. This framework naturally integrates both data-dependent learning and prior knowledge, yielding sharp, non-vacuous guarantees—even for complex or high-capacity models such as deep neural networks.

14.1 PAC-Bayes Bounds for Generalization

Let $\mathscr{H}$ be a hypothesis class from X to Y. Let P be some prior distribution over $\mathscr{H}$. Instead of learning a single hypothesis, we aim to learn a distribution over $\mathscr{H}$ from data. As before, we assume that there exists a loss function L such that, given a sample $z = (x, y) \in Z := X \times Y$, we can evaluate $L(h(x), y)$. Let $S = \{(x_i, y_i)\}_{i=1}^m$ be a set of samples, and let $\mathscr{D}$ denote the data distribution over Z. We define the expected and empirical risk, respectively, as

$$R_{\mathscr{D}}(h) = \mathbb{E}_{z\sim\mathscr{D}}[L(h(x), y)], \quad \hat{R}_S(h) = \frac{1}{m}\sum_{i=1}^{m} L(h(x_i), y_i).$$

Let Q be any distribution over $\mathscr{H}$. With a slight abuse of notation, we define

$$R_{\mathscr{D}}(Q) = \mathbb{E}_{h\sim Q}[R_{\mathscr{D}}(h)], \quad \hat{R}_S(Q) = \mathbb{E}_{h\sim Q}[\hat{R}_S(h)].$$

Theorem 14.1. *Let $\mathscr{H}$ be a hypothesis class from X to Y. Let $\mathscr{D}$ be a data distribution over $X \times Y$. Assume that the loss function L takes value in $[0,1]$. Let P be a prior distribution over $\mathscr{H}$ and $\delta \in (0,1)$. Let $S = \{(x_i, y_i)\}_{i=1}^m$ be a set of samples. Then, with probability at least $1-\delta$ over drawing of m samples according to $\mathscr{D}^m$, we have, for all distributions Q over $\mathscr{H}$,*

$$R_{\mathscr{D}}(Q) \le \hat{R}_S(Q) + \sqrt{\frac{D_{KL}(Q \parallel P) + \log\frac{1}{\delta}}{2m}}, \tag{14.1}$$

where $D_{KL}(Q \parallel P)$ is the Kullback-Leibler (KL) divergence defined as

$$D_{KL}(Q \parallel P) = \mathbb{E}_{h\sim Q}\left[\log \frac{dQ}{dP}(h)\right], \tag{14.2}$$

where $\frac{dQ}{dP}$ is the Radon–Nikodym derivative[1] of Q with respect to P.

Proof. For each $h \in \mathscr{H}$, define

$$\Delta(h) = R_{\mathscr{D}}(h) - \hat{R}_S(h).$$

Then, for each $\lambda > 0$ and all distributions Q over $\mathscr{H}$, we have

$$\begin{aligned}
\mathbb{E}_{h\sim Q}[\lambda\Delta(h)] - D_{\mathrm{KL}}(Q \parallel P) &= \mathbb{E}_{h\sim Q}\left[\lambda\Delta(h) - \log\frac{dQ}{dP}(h)\right] \\
&= \mathbb{E}_{h\sim Q}\log\left(e^{\lambda\Delta(h)}\frac{dQ}{dP}(h)\right) \\
&\le \log\mathbb{E}_{h\sim Q}\left(e^{\lambda\Delta(h)}\frac{dQ}{dP}(h)\right) \\
&= \log\mathbb{E}_{h\sim P}e^{\lambda\Delta(h)},
\end{aligned} \tag{14.3}$$

[1] The Radon–Nikodym derivative $\frac{dQ}{dP}$ is a generalization of the density ratio $\frac{q}{p}$, used to relate two probability measures Q and P. It is well-defined when Q is absolutely continuous with respect to P (i.e., $P(A) = 0 \Rightarrow Q(A) = 0$). It allows change of measure: expectations under Q can be computed as weighted expectations under P. If both have densities, then $\frac{dQ}{dP}(h) = \frac{q(h)}{p(h)}$.

where the inequality is Jensen's inequality applied to the concave function log, and the last equality is a change of probability measure.

We now aim to find a high-probability bound for $\mathbb{E}_{h\sim P} e^{\lambda\Delta(h)}$. Since P is a prior distribution that does not depend on S, we have

$$\mathbb{E}_S \mathbb{E}_{h\sim P} e^{\lambda\Delta(h)} = \mathbb{E}_{h\sim P} \mathbb{E}_S [e^{\lambda\Delta(h)}]. \tag{14.4}$$

By Hoeffding's inequality (Lemma 10.1), we have

$$\mathbb{P}(|\Delta(h)| \geq t) \leq 2\exp\left(-\frac{2t^2}{m}\right) = 2\exp\left(-\frac{t^2}{2(m/4)}\right) \tag{14.5}$$

for all $t \geq 0$. By the characterization of sub-Gaussian random variables (Proposition 13.1), $\Delta(h)$ is a sub-Gaussian random variable with variance proxy $\sigma^2 = m/4$. Consequently, we have

$$\mathbb{E}_S[e^{\lambda\Delta(h)}] \leq \exp\left(\frac{\sigma^2\lambda^2}{2}\right) = \exp\left(\frac{\lambda^2}{8m}\right), \tag{14.6}$$

which, together with (14.4), implies

$$\mathbb{E}_S \mathbb{E}_{h\sim P} e^{\lambda\Delta(h)} \leq \exp\left(\frac{\lambda^2}{8m}\right). \tag{14.7}$$

By Markov's inequality (Proposition C.2 in Appendix C), we have, for any $\varepsilon > 0$,

$$\mathbb{P}(\mathbb{E}_{h\sim P} e^{\lambda\Delta(h)} \geq \varepsilon) \leq \frac{\mathbb{E}_S \mathbb{E}_{h\sim P} e^{\lambda\Delta(h)}}{\varepsilon} \leq \frac{\exp\left(\frac{\lambda^2}{8m}\right)}{\varepsilon}$$

It follows that, for any $\delta \in (0,1)$, setting $\varepsilon = \frac{1}{\delta}\exp\left(\frac{\lambda^2}{8m}\right)$, we have

$$\mathbb{E}_{h\sim P} e^{\lambda\Delta(h)} \leq \frac{1}{\delta}\exp\left(\frac{\lambda^2}{8m}\right)$$

with probability at least $1-\delta$ over the choice of S. Substituting this back to (14.3)

and by rearrangement, we obtain

$$\mathbb{E}_{h\sim Q}[\Delta(h)] \leq \frac{D_{\mathrm{KL}}(Q \parallel P) + \log \frac{1}{\delta}}{\lambda} + \frac{\lambda}{8m}$$

holds for all $\lambda > 0$ and all distributions Q over $\mathscr{H}$. Picking $\lambda = \sqrt{8m(D_{\mathrm{KL}}(Q \parallel P) + \log \frac{1}{\delta})}$ leads to the desired bound (14.1). □

14.2 Non-Vacuous PAC-Bayes Bounds for Deep Neural Networks

With the PAC-Bayes bound (14.1), the learning objective becomes: given a prior P, return a posterior Q that minimizes the function

$$\hat{R}_S(Q) + \sqrt{\frac{D_{\mathrm{KL}}(Q \parallel P) + \log \frac{1}{\delta}}{2m}}.$$

In other words, we aim to jointly minimize the empirical loss of Q on the sample and the KL divergence between Q and P. This idea of optimizing the PAC-Bayes bound directly is explored by Dziugaite and Roy (2017), where it is shown that non-vacuous generalization bounds can be obtained for deep neural networks. This contrasts sharply with classical complexity measures such as VC dimension, which often yield vacuous bounds in overparameterized settings. Their approach involves parameterizing Q as a Gaussian distribution over network weights and minimizing the PAC-Bayes bound using stochastic optimization, demonstrating that principled, data-dependent generalization guarantees are feasible even in large-scale models.

Appendix A

Useful Results for Approximation

A.1 Weierstrass Approximation Theorem

Lemma A.1 (Weierstrass Approximation Theorem)**.** *Suppose f is a continuous real-valued function defined on the real interval $[a, b]$. For every $\epsilon > 0$, there exists a polynomial p such that for all $x \in [a, b]$, we have*

$$|f(x) - p(x)| < \epsilon.$$

For a proof, see (Davidson and Donsig, 2010, Chapter 10.3).

A.2 Approximations of Continuous Function by Mollification

Define $\phi \in \mathscr{C}^\infty(\mathbb{R}^n)$ by

$$\phi(x) := \begin{cases} C \exp\left(\dfrac{1}{|x|^2 - 1}\right), & |x| < 1, \\ 0, & |x| \geq 1. \end{cases}$$

The constant $C > 0$ is such that $\int_{\mathbb{R}^n} \phi(x)dx = 1$. This is called a *standard mollifier*. For each $\varepsilon > 0$, define

$$\phi_\varepsilon(x) := \frac{1}{\varepsilon^n}\phi\left(\frac{x}{\varepsilon}\right).$$

Then $\phi_\varepsilon \in \mathscr{C}^\infty(\mathbb{R}^n)$, $\int_{\mathbb{R}^n} \phi_\varepsilon(x)dx = 1$, and ϕ_ε vanishes outside of $B_\varepsilon = \{x \in \mathbb{R}^n : |x| < \varepsilon\}$.

The mollification of a function $\sigma : \mathbb{R}^n \to \mathbb{R}$ by ϕ_ε (i.e., the convolution of σ with the mollifier ϕ_ε) is defined as

$$\sigma_{\phi_\varepsilon}(x) = (\sigma * \phi_\varepsilon)(x) = \int_{-\infty}^{\infty} \sigma(x-y)\phi(y)dt,$$

which is well defined if σ is locally integrable (i.e., $\int_M |\sigma(x)|\, dx < \infty$ on every compact set $M \subseteq \mathbb{R}^n$). This is clearly satisfied by $\sigma \in \mathscr{C}(\mathbb{R}^n)$.

Theorem A.1 (Approximation of Continuous Function by Mollification)**.** *Let $\sigma \in \mathscr{C}(\mathbb{R}^n)$. We have $\sigma_{\phi_\varepsilon} \to \sigma$ uniformly on compact subsets of $\mathbb{R}^n$ as $\varepsilon \to 0$.*

For a proof, see (Evans, 2010, Theorem 7(iii), Appendix C.5).

Appendix B

Taylor's Theorem

We recall the multi-index notation as follows. An n-dimensional multi-index is an n-tuple of nonnegative integers $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_n) \in (\mathbb{Z}_{\geq 0})^n$. A multi-index α can be used to denote the corresponding partial derivative of a function f with n variables as

$$D^\alpha f = \frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2} \ldots \partial x_n^{\alpha_n}},$$

where $|\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_n$ is the order of differentiation. We can also use the multi-index α to define the factorial

$$\alpha! = \alpha_1! \alpha_2! \ldots \alpha_n!,$$

and the power

$$x^\alpha = x_1^{\alpha_1} x_2^{\alpha_2} \ldots x_n^{\alpha_n},$$

for $x = (x_1, x_2, \ldots, x_n)$.

Let $D \subseteq \mathbb{R}^n$ be an open set. Let $f : D \to \mathbb{R}^m$ be a function that is k-times differentiable at a point $a \in D$. The **Taylor polynomial** of f at a is given by

$$P_k(x, a) = \sum_{|\alpha| \leq k} \frac{D^\alpha f(a)}{\alpha!} (x - a)^\alpha.$$

Define the remainder of the Taylor approximation as

$$R_k(x,a) = f(x) - P_k(x,a).$$

Theorem B.1 (Approximation by Taylor Polynomial)**.** *Let $D \subseteq \mathbb{R}^n$ be an open set. Let $f : D \to \mathbb{R}^m$ be a function that is k-times differentiable at $a \in D$. We have*

$$\lim_{x \to a} \frac{R_k(x,a)}{|x-a|^k} = 0.$$

This result is also known as Peano's theorem. For a direct proof, see Facenda Aguirre (1989).

We also quote the following version of Taylor's theorem under slightly stronger assumptions; see Folland (2005).

Theorem B.2. *Suppose that $f : \mathbb{R}^n \to \mathbb{R}^m$ is of class C^{k+1} on an open convex set $S \subseteq \mathbb{R}^n$. Then, for any $x, a \in S$, we have*

$$f(x) = P_k(x,a) + R_k(x,a) = \sum_{|\alpha| \le k} \frac{D^\alpha f(a)}{\alpha!}(x-a)^\alpha + R_k(x,a), \tag{B.1}$$

where

$$R_k(x,a) = (k+1) \sum_{|\alpha|=k+1} \frac{(x-a)^\alpha}{\alpha!} \int_0^1 (1-t)^k D^\alpha f(a + t(x-a))dt.$$

We state two special cases of the theorems above as follows.

Corollary B.1. *Let $f : \mathbb{R}^n \to \mathbb{R}$ be twice differentiable at $a \in \mathbb{R}^n$. Then we have*

$$\lim_{x \to a} \frac{f(x) - f(a) - \nabla f(a) \cdot (x-a) - \frac{1}{2}(x-a)^\top \nabla^2 f(a)(x-a)}{|x-a|^2} = 0.$$

This follows Theorem B.1 with $k = 2$.

Corollary B.2. *Let $f : \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable on $\mathbb{R}^n$. Then we have*

$$f(x) = f(a) + \nabla f(a) \cdot (x - a) + \int_0^1 (x - a)^\top (1 - t) \nabla^2 f(a + t(x - a))(x - a) dt,$$

and

$$\nabla f(x) = \nabla f(a) + \int_0^1 \nabla^2 f(a + t(x - a))(x - a) dt.$$

This follows Theorem B.2 with $k = 1$ for f and with $k = 0$ for ∇f, respectively.

Appendix C

Probability and Concentration Inequalities

C.1 Basic Probability Facts

We shall not introduce the measure-theoretic foundation of probability but instead recall the following basic facts.

C.1.1 Probability Space

A **probability space** consists of three components:

$$(\Omega, \mathcal{F}, P),$$

where:

- Ω is the *sample space*, the set of all possible outcomes.
- $\mathcal{F}$ is a collection of subsets of Ω, called *events*, for which probabilities are defined.
- $P : \mathcal{F} \to [0, 1]$ is a *probability function* satisfying:
  (1) $P(\Omega) = 1$ (certainty),
  (2) $P(A) \geq 0$ for all $A \in \mathcal{F}$ (non-negativity),

(3) For any disjoint events $A_1, A_2, \ldots$, we have countable additivity:

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

C.1.2 Basic Properties of Probability

We can derive several fundamental properties of probability from its definition:

- **Monotonicity:** If $A \subseteq B$, then

$$P(A) \leq P(B).$$

- **Union Bound (Boole's Inequality):** For any events $A_1, A_2, \ldots, A_n$,

$$P\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} P(A_i).$$

C.1.3 Random Variables and Expectation

A **random variable** is a function $X : \Omega \to \mathbb{R}$ that assigns a numerical value to each outcome in Ω. The **expectation** of X, denoted $\mathbb{E}[X]$, is defined as

$$\mathbb{E}[X] = \int_{\Omega} X \, dP.$$

For discrete random variables, this corresponds to

$$\mathbb{E}[X] = \sum_{x} x P(X = x),$$

while for continuous random variables with density f_X, it takes the form

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx.$$

The **variance** of X, which measures its spread, is given by

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

C.1.4 Basic Probability Inequalities

We first sate Jensen's inequality.

Proposition C.1 (Jensen's Inequality)**.** *Let $\phi : \mathbb{R} \to \mathbb{R}$ be a convex function and let X be a random variable such that $\mathbb{E}[X]$ exists. Then*

$$\phi(\mathbb{E}[X]) \leq \mathbb{E}[\phi(X)].$$

If ϕ is concave, the inequality is reversed:

$$\phi(\mathbb{E}[X]) \geq \mathbb{E}[\phi(X)].$$

We now state some fundamental probability inequalities, which will be used later for proving concentration bounds.

Proposition C.2 (Markov's Inequality)**.** *Let X be a non-negative random variable, and let $a > 0$. Then,*

$$\Pr(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Proposition C.3 (Chebyshev's Inequality)**.** *Let X be a random variable with finite expectation and variance. Then, for any $b > 0$,*

$$\Pr(|X - \mathbb{E}[X]| \geq b) \leq \frac{Var(X)}{b^2}.$$

Proposition C.4 (Hoeffding's Inequality)**.** *Let X_i, $i = 1, \ldots, m$, be independent random variables taking values in $[a_i, b_i]$. Then, for any $t > 0$,*

$$\mathbb{P}\left(S_m - \mathbb{E}\left[S_m\right] \geq t\right) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^m (b_i - a_i)^2}\right), \tag{C.1}$$

$$\mathbb{P}\left(S_m - \mathbb{E}\left[S_m\right] \leq -t\right) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^m (b_i - a_i)^2}\right), \tag{C.2}$$

$$\mathbb{P}\left(|S_m - \mathbb{E}\left[S_m\right]| \geq t\right) \leq 2\exp\left(-\frac{2t^2}{\sum_{i=1}^m (b_i - a_i)^2}\right), \tag{C.3}$$

where $S_m = \sum_{i=1}^m X_i$ and $\mathbb{E}[S_m]$ is the expectation of S_m.

Proposition C.5 (McDiarmid's inequality)**.** *Let $X_1, \ldots, X_n$ be independent random variables taking values in some space $\mathcal{X}$, and let $f : \mathcal{X}^n \to \mathbb{R}$ be a function satisfying the bounded differences property: for some constants c_i,*

$$\sup_{x_1,\ldots,x_n,x_i'} \left| f(x_1, \ldots, x_i, \ldots, x_n) - f(x_1, \ldots, x_i', \ldots, x_n) \right| \leq c_i, \quad \forall i \in \{1, \ldots, n\}.$$

Define the expectation $\mu = \mathbb{E}[f(X_1, \ldots, X_n)]$. Then, for any $\epsilon > 0$,

$$\mathbb{P}\left(f(X_1, \ldots, X_n) - \mu \geq \epsilon\right) \leq \exp\left(\frac{-2\epsilon^2}{\sum_{i=1}^n c_i^2}\right),$$

$$\mathbb{P}\left(f(X_1, \ldots, X_n) - \mu \leq -\epsilon\right) \leq \exp\left(\frac{-2\epsilon^2}{\sum_{i=1}^n c_i^2}\right),$$

$$\mathbb{P}\left(|f(X_1, \ldots, X_n) - \mu| \geq \epsilon\right) \leq 2\exp\left(\frac{-2\epsilon^2}{\sum_{i=1}^n c_i^2}\right).$$

Bibliography

Anthony, M. and Bartlett, P. L. (2009), *Neural Network Learning: Theoretical Foundations*, cambridge University Press.

Bach, F. (2017), 'Breaking the curse of dimensionality with convex neural networks', *Journal of Machine Learning Research* **18**(19), 1–53.

Bottou, L., Curtis, F. E. and Nocedal, J. (2018), 'Optimization methods for large-scale machine learning', *SIAM Review* **60**(2), 223–311.

Cauchy, A. et al. (1847), 'Méthode générale pour la résolution des systemes d'équations simultanées', *Comp. Rend. Sci. Paris* **25**(1847), 536–538.

Cybenko, G. (1989), 'Approximation by superpositions of a sigmoidal function', *Mathematics of Control, Signals and Systems* **2**(4), 303–314.

Davidson, K. R. and Donsig, A. P. (2010), *Real Analysis and Applications: Theory in Practice*, Springer Science & Business Media.

DeVore, R. A., Howard, R. and Micchelli, C. (1989), 'Optimal nonlinear approximation', *Manuscripta mathematica* **63**, 469–478.

DeVore, R., Hanin, B. and Petrova, G. (2021), 'Neural network approximation', *Acta Numerica* **30**, 327–444.

Donoghue, W. F. (1969), *Distributions and Fourier transforms*, Vol. 32, Academic Press.

Dziugaite, G. K. and Roy, D. M. (2017), Computing nonvacuous generalization bounds for deep (stochastic) neural networks via pac-bayes, *in* 'Proceedings of the Thirty-Third Conference on Uncertainty in Artificial Intelligence (UAI 2017)'.

Evans, L. C. (2010), *Partial Differential Equations*, 2nd edn, American Mathematical Society.

Facenda Aguirre, J. A. (1989), 'A note on taylor's theorem', *The American Mathematical Monthly* **96**(3), 244–247.

Folland, G. B. (2005), 'Higher-order derivatives and taylor's formula in several variables', *Preprint* **1**.

Funahashi, K.-I. (1989), 'On the approximate realization of continuous mappings by neural networks', *Neural Networks* **2**(3), 183–192.

Ghadimi, E., Feyzmahdavian, H. R. and Johansson, M. (2015), Global convergence of the heavy-ball method for convex optimization, *in* 'Proc. of European Control Conference (ECC)', IEEE, pp. 310–315.

Golowich, N., Rakhlin, A. and Shamir, O. (2020), 'Size-independent sample complexity of neural networks', *Information and Inference: A Journal of the IMA* **9**(2), 473–504.

Goodfellow, I., Bengio, Y. and Courville, A. (2016), *Deep Learning*, MIT Press, Cambridge, MA.

Gordon, Y., Maiorov, V., Meyer, M. and Reisner, S. (2002), 'On the best approximation by ridge functions in the uniform norm', *Constructive approximation* **18**, 61–85.

Hanin, B. and Sellke, M. (2017), 'Approximating continuous functions by relu nets of minimal width', *arXiv preprint arXiv:1710.11278* .

Horn, R. A. and Johnson, C. R. (2013), *Matrix Analysis*, Cambridge University Press.

Hornik, K., Stinchcombe, M. and White, H. (1989), 'Multilayer feedforward networks are universal approximators', *Neural Networks* **2**(5), 359–366.

Hwang, G. (2023), 'Minimum width for deep, narrow mlp: A diffeomorphism and the whitney embedding theorem approach', *arXiv preprint arXiv:2308.15873* .

Jang, U. and Ryu, E. K. (2025), 'Point convergence of Nesterov's accelerated gradient method: An AI-assisted proof', *arXiv preprint arXiv:2510.23513* .

Kainen, P. C., Kůrková, V. and Vogt, A. (1999), 'Approximation by neural networks is not continuous', *Neurocomputing* **29**(1-3), 47–56.

Khaled, A. and Richtárik, P. (2020), 'Better theory for SGD in the nonconvex world', *arXiv preprint arXiv:2002.03329* .

Khalil, H. K. (2002), *Nonlinear Systems*, Prentice Hall.

Kidger, P. and Lyons, T. (2020), Universal approximation with deep narrow networks, *in* 'Conference on learning theory', PMLR, pp. 2306–2327.

Kim, N., Min, C. and Park, S. (2024), Minimum width for universal approximation using relu networks on compact domain, *in* 'The Twelfth International Conference on Learning Representations'.

Kolmogorov, A. N. (1957), On the representation of continuous functions of many variables by superposition of continuous functions of one variable and addition, *in* 'Doklady Akademii Nauk', Vol. 114, Russian Academy of Sciences, pp. 953–956.

Ledoux, M. and Talagrand, M. (1991), *Probability in Banach Spaces: Isoperimetry and Processes*, Vol. 23, Springer Science & Business Media.

Lee, J. D., Simchowitz, M., Jordan, M. I. and Recht, B. (2016), Gradient descent only converges to minimizers, *in* 'Conference on Learning Theory', PMLR, pp. 1246–1257.

Leshno, M., Lin, V. Y., Pinkus, A. and Schocken, S. (1993), 'Multilayer feedforward networks with a nonpolynomial activation function can approximate any function', *Neural Networks* **6**(6), 861–867.

Lessard, L., Recht, B. and Packard, A. (2016), 'Analysis and design of optimization algorithms via integral quadratic constraints', *SIAM Journal on Optimization* **26**(1), 57–95.

Lin, V. Y. and Pinkus, A. (1993), 'Fundamentality of ridge functions', *Journal of Approximation Theory* **75**(3), 295–311.

Liu, J. (2021), 'On the convergence of reinforcement learning with monte carlo exploring starts', *Automatica* **129**, 109693.

Liu, J. and Yuan, Y. (2022), On almost sure convergence rates of stochastic gradient methods, *in* 'Conference on Learning Theory', PMLR, pp. 2963–2983.

Liu, J. and Yuan, Y. (2024), 'Almost sure convergence rates analysis and saddle avoidance of stochastic gradient methods', *Journal of Machine Learning Research* **25**(271), 1–40.

Lorentz, G. G., von Golitschek, M. and Makovoz, Y. (1996), *Constructive Approximation: Advanced Problems*, Springer.

Lu, J., Shen, Z., Yang, H. and Zhang, S. (2021), 'Deep network approximation for smooth functions', *SIAM Journal on Mathematical Analysis* **53**(5), 5465–5506.

Lu, Z., Pu, H., Wang, F., Hu, Z. and Wang, L. (2017), 'The expressive power of neural networks: A view from the width', *Advances in neural information processing systems* **30**.

Maiorov, V. (2010), 'Best approximation by ridge functions in l p-spaces', *Ukrainian Mathematical Journal* **62**, 452–466.

Maiorov, V. E. (1999), 'On best approximation by ridge functions', *Journal of Approximation Theory* **99**(1), 68–94.

Maiorov, V. and Meir, R. (2000), 'On the near optimality of the stochastic approximation of smooth functions by neural networks', *Advances in Computational Mathematics* **13**, 79–103.

Maiorov, V. and Pinkus, A. (1999), 'Lower bounds for approximation by mlp neural networks', *Neurocomputing* **25**(1-3), 81–91.

McCulloch, W. S. and Pitts, W. (1943), 'A logical calculus of the ideas immanent in nervous activity', *The Bulletin of Mathematical Biophysics* **5**(4), 115–133.

Mhaskar, H. N. (1996), 'Neural networks for optimal approximation of smooth and analytic functions', *Neural Computation* **8**(1), 164–177.

Minsky, M. and Papert, S. (1969), *Perceptrons: An Introduction to Computational Geometry*, MIT Press, Cambridge, MA.

Mohri, M., Rostamizadeh, A. and Talwalkar, A. (2018), *Foundations of Machine Learning*, second edn, The MIT Press.

Nesterov, Y. (1983), A method for unconstrained convex minimization problem with the rate of convergence $o(1/k^2)$, *in* 'Dokl. Akad. Nauk. SSSR', Vol. 269, p. 543.

Nesterov, Y. (2013), *Introductory Lectures on Convex Optimization: A Basic Course*, Springer Science & Business Media.

Ngo, H. Q. (2010), 'Three proofs of sauer-shelah lemma', *lecture notes, http://www. cse. buffalo. edu/~ hungngo/classes/2010/711/lectures/sauer. pdf* **199**.

Park, S., Yun, C., Lee, J. and Shin, J. (2021), Minimum width for universal approximation, *in* 'International Conference on Learning Representations'.

Pinkus, A. (1999), 'Approximation theory of the mlp model in neural networks', *Acta numerica* **8**, 143–195.

Polyak, B. T. (1964), 'Some methods of speeding up the convergence of iteration methods', *Ussr computational mathematics and mathematical physics* **4**(5), 1–17.

Robbins, H. and Siegmund, D. (1971), A convergence theorem for non negative almost supermartingales and some applications, *in* 'Optimizing methods in statistics', Elsevier, pp. 233–257.

Sellke, M. (2024), 'On size-independent sample complexity of relu networks', *Information Processing Letters* **186**, 106482.

Shalev-Shwartz, S. and Ben-David, S. (2014), *Understanding Machine Learning: From Theory to Algorithms*, Cambridge university press.

Shen, Z., Yang, H. and Zhang, S. (2019), 'Nonlinear approximation via compositions', *Neural Networks* **119**, 74–84.

Su, W., Boyd, S. and Candes, E. J. (2016), 'A differential equation for modeling nesterov's accelerated gradient method: Theory and insights', *Journal of Machine Learning Research* **17**(153), 1–43.

Telgarsky, M. (2015), 'Representation benefits of deep feedforward networks', *arXiv preprint arXiv:1509.08101* .

Valiant, L. G. (1984), 'A theory of the learnable', *Communications of the ACM* **27**(11), 1134–1142.

Vapnik, V. and Chervonenkis, A. Y. (1971), 'On the uniform convergence of relative frequencies of events to their probabilities', *Theory of Probability & Its Applications* **16**(2), 264–280.

Vostrecov, B. and Kreines, M. (1961), Approximation of continuous functions by superpositions of plane waves, *in* 'Dokl. Akad. Nauk SSSR', Vol. 140, pp. 1237–1240.

Wainwright, M. J. (2019), *High-dimensional statistics: A non-asymptotic viewpoint*, Vol. 48, Cambridge University Press.

Wilson, A. C., Recht, B. and Jordan, M. I. (2021), 'A Lyapunov analysis of accelerated methods in optimization', *Journal of Machine Learning Research* **22**(113), 1–34.

Yarotsky, D. (2017), 'Error bounds for approximations with deep relu networks', *Neural networks* **94**, 103–114.

Yarotsky, D. (2018), Optimal approximation of continuous functions by very deep relu networks, *in* 'Conference on learning theory', PMLR, pp. 639–649.

Zhang, S., Lu, J. and Zhao, H. (2024), 'Deep network approximation: Beyond relu to diverse activation functions', *Journal of Machine Learning Research* **25**(35), 1–39.